



SAS® Visual Text Analytics in SAS® Viya®

Slides and Notes

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SAS® Visual Text Analytics in SAS® Viya® Slides and Notes

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Course Overview



Course Overview

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Results in SAS Viya

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Lesson 1: Introduction to SAS Visual Text Analytics



1.0 Lesson Overview



Lesson Overview

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1.1 SAS Visual Text Analytics



Overview of Text Analytics



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SAS® Visual Text Analytics



- Sepsis indicators
- Painkiller addict
- Environmental influences

- Is product meeting quality expectations?

- Terrorism
- Cyber bullying
- At-risk persons

- What amenities influence bookings?

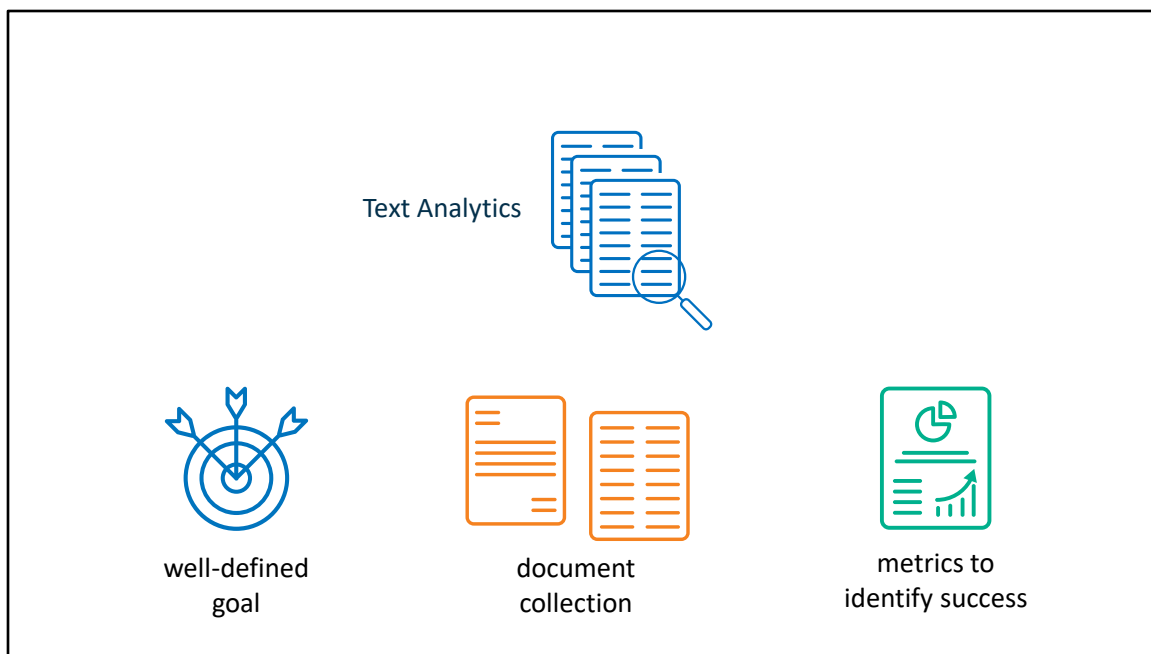
Why is SAS Visual Text Analytics important? SAS Visual Text Analytics automatically converts unstructured data into meaningful insights. Considering the staggering amount of unstructured data that's generated every day, automation is critical to fully analyze text and speech data efficiently.

Sepsis is the leading cause of death in hospitals, and it must be detected early or it can be life threatening. Sepsis is an inflammatory response to infection. The toxins from bacteria in the body release into the bloodstream to fight the threat, causing systemic inflammation that can slow blood flow and damage organs. If sepsis progresses to the later stages of septic shock, it can be life threatening. If sepsis is properly treated in a timely manner, we've seen that we can save up to 80 percent of the lives that we're losing. Taking advantage of data from the electronic medical records of all patients in its facilities, the solution uses natural language processing and a rules engine to continually monitor factors that could indicate a sepsis infection.

When it comes to customer reviews, one could simply report in terms of a numerical metric "Average Customer Review: 4 stars". However, when customer comments are analyzed, or patterns in those comments are extracted, a better picture of the product is obtained.

Consider social media like Facebook and Twitter. Unstructured data has a great amount of information. If text patterns are extracted and series of behaviors detected, then interventions for cyber bullying, sexual predatory behavior, and terrorism planning could be done timely.

Think about web sites like TripAdvisor. Reviews from Airbnb from specific geographic areas can help determine how important amenities are to securing bookings and customer satisfaction.

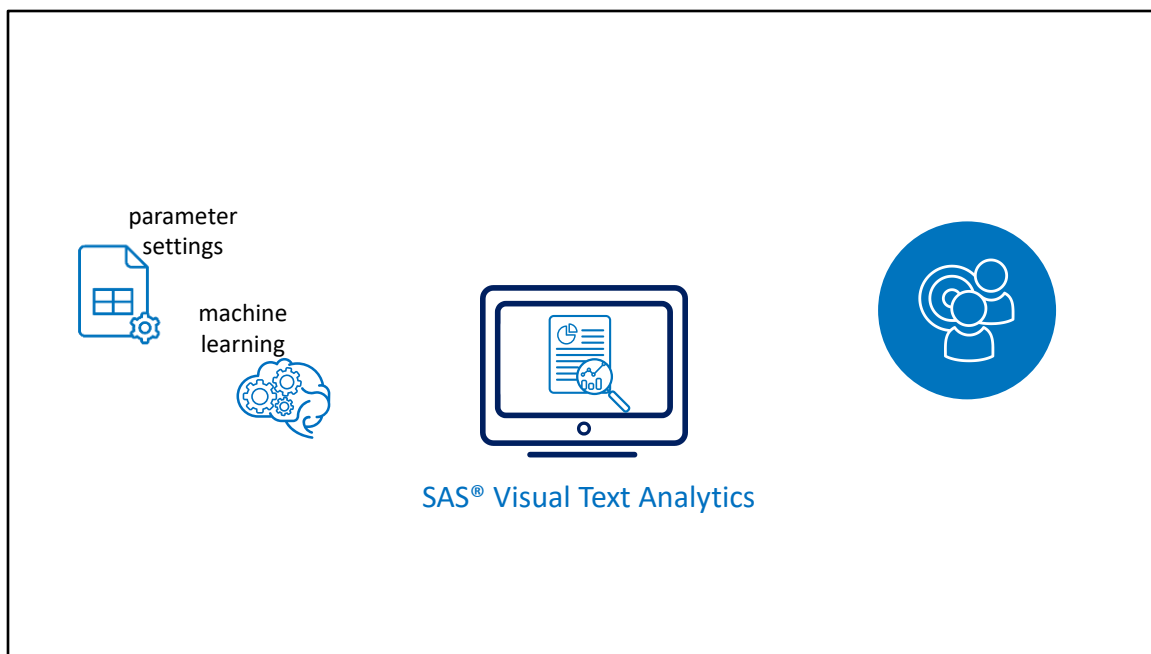


Text analytics project requirements include a well-defined goal, document collection, and metrics to identify success.

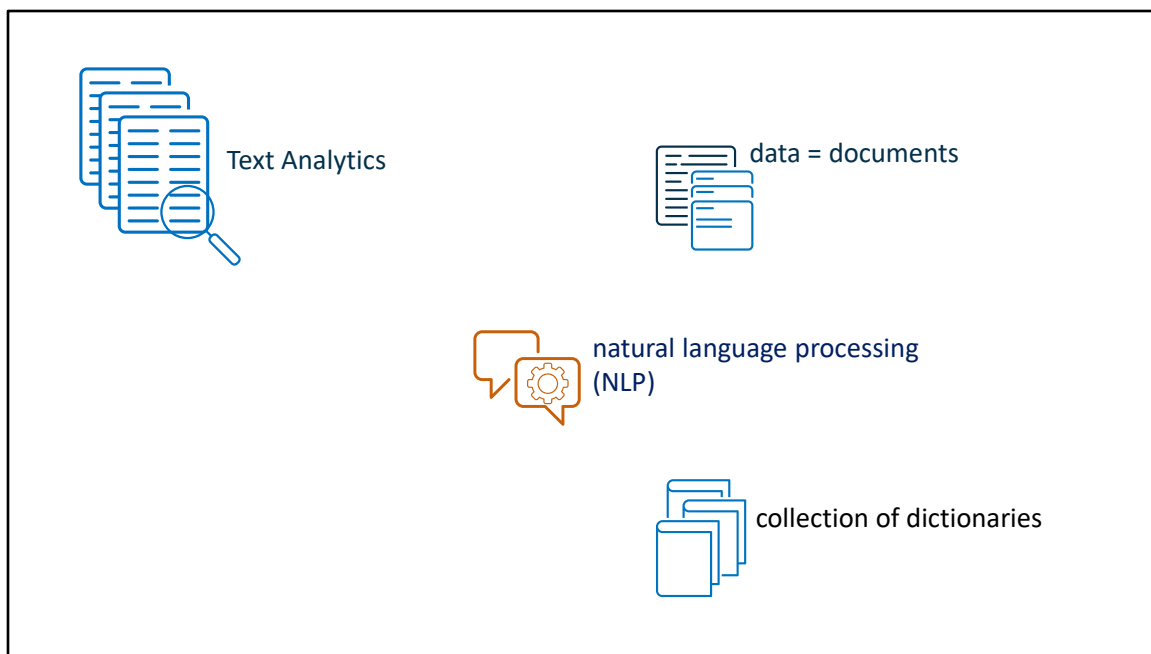


SAS® Visual Text Analytics

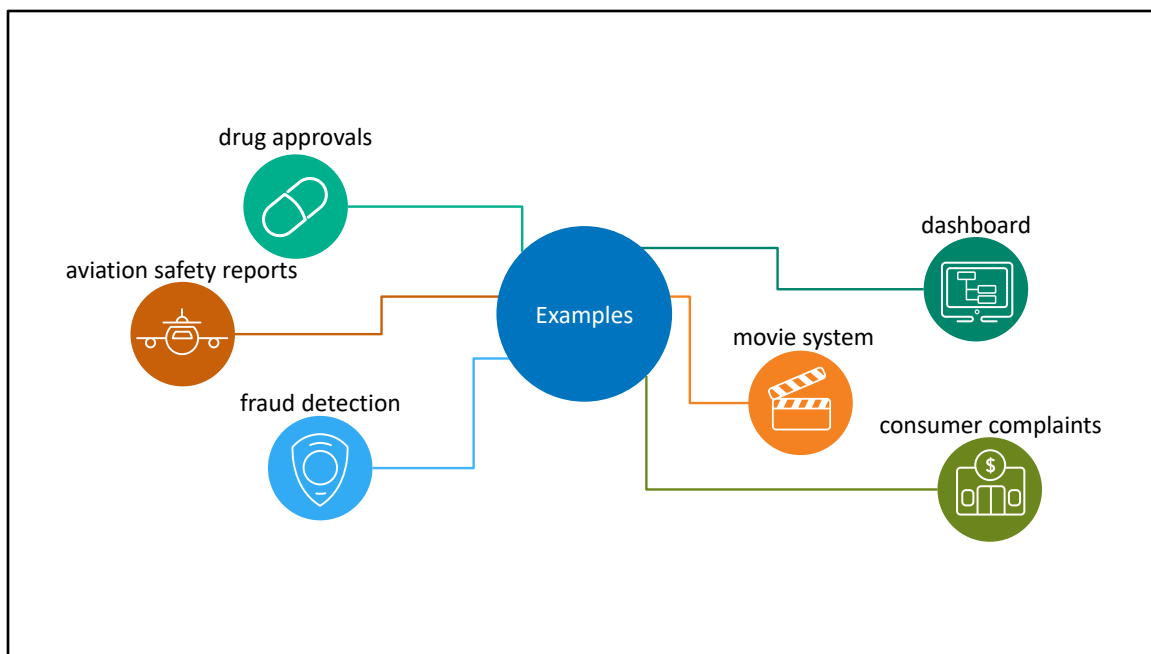
SAS Visual Text Analytics is a high-end software product that simplifies the use of text analytics for many different types of users, from business analysts to document librarians.



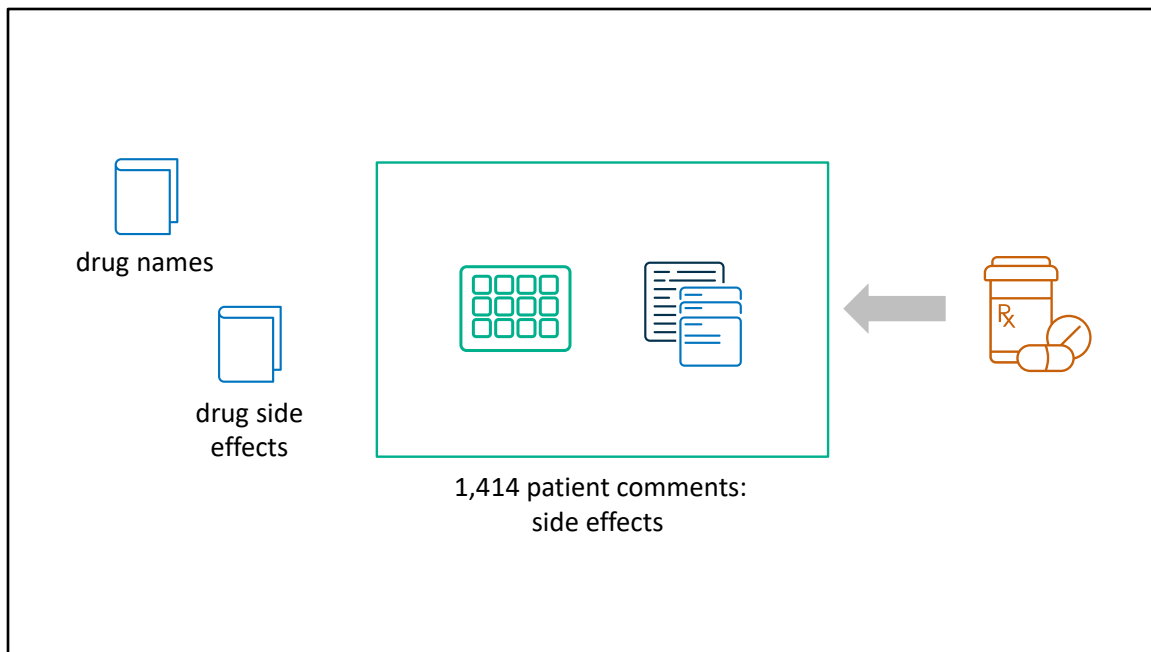
By choosing intelligent parameter settings, or by using machine-learning tools to “learn” an appropriate setting, Visual Text Analytics frees the user to concentrate on the immediate task.



The principal analysis object in text analytics is a document. Data for a text analytics project consists of a collection of documents. A feature of SAS Visual Text Analytics is the ability to convert document collections stored in a variety of forms into a form that can be processed using the software. A key algorithmic component of text analytics is natural language processing. In addition, projects are aided by dictionaries that encapsulate domain expertise. They exploit human knowledge that is related to the documents. A key to success is software that can rapidly and consistently process, analyze, and score documents.



Many examples of successful text analytics projects exist. In regulatory filings for drug approvals, you can find all dosage amounts for medications that are mentioned in the application. In order to promote customer satisfaction, you can populate a dashboard with the top 10 customer complaints for the week. To expedite processing aviation safety reports, you can identify all reports that involve a runway incursion. You can enhance the accuracy of a movie recommendation system by exploiting the relationships between movie synopses and customer movie ratings. You can enhance the accuracy of a fraud and abuse detection system for workers' compensation insurance by exploiting the relationships between adjuster notes, policy-based model inputs, claim-based model inputs, and a fraud target variable. You could rapidly and consistently identify all consumer complaints that are related to home loans.



In our first demonstration, we're going to consider a text analytics project that seeks to extract drug dosages from a collection of customer feedback documents. The data set contains 1,414 patient comments about drug side effects. The documents are in the English language. Two dictionaries are available, one that contains drug names, and another that contains a drug's side effects. The software employed to analyze the documents is SAS Visual Text Analytics.

Demo: Extracting Drug Dosages from User-Feedback Documents, Part 1

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Demo: Extracting Drug Dosages from User-Feedback Documents, Part 2

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Demo: Extracting Drug Dosages from User-Feedback Documents, Part 3

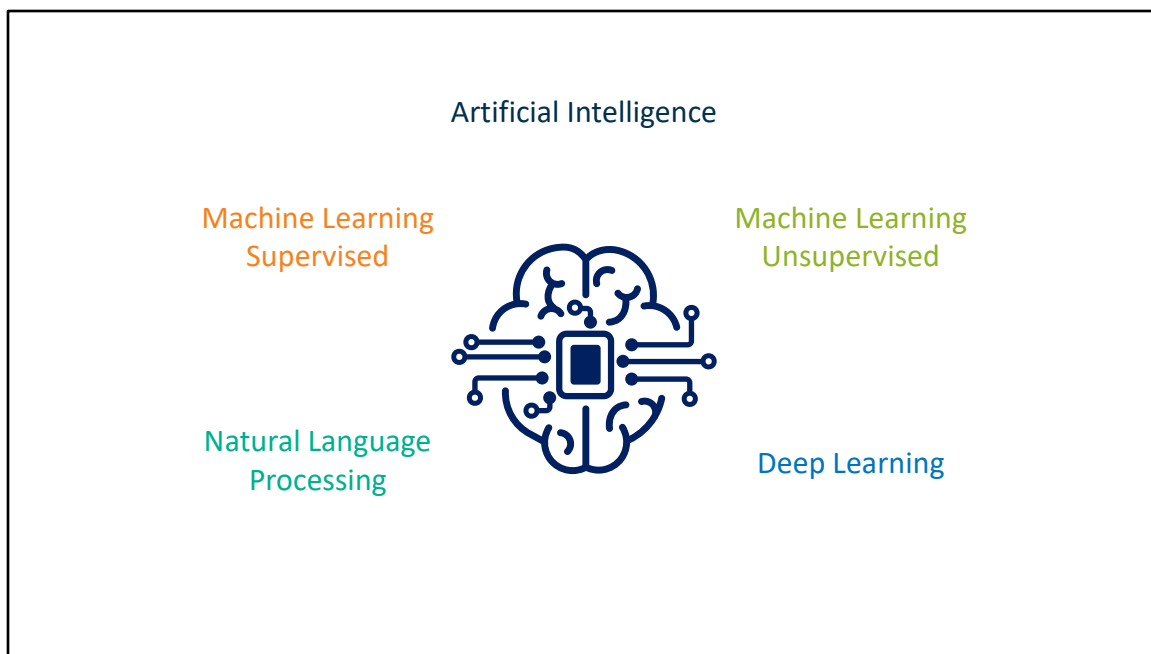
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Overview of Natural Language Processing

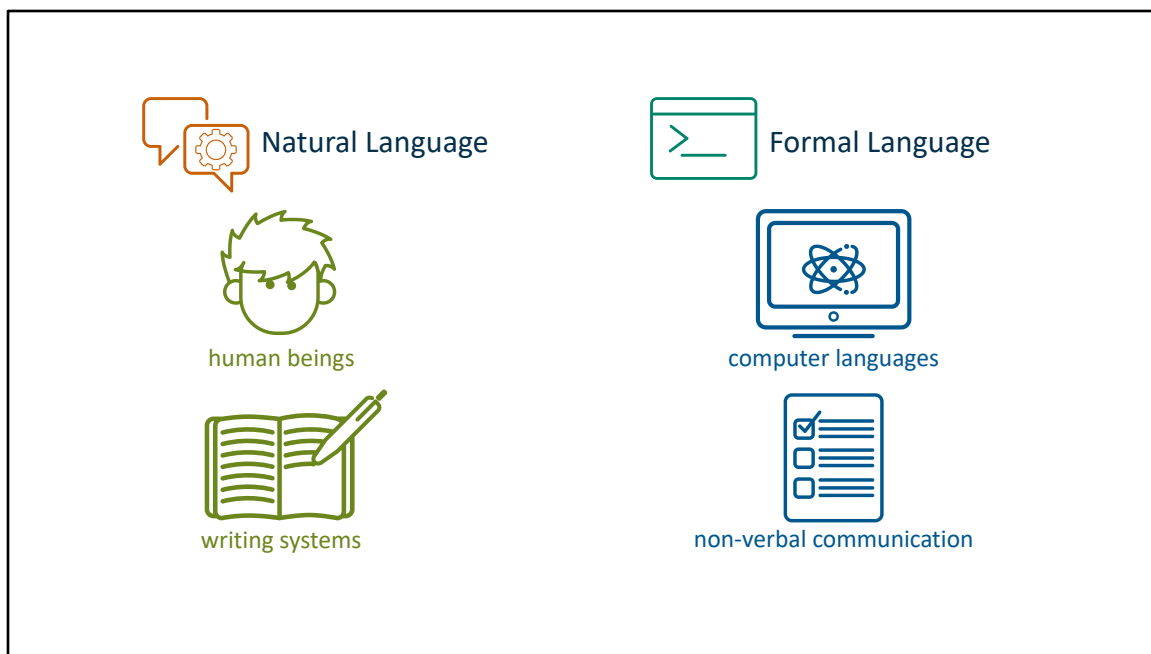


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Artificial Intelligence is a scientific discipline that explores how human intelligence functions, with the goal of replicating human intelligence using machines. There are many subfields of artificial intelligence, including machine learning, natural language processing, and deep learning. Natural language processing, abbreviated NLP, benefits from research in computer science, computational linguistics, and other areas to bridge the gap between human communication and computer understanding.

While supervised and unsupervised learning, and specifically deep learning, are now widely used for modeling human language, there's also a need for syntactic and semantic understanding and domain expertise that are not necessarily present in these machine learning approaches.



Natural language processing is the key algorithmic component to SAS Visual Text Analytics. *Natural language* refers to verbal language spoken by human beings, for example, English, Spanish, French, and so on. Most human languages have been encoded using a writing system. *Formal language* refers to C, C++, C#, Java, and other computer languages, or any man-made language designed for any purpose other than verbal communication between human beings.

A formal language has rigid syntactical rules to avoid ambiguities. Some references use the term *artificial language* rather than formal language. However, artificial language can also refer to languages constructed for purposes other than communication.

Artificial Intelligence



Natural Language Processing (NLP)



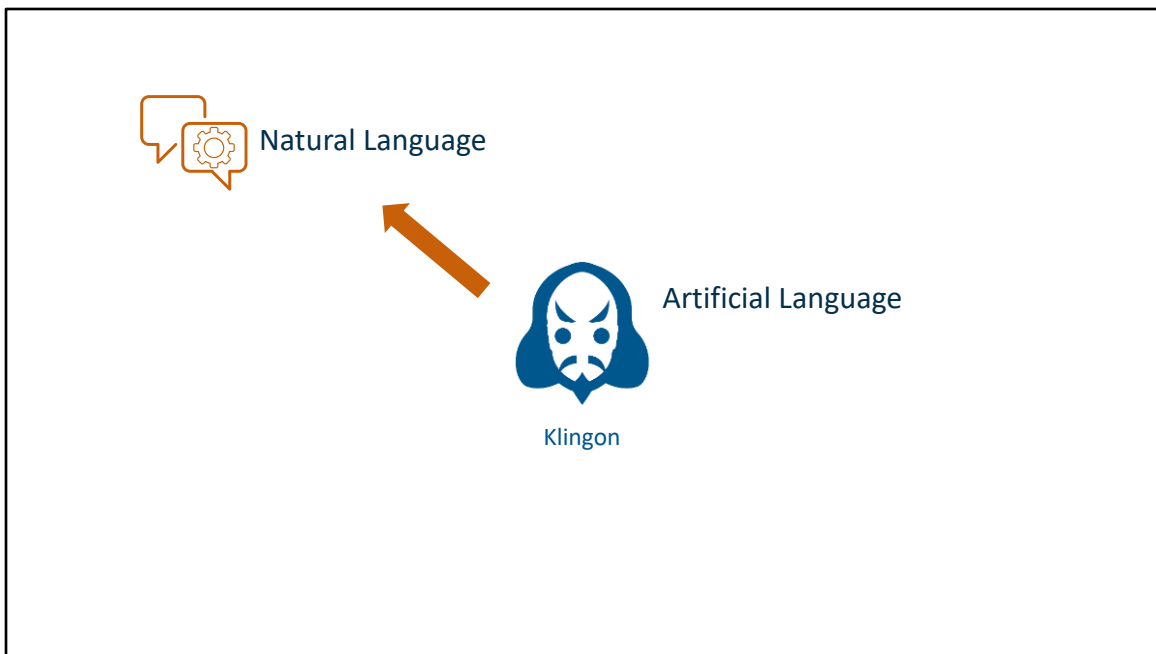
natural language



specific purposes



Natural language processing uses computers to process natural language for specific purposes. For example, one specific purpose is to identify components of a language, such as parts-of-speech in the English language.



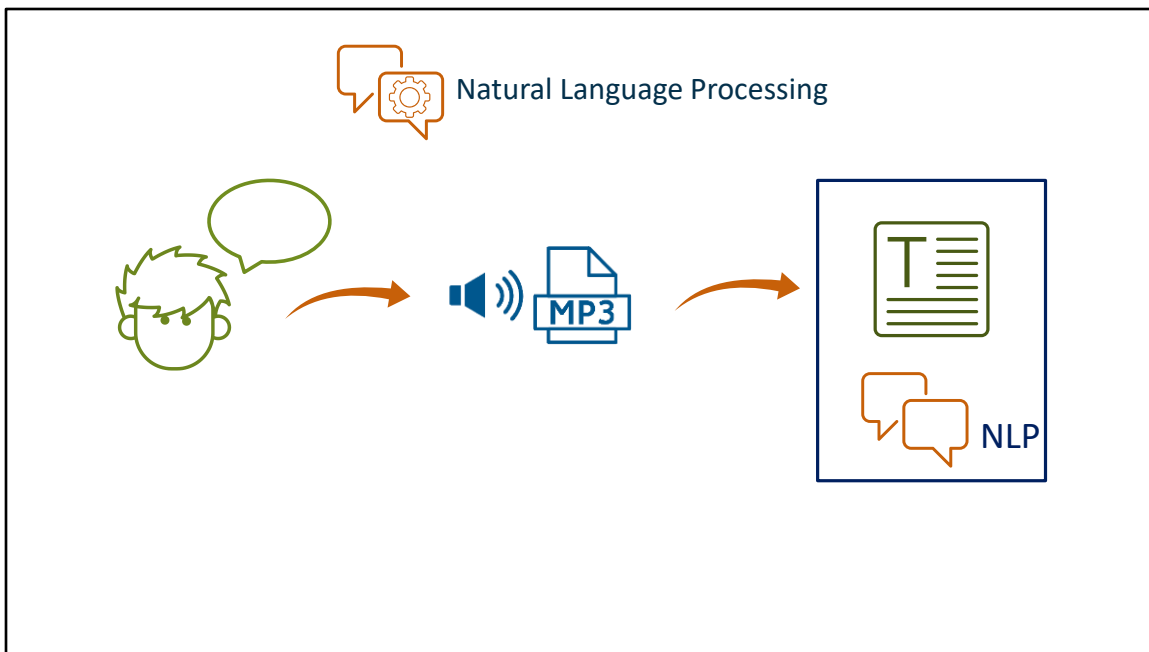
There are invented languages like Klingon developed for the Star Trek science fiction television and movie series. Such languages are often called artificial languages, but in our taxonomy, an artificial language would be treated just like a natural language.



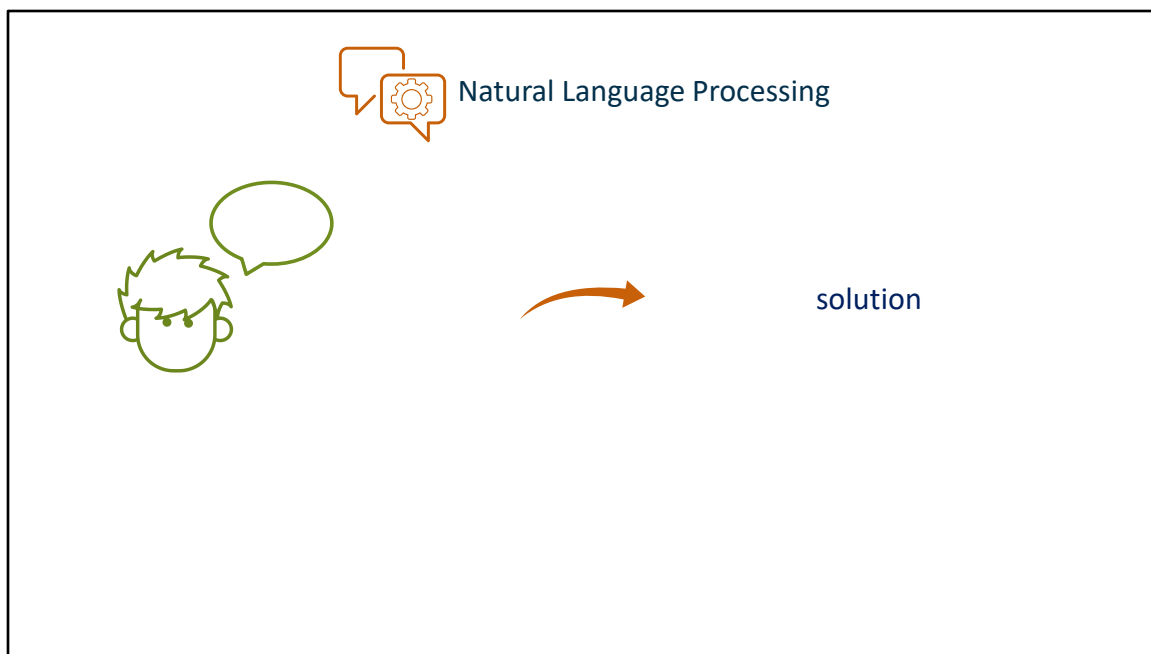
Natural Language Processing



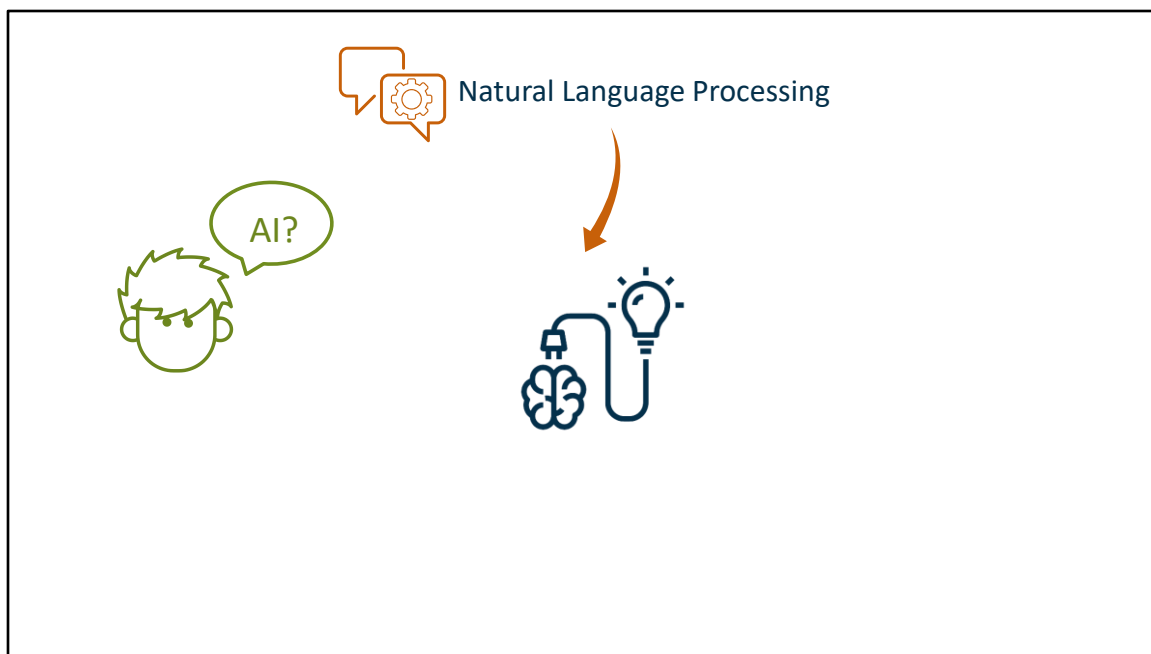
The goal of NLP is to process both spoken and written language.



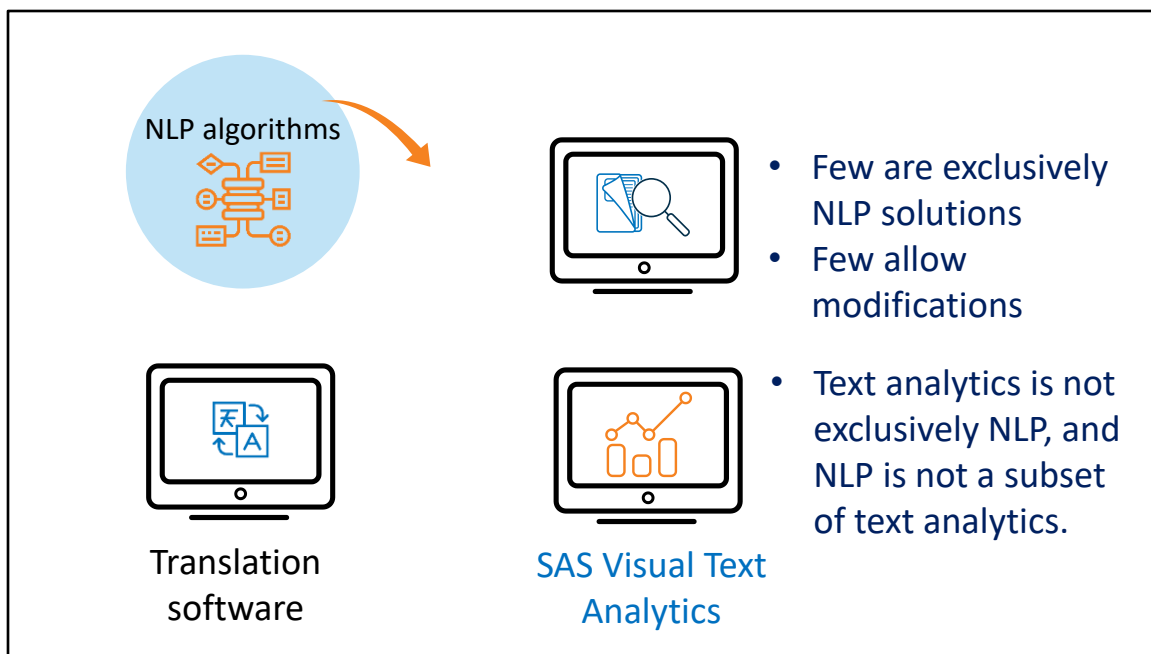
If you think about the process of going from spoken language to text analytics, the first step is to convert spoken language to digitized audio. For example, you could record a conversation as an MP3 file. The second step is to convert the digitized audio of the spoken language to written language. Speech recognition software performs this conversion. The last step is to process the written language using Natural Language Processing.



Some complete NLP solutions, like Apple Siri and Amazon Alexa, proceed directly from spoken language to solution.



One goal of SAS Visual Text Analytics is to unleash the power of natural language processing without burdening the user with prerequisite knowledge of artificial intelligence.



Text analytics software includes integrated NLP algorithms. NLP algorithms are also used in text processing software not intended for analytical use, such as translation software. Many advanced software products use NLP, and some of these would not be classified as text analytics software products.

Few software products are exclusively NLP solutions. Many software products that use NLP do not permit users to modify hyperparameters that are employed by the NLP algorithms.

SAS Visual Text Analytics and other SAS text analytics products use NLP algorithms embedded within the software. Text analytics is not exclusively NLP, and NLP is not a subset of text analytics.

Natural Language Processing

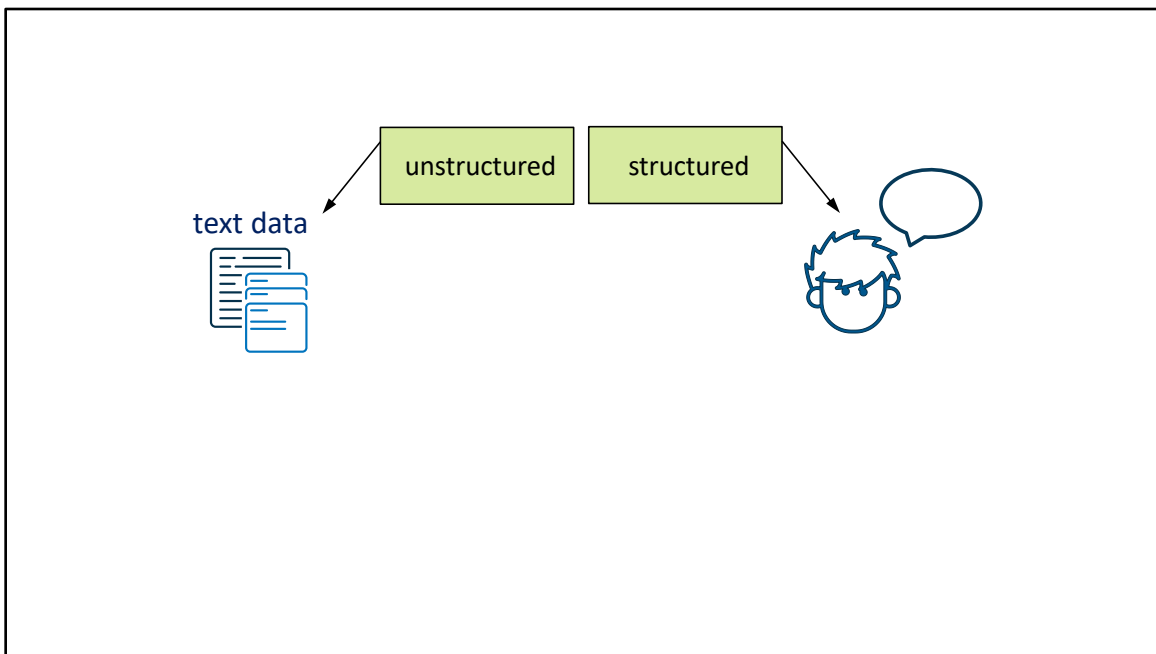
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Unstructured Text to Structured Data



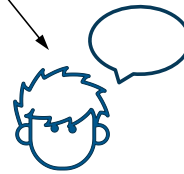
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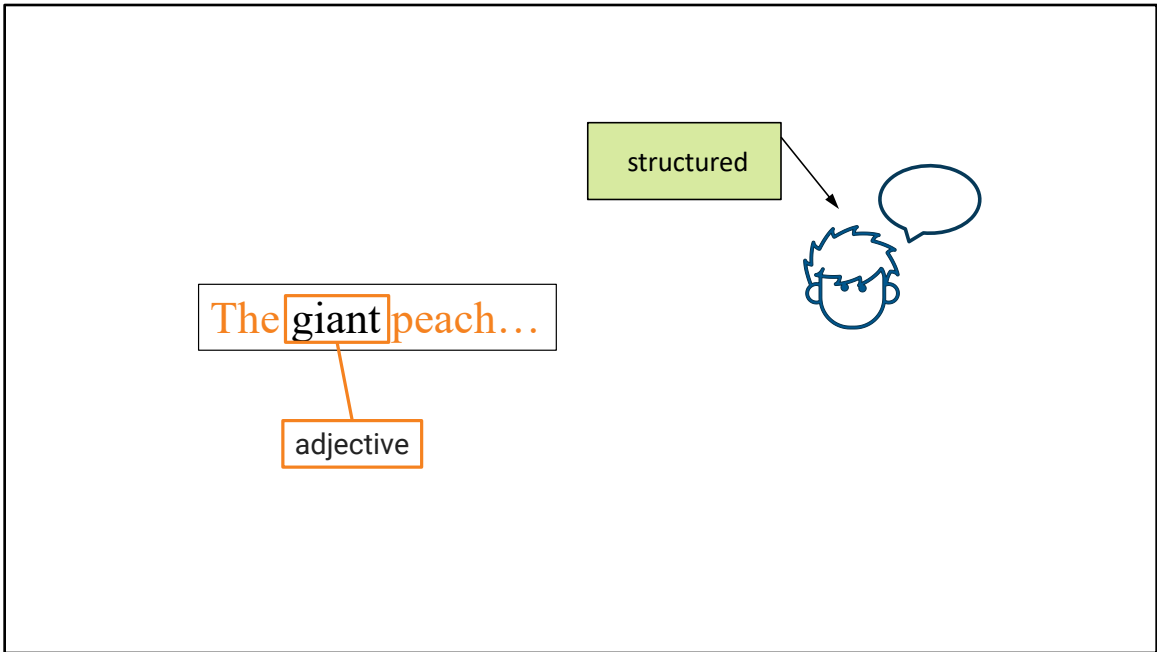
Text data are often described as unstructured data. Human language actually has a lot of structure.

The giant peach...

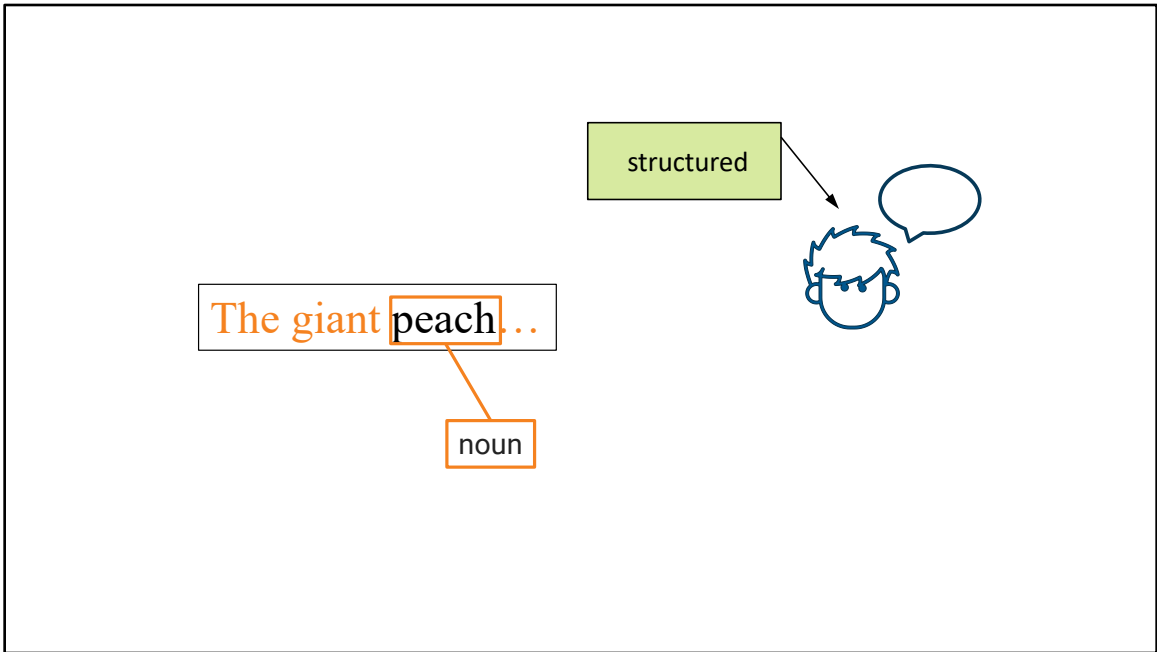
structured



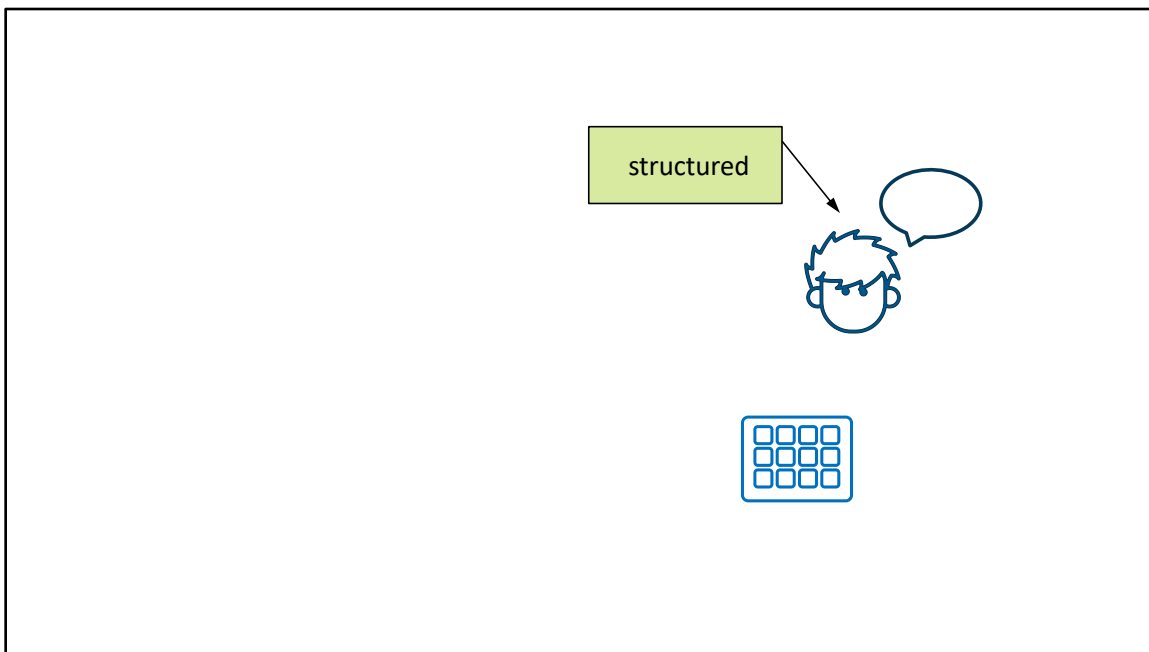
In the English language,



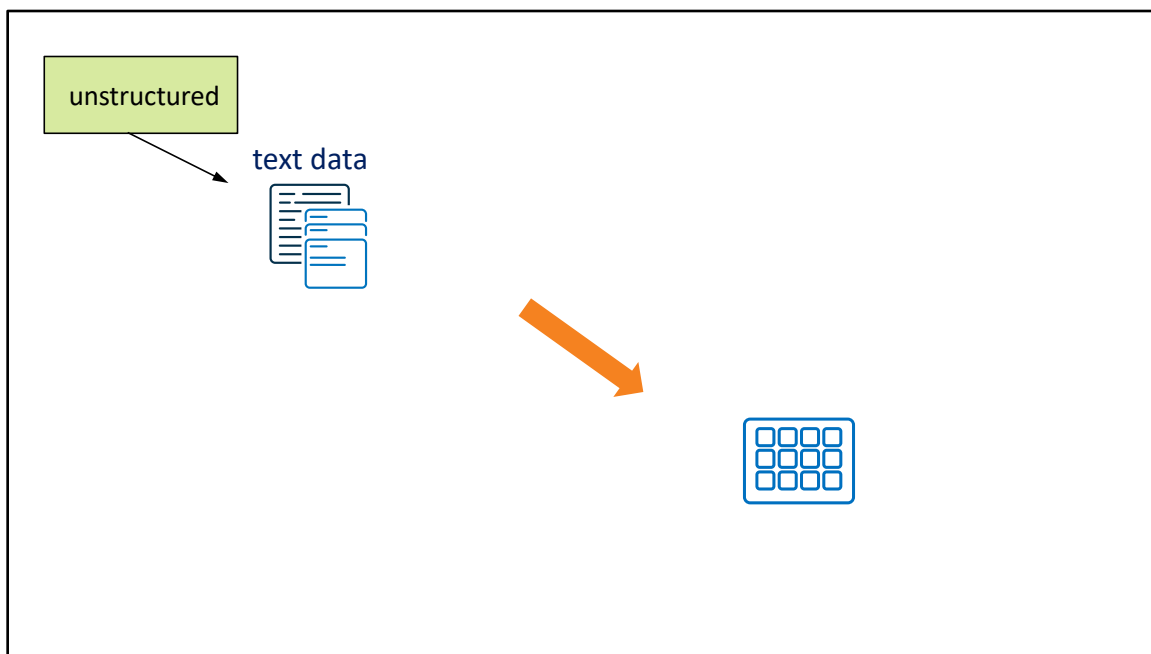
if you detect an adjective,



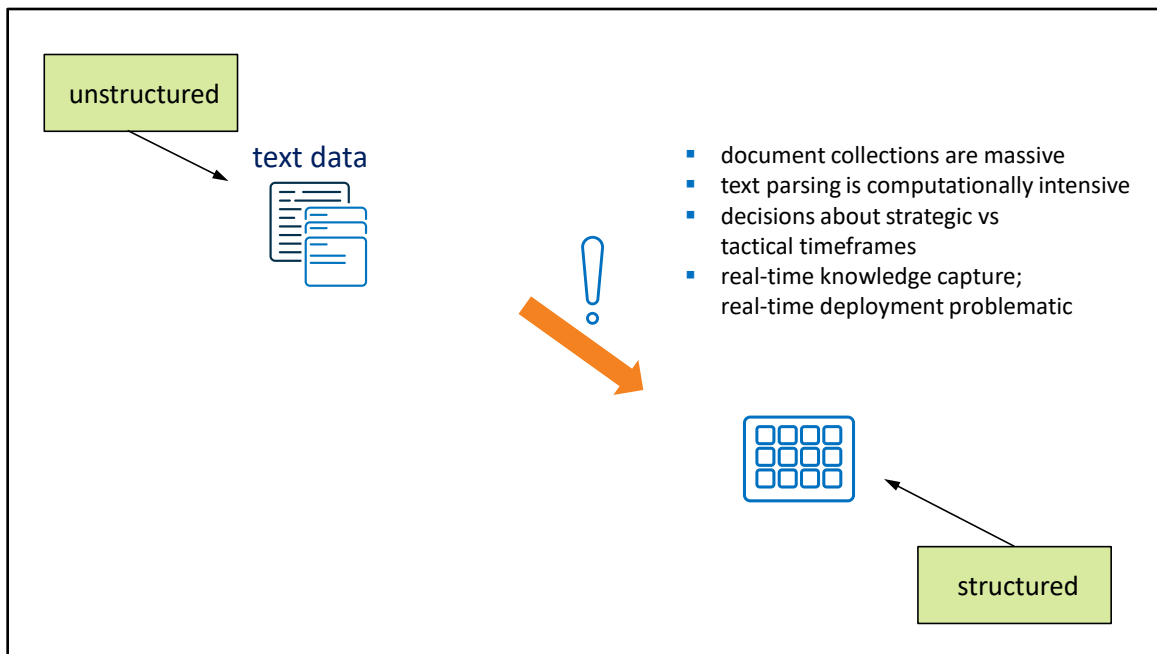
you can expect a noun to follow. In the Spanish language, an adjective comes after a noun.



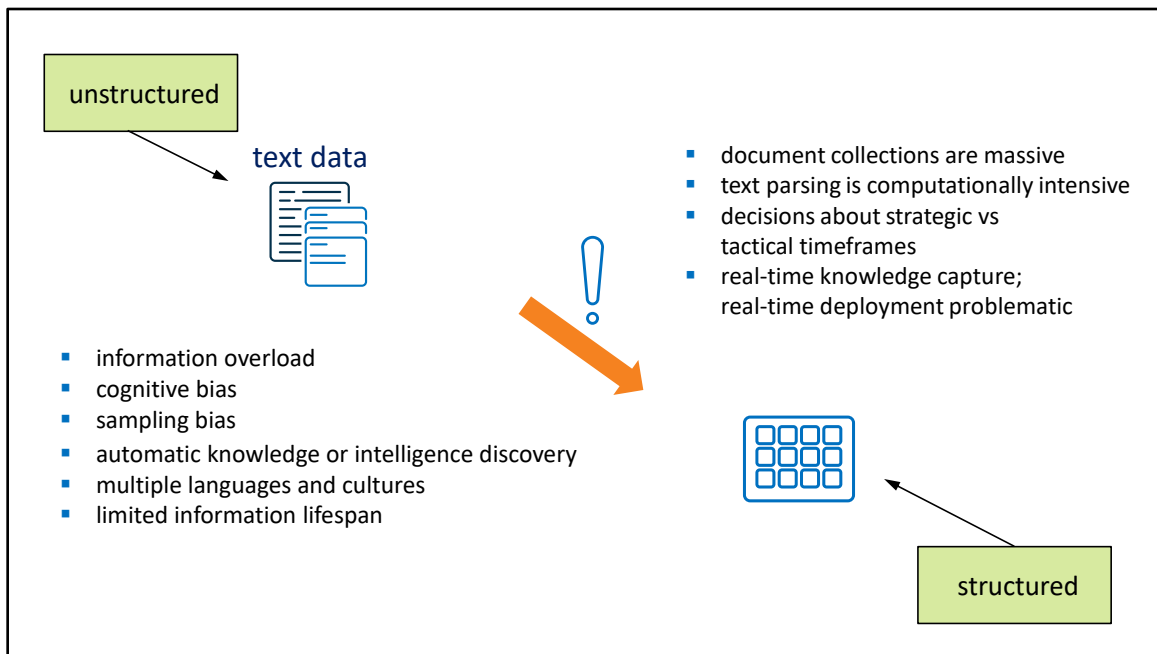
In the field of analytics, structure refers to a data set. Data sets are usually defined by simple row-column structures, or by hierarchical structures. Such data set structures are rigid and well-defined.



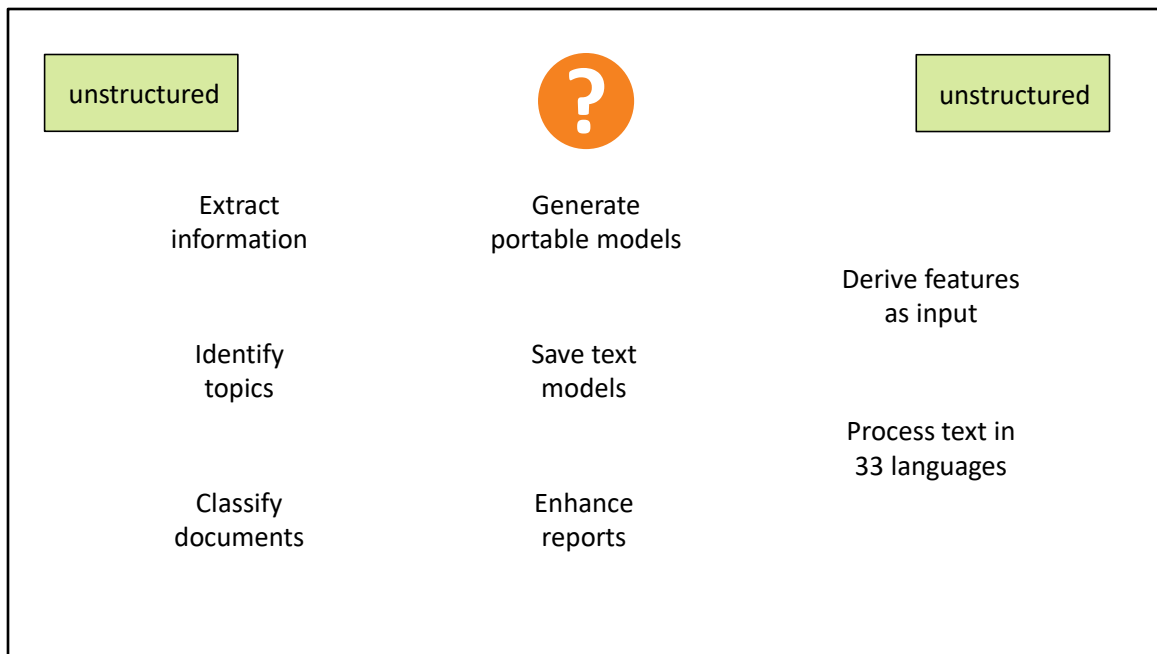
One goal of text analytics is to convert the weakly structured text data into rigidly structured analytics data.



There are many challenges to this conversion. Often document collections are massive. Text parsing is very computationally intensive. Consequently, processing times can be very lengthy. This leads to decisions about strategic versus tactical time frames. For data collection, you may require real-time knowledge capture. However, real-time deployment may be problematic. This is because conversions and translations are difficult to perform in real time.



The practical challenges to this conversion are information overload, cognitive bias, sampling bias (for example, social media), automatic knowledge or intelligence discovery, multiple languages and cultures, and limited information lifespan.



What can you do with unstructured and semi-structured text?

Extract information (including feature-level sentiment). Identify topics. Classify documents (including document-level sentiment). Generate portable models for contextual extraction, topics, categorization and sentiment. Save text models for deployment in batch, via API, in stream, or in Hadoop. Enhance business reports via native integration with data visualization. Derive new features as input for predictive modeling or artificial intelligence. Process text in 33 languages.

Unstructured Text to Structured Data: Practical Challenges

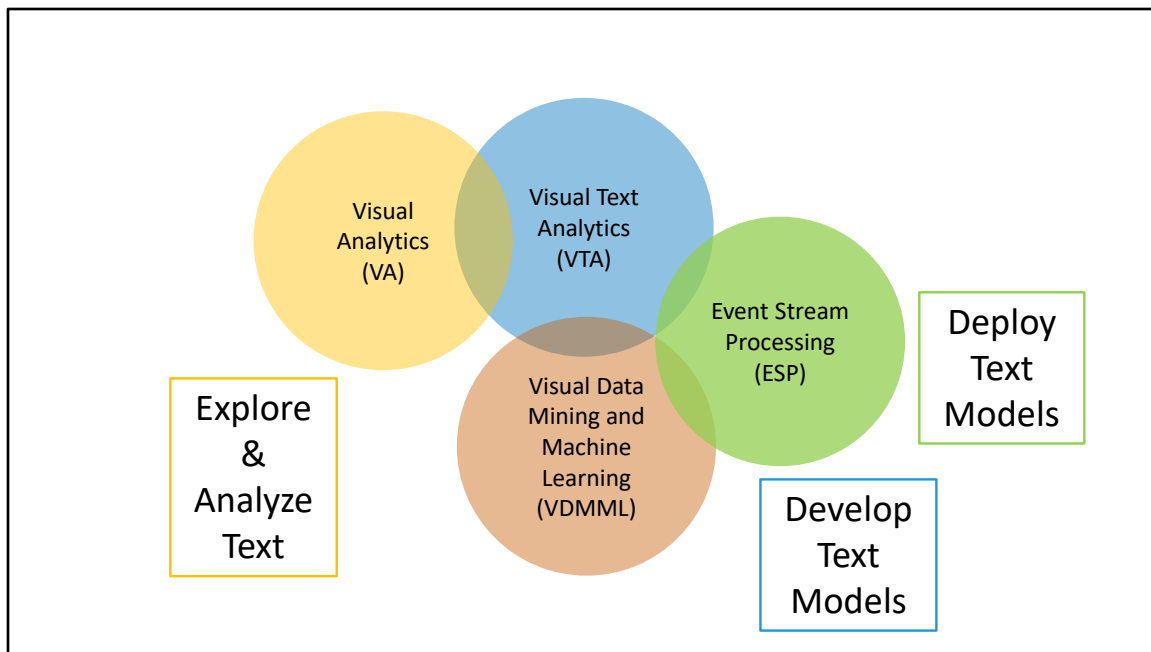
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SAS Products with Embedded Text Analytics Capabilities



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Text Analytics capabilities are found across multiple solutions at SAS, so how do you know which to use for a given situation?

Visual Text Analytics is our core solution for analyzing text, and contains the **most comprehensive** set of capabilities for exploring a document collection and developing text models for production.

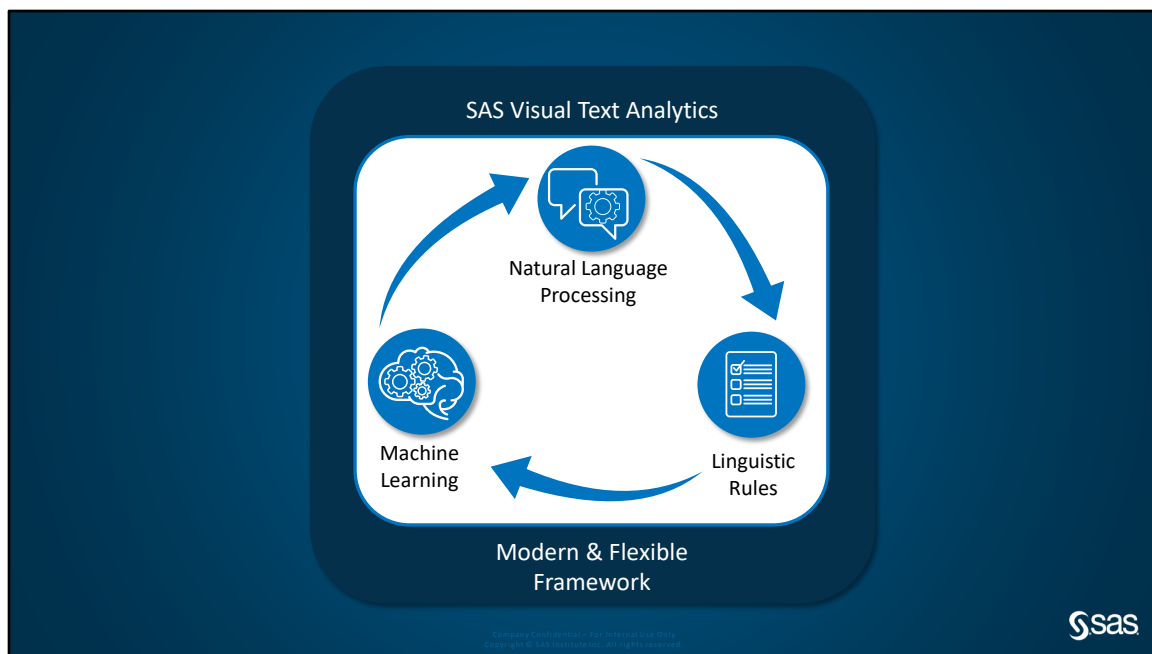
However, text capabilities have been embedded into other solutions in a limited fashion as well, and we'll touch on each of those intersections in a moment. The key thing to take away is that you can explore and analyze text collections in VA, but it's not intended for model creation...that is best left to VTA and VDMML. While VDMML contains only a subset of the text functionality that VTA does, it is well suited for incorporating text into other advanced analytics projects.

Finally, ESP can apply your text models to streaming data and thus enrich it, but ESP is not the interface where you would actually perform any text analytics itself.

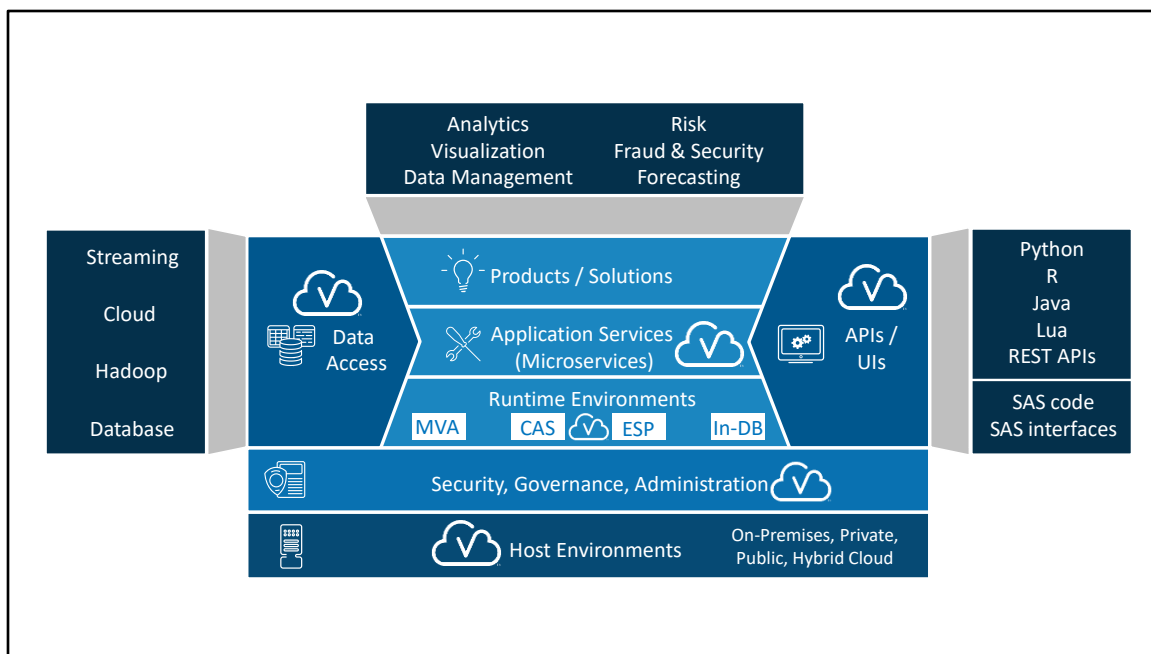
What Is SAS Visual Text Analytics?



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SAS Visual Text Analytics is the SAS offering designed to effectively extract insights from unstructured data in large scale. Offered on the SAS Viya architecture, Visual Text Analytics combines the power of natural language processing (NLP), machine learning (ML), and linguistic rules (algorithmically derived or human supplied).



Visual Text Analytics has both a Visual and Programming Interface. SAS solutions sit atop a stacked framework of architectural components built up including: common hosting environments, standardized implementations of security, governance, and administration, SAS runtime environments to process different kinds of analytical workloads, new SAS Viya software infrastructure to provide access to core SAS analytic engines, support for access to data from a range of sources, including standard DBMS, Hadoop, Cloud, and streaming (like IOT), support for client access, including SAS-built clients as well as open APIs for programming languages like R, Lua, Java, etc.

- [illegible]

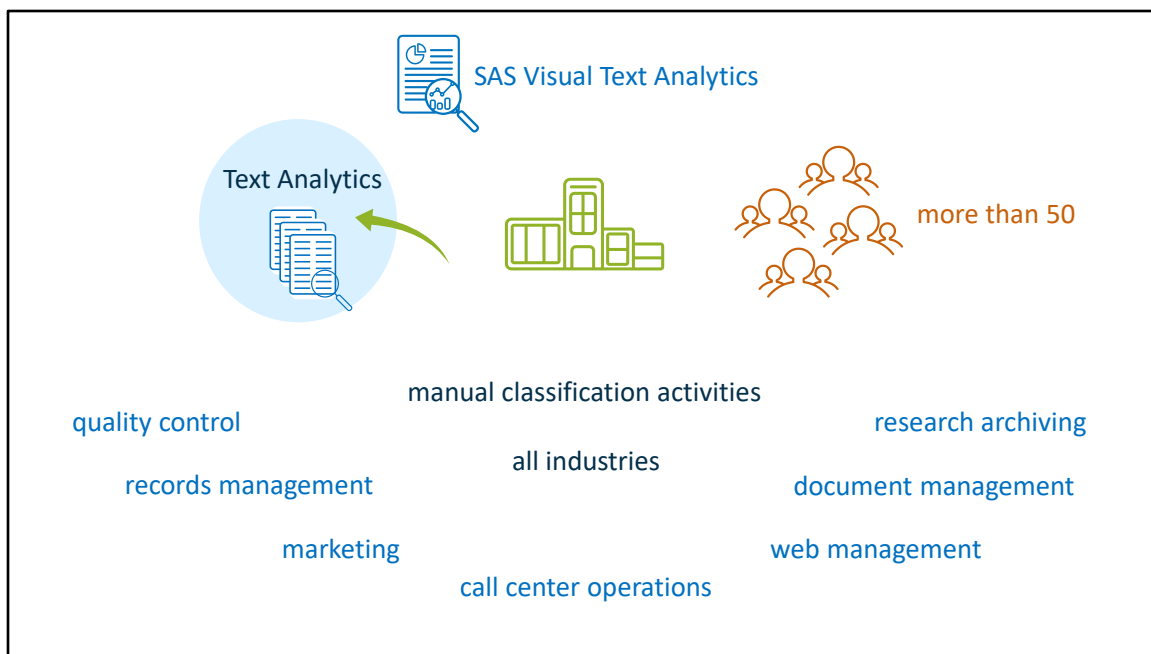
If you understand the text data well, then you could write linguistic rules to detect slang and sarcasm.

Stemming and part-of-speech tagging are relevant for many European languages. Stemming means detecting dictionary form and its variants, such as detecting singular and plural versions of a noun. Part-of-speech tagging identifies parts of speech such as verbs, nouns, and adjectives. Compound Decomposition is more relevant for languages like German, which accommodates combining words together to create a new word with a new meaning. Segmentation refers to breaking words into segments, for example when hyphens or slashes are used. Named entity recognition recognizes names of persons, organizations, locations, and products through predefined concepts. Semantic Parsing detects relationships between words in the sentence, for example, determining if a pronoun is the object or subject of a verb. Tokenization and sentence breaking identifies terms, sentences, and paragraphs based on punctuation and formatting.

Overview of SAS Visual Text Analytics



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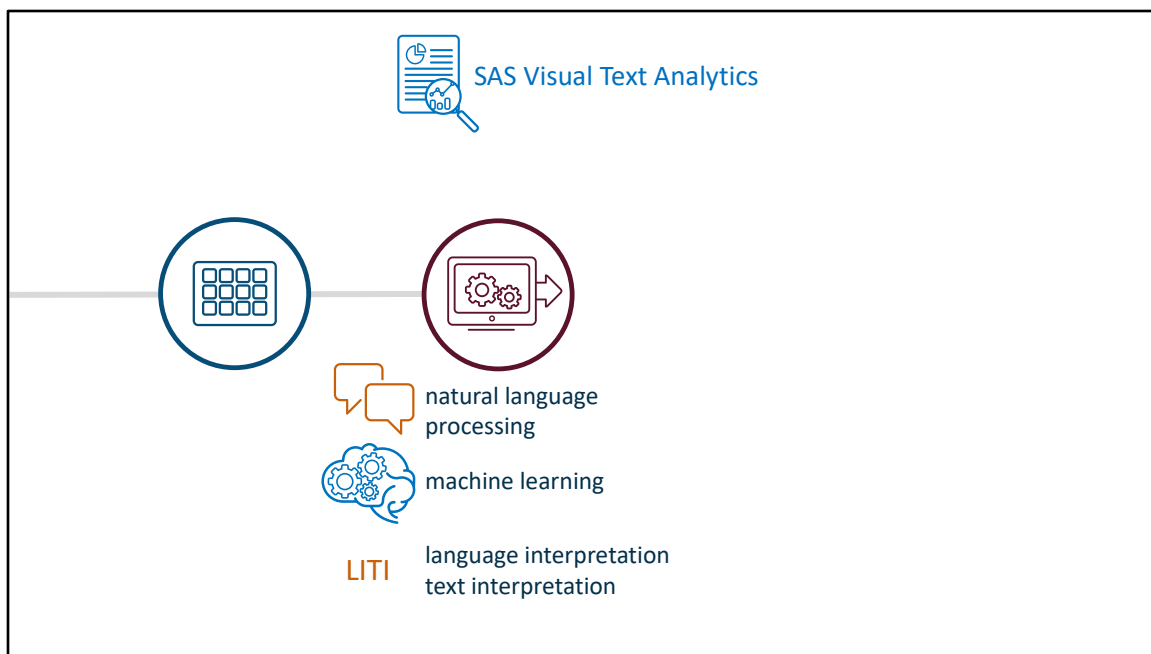
SAS Visual Text Analytics users come from enterprises that are often new to text analytics. These organizations often have more than 50 employees. Current operations include manual classification activities for large document collections. All industries are represented. Common business areas include quality control, records management, marketing, call center operations, research archiving, document management, web management, IT, or any other area that needs to classify documents or extract information.



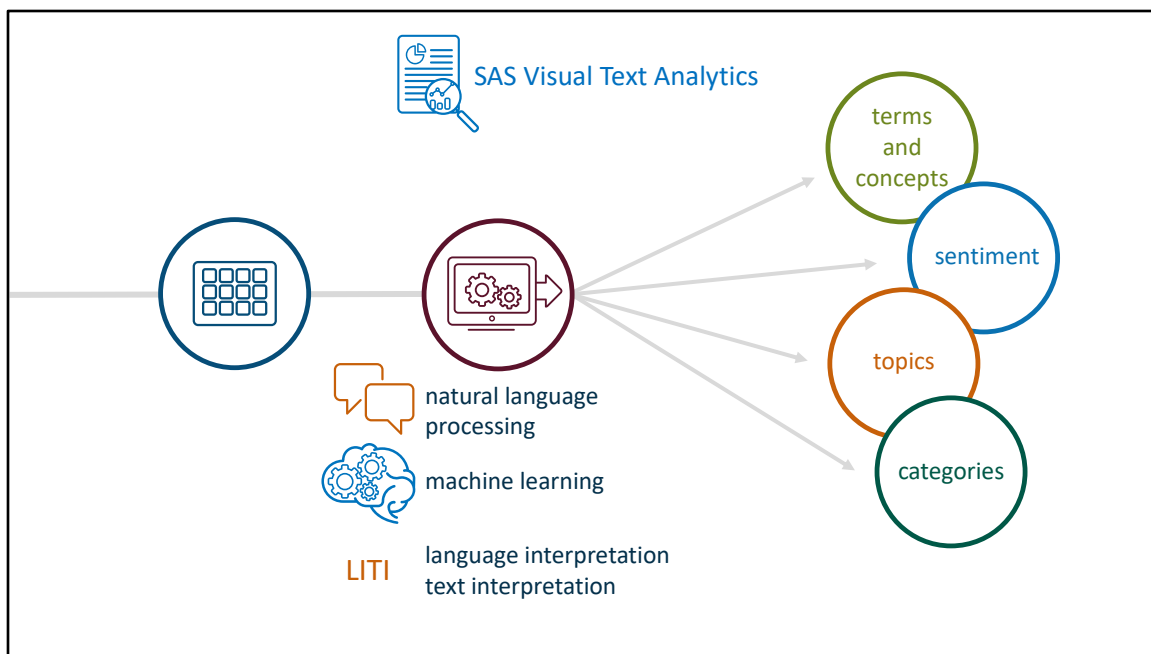
SAS Visual Text Analytics



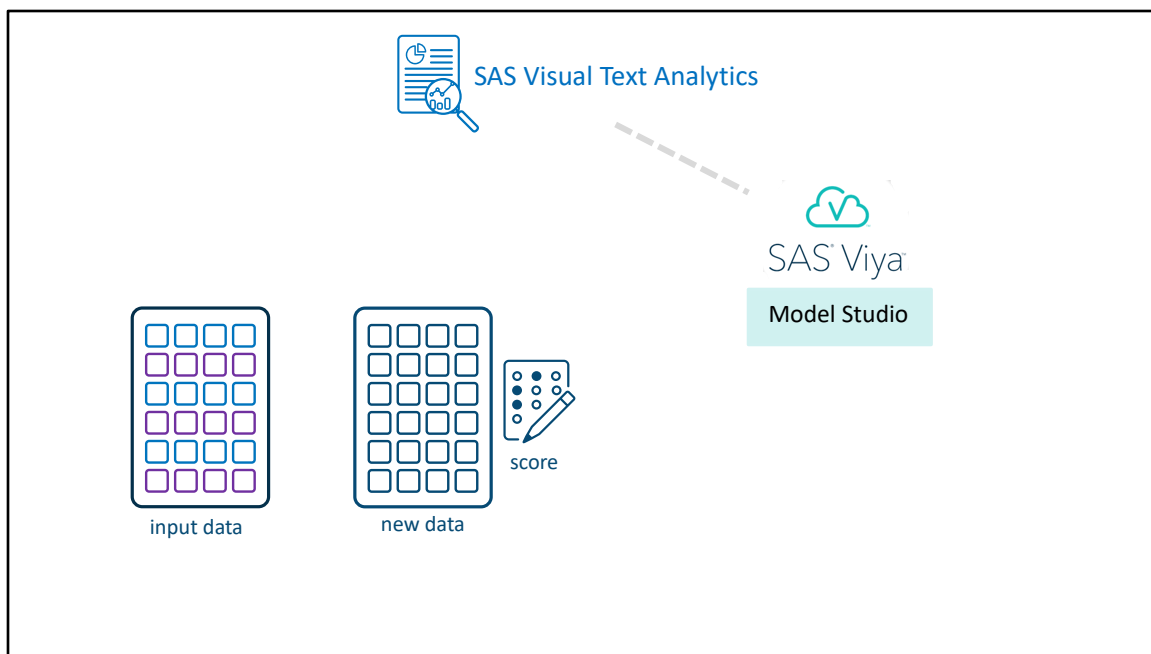
The SAS Visual Text Analytics process can be



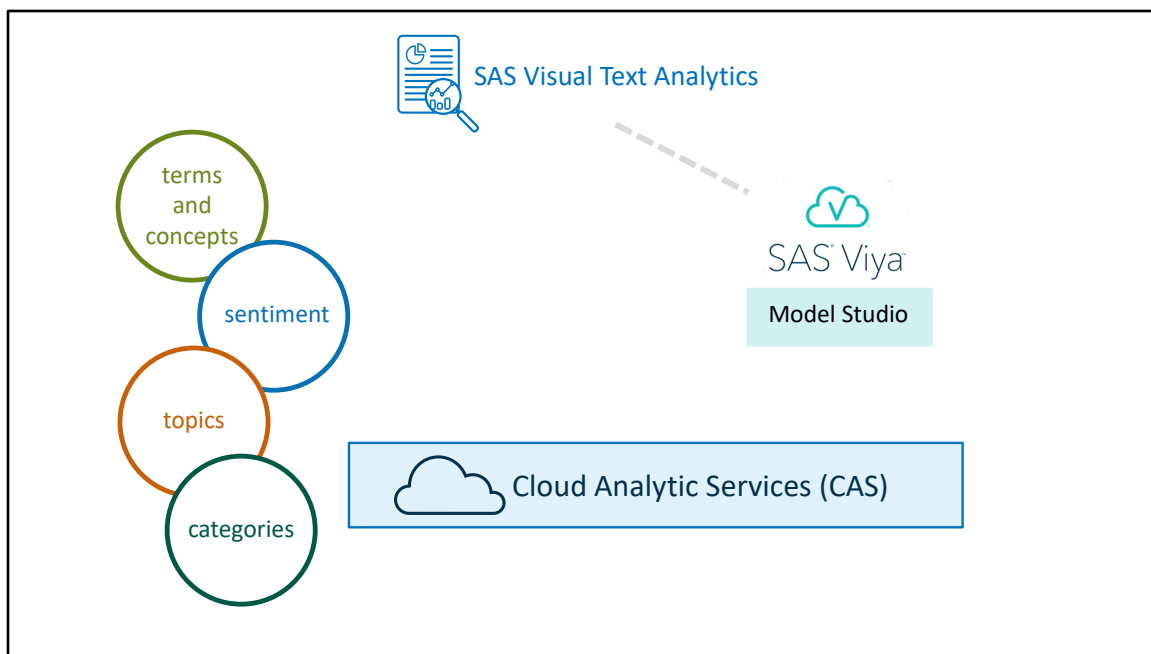
diagrammed as data being processed by the software using natural language processing, machine-learning, and the LITI (language interpretation for textual information) computing language. This process is represented



by a division of labor into four primary results: terms and concepts obtained from the Concepts node and the Text Parsing node, sentiment from the Sentiment node, topics from the Topics node, and categories from the Categories node.



When evaluating this as an enterprise solution, SAS Visual Text Analytics is fully integrated with SAS software. It's accessible via Model Studio in SAS Viya. SAS Visual Text Analytics can prepare input data and can score new data. And the interface makes it easy to manage multiple projects and to perform analyses on large or complex text data. You can include data from a SAS table, a CAS table, or documents converted into a CAS table using SAS Data Explorer.

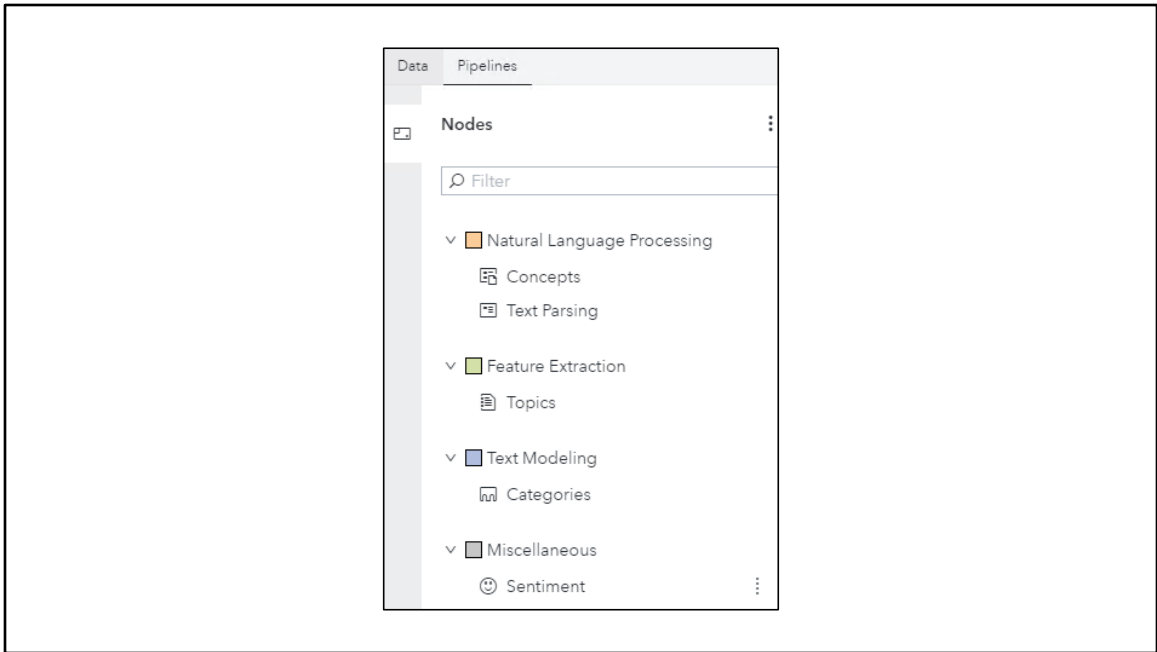


The analytic processing related to terms and concepts, sentiment, topics, and categories takes place on CAS tables in memory.

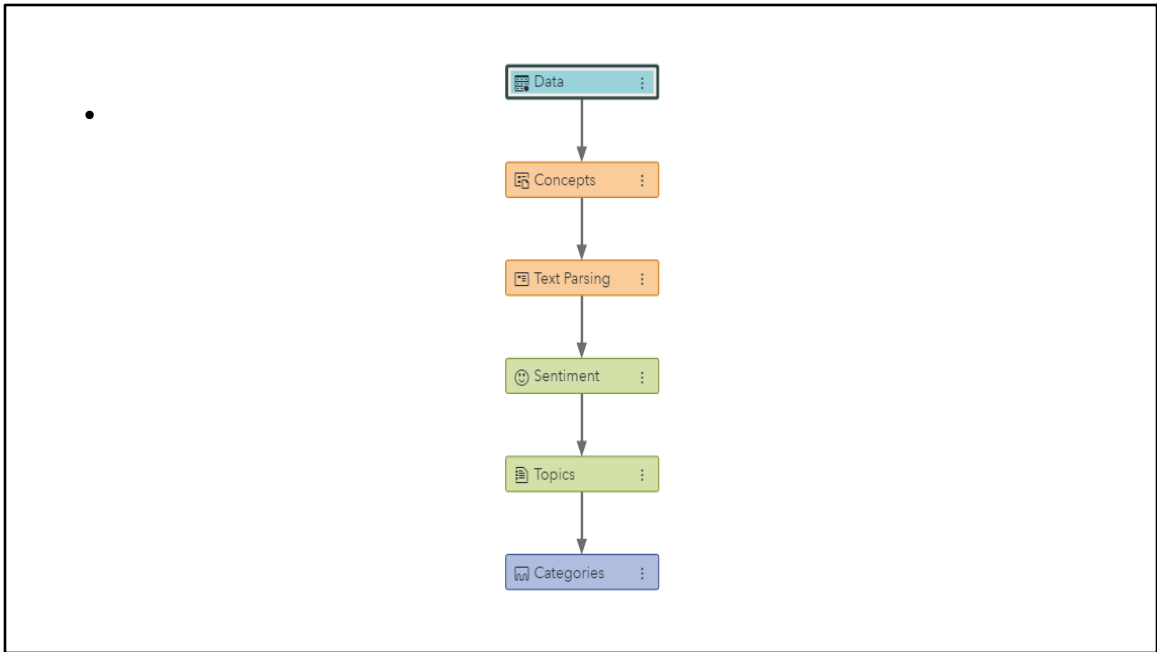
SAS Visual Text Analytics Pipeline



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SAS Visual Text Analytics pipelines have four major areas and five nodes. The four areas are Natural Language Processing, Feature Extraction, Text Modeling, and Miscellaneous, and the five nodes are Concepts, Text Parsing, Topics, Categories, and Sentiment.

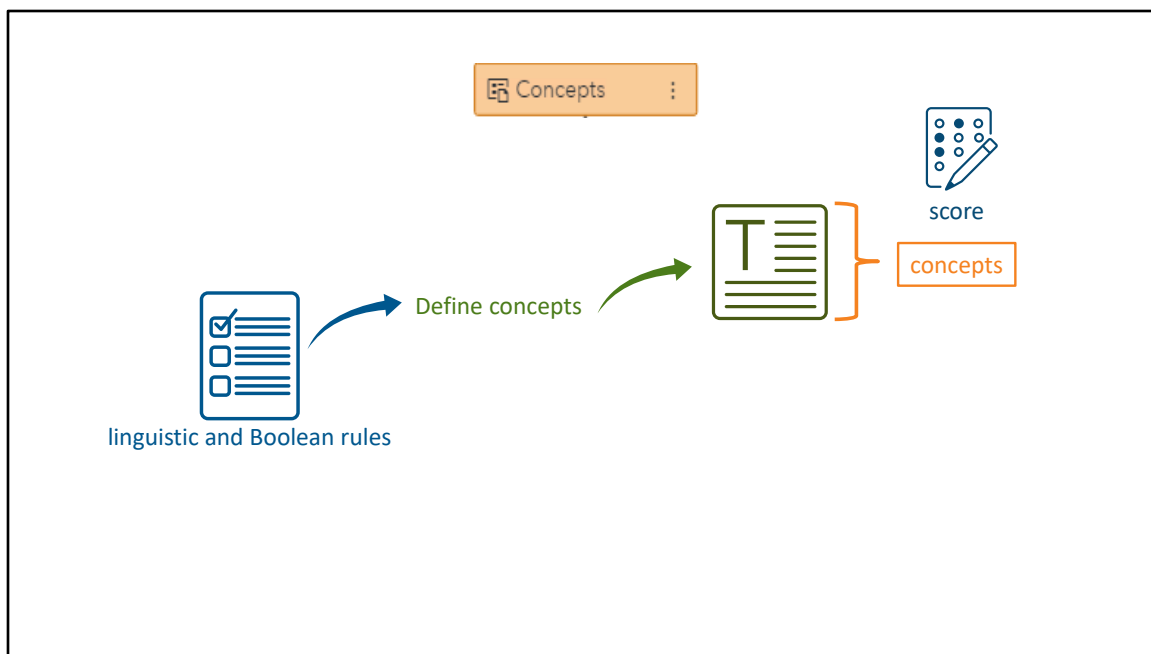


The default visual text analytics pipeline starts with the Data node and progresses through the five visual text analytics nodes. Note that the Concepts node appears before the Text Parsing node. This ensures that concepts will be used in topic derivation, category rule derivation, and sentiment analysis.

The Concepts Node: Concept Extraction



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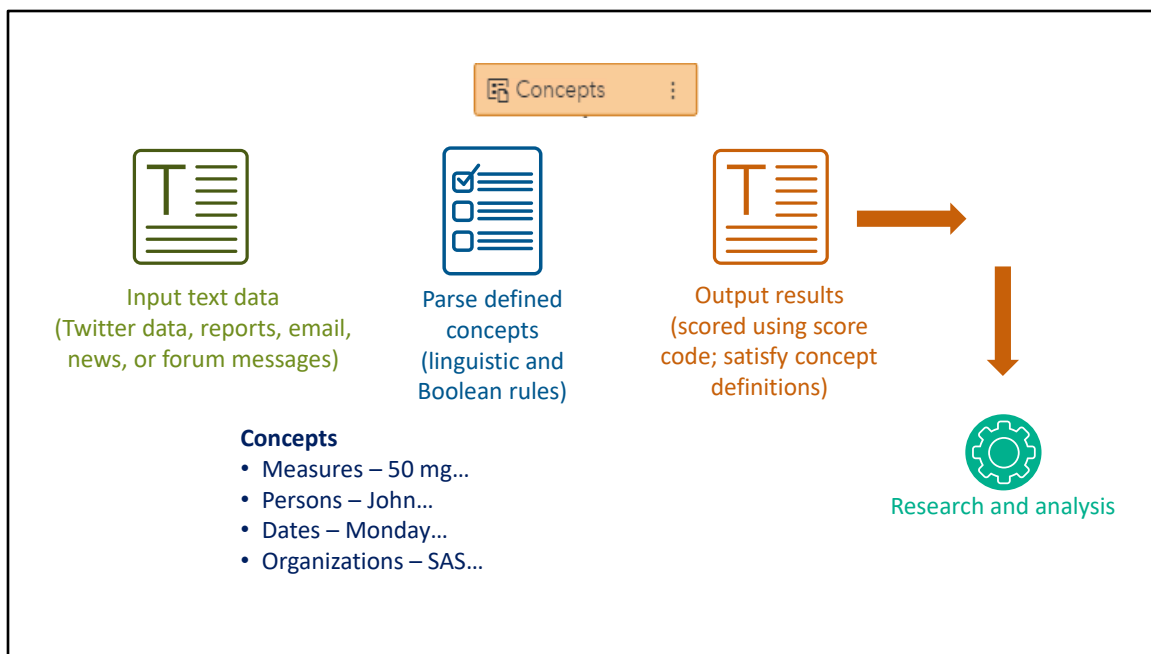
The Concepts node allows you to define concepts using linguistic and Boolean rules. Concepts are identified within the document, and a document can contain more than one concept. Concepts are extracted using the score code produced by the Concepts node.

The screenshot shows the SAS Concepts interface. At the top, there is a 'Concepts' button. Below it, a 'Custom Concepts (1)' folder is expanded, showing the '_Dosage' concept. A table defines the REGEX for this concept:

1	REGEX: [0-9]+[\.][0-9]+\s?mg\.
2	REGEX: [\d]+\s?mg\.

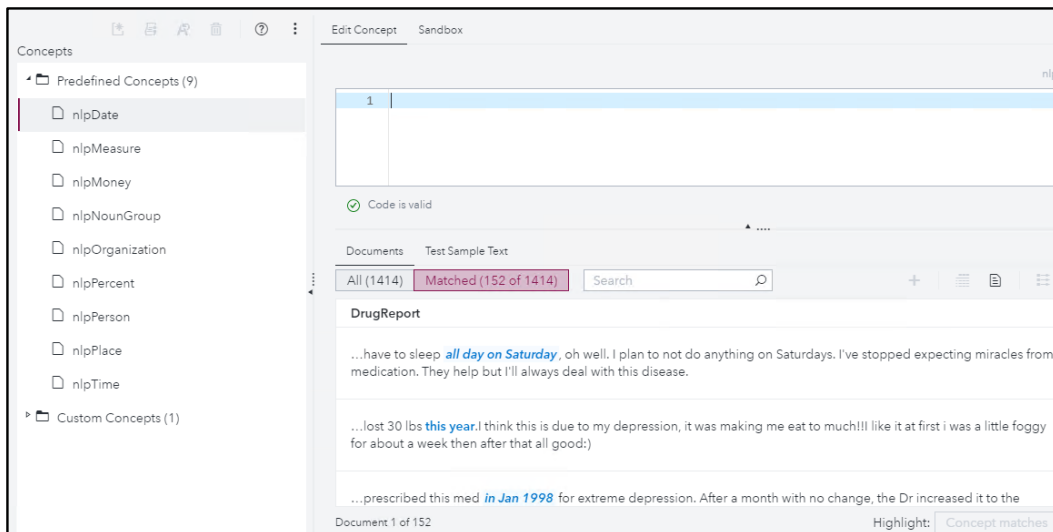
Below the table, a text box contains the example sentence: "I have been on **10mg** prexifan for 6 months along with **2000mg** noderal| **1mg** gemulex and fortifex when needed."

For example, the _DOSAGE concept is defined using regular expressions so that individual dosage values can be extracted from documents.

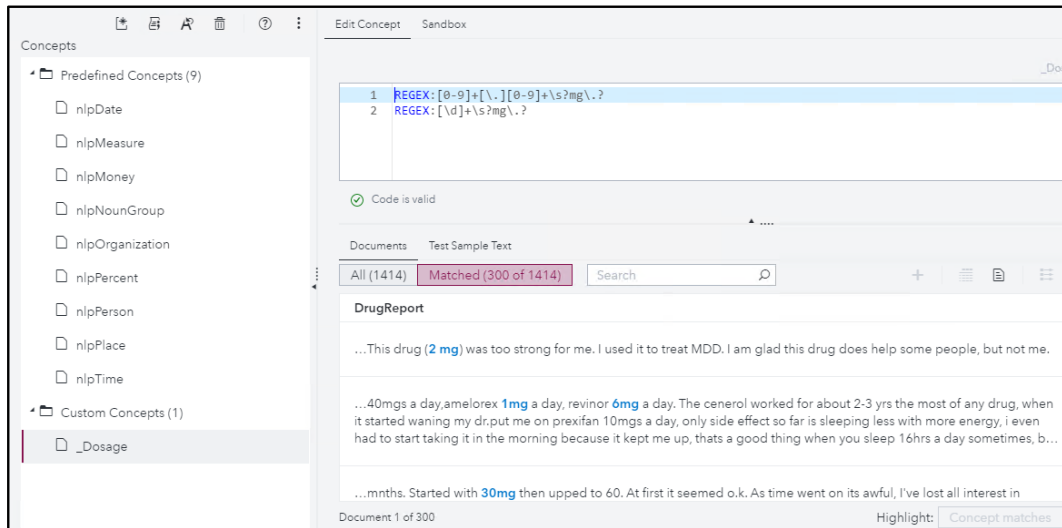


Concepts are the entities, facts, events, or key pieces of information in the input text data. The Concepts node parses, or extracts, concepts from the incoming documents by applying the concept definitions that you developed using linguistic and Boolean rules. Nine predefined concepts are supported by the Concepts node. These include concepts like measures, persons, dates, and organizations.

When a document is scored using the concept score code, the output contains the individual textual components that satisfy the concept definitions. Results are indexed, or fed, into existing systems for research and analysis.



If you request predefined concepts, and you open the Concepts node after it has run successfully, you will see the nine predefined concepts. Each predefined concept is named beginning with the letters NLP for natural language processing. When you select a predefined concept, for example, **nlpNounGroup**, you see the terms that satisfy the concept highlighted in the Documents pane under the Matched tab. You will not be able to see the source definition of the predefined concept in the Edit pane.

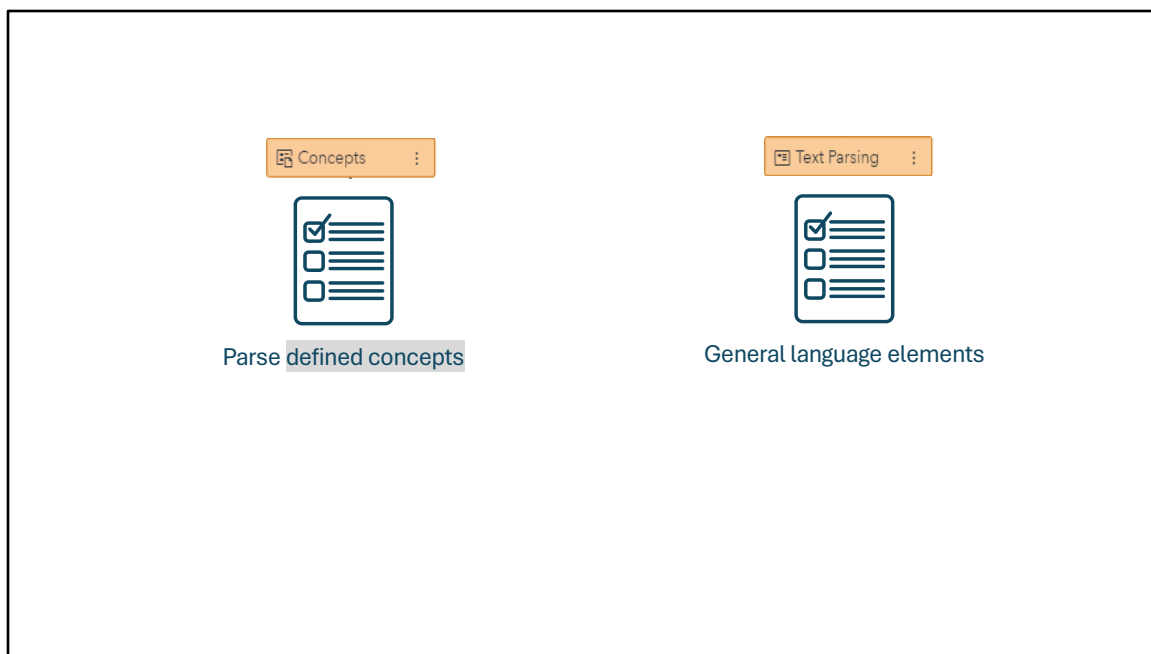


You can create a custom concept by supplying the rules for the concept in the Edit pane. The rules are taken from the LITI language. When you select the Matched tab in the Documents pane, you see highlighted concepts in the documents.

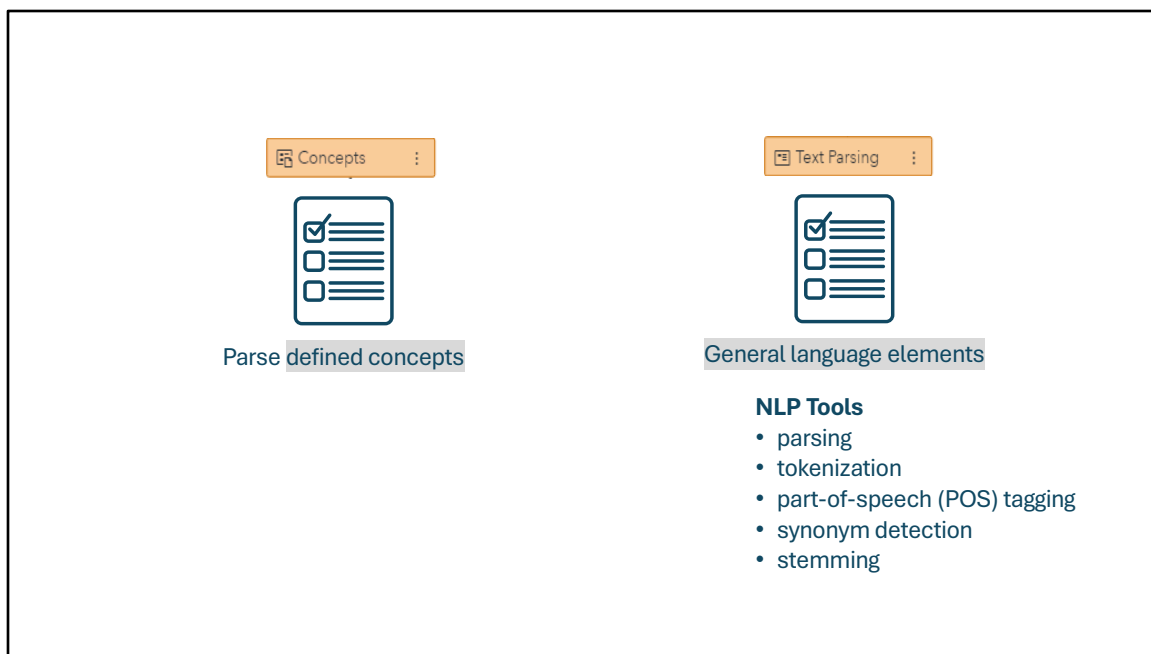
The Text Parsing Node: Term Extraction



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Although the Concepts node only parses for the defined concepts in the node,



the Text Parsing node parses for general language elements. The NLP tools supported by the Text Parsing node include parsing, tokenization, part of speech tagging, synonym detection, and stemming. In addition, if the Text Parsing node is preceded by the Concepts node, it will also parse with respect to the concepts, and any concepts found in the document collection will appear in the term table.

Kept Terms (1522)

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☒ take	V	677	1105
☐ depression	N	492	616
☐ feel	V	371	518
☐ year	N	395	502
☐ drug	N	341	485

Dropped Terms (6849)

Term	Role	Documents	Frequency
☐ i	PRO	1081	5003
☐ be	V	1129	4183
☐ have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386

Documents

All (1414) Matched

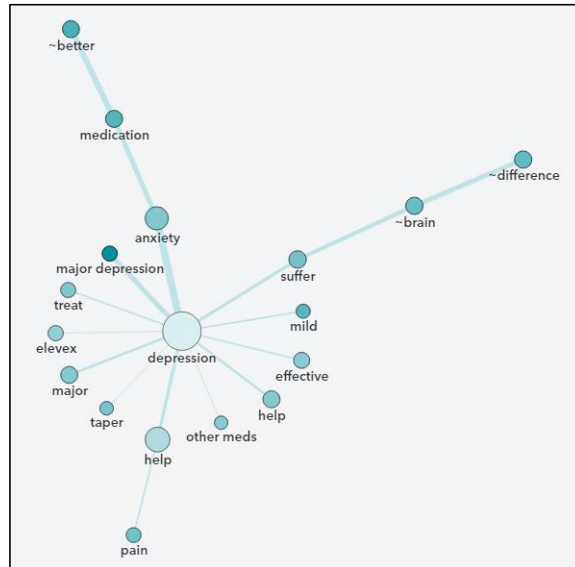
DrugReport

This medication made me gain 40 pounds it has been 2 years and I have only lost 10 pounds. Beware and watch your weight.

causing extreme anger, to the point that my family has become afraid of me. Doing things i would have considering doing before, like challanging the police, starting arguments, really, really wanting to beat the crap out of someone. Enough to the point that it scares me. But my doctor at this time refuses to take me off this medication. I am scared I will end up in jail for severly hurting someone.

Document 1 of 1414

When you open a Text Parsing node that has been run successfully, you see a Kept Terms pane, a Dropped Terms pane, and a Documents pane. You can select a term from either term table and find documents that contain the term by selecting the Matched tab in the Documents pane.



If you select a term in one of the tables, and click on the **Term Map** icon, you can see a term map with term associations related to the selected term. So, for example, if you select the term *depression*, associated terms such as *anxiety*, *help*, *treat*, and *suffer* are displayed.

The Topics, Categories, and Sentiments Nodes



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Topics (10)

☐	Topic	Created by	Docume... ↓
☒	+symptom, +withdrawal, off of, horrible, +dose	System	224
☐	+effect, +side effect, side, +side, sexual	System	223
☐	+work, ecstapin, best, +year, +try	System	223
☐	+day, +feel, mg, a day, +start	System	217
☐	abidal, +pain, just, +start, +work	System	208

Terms

All (1522) Matched Filter

☐	Term	Role	Documents	Frequency ↓
☐	not	ADV	658	1174
☐	take	V	677	1105
☐	depression	N	492	616
☐	feel	V	371	518
☐	year	N	395	502

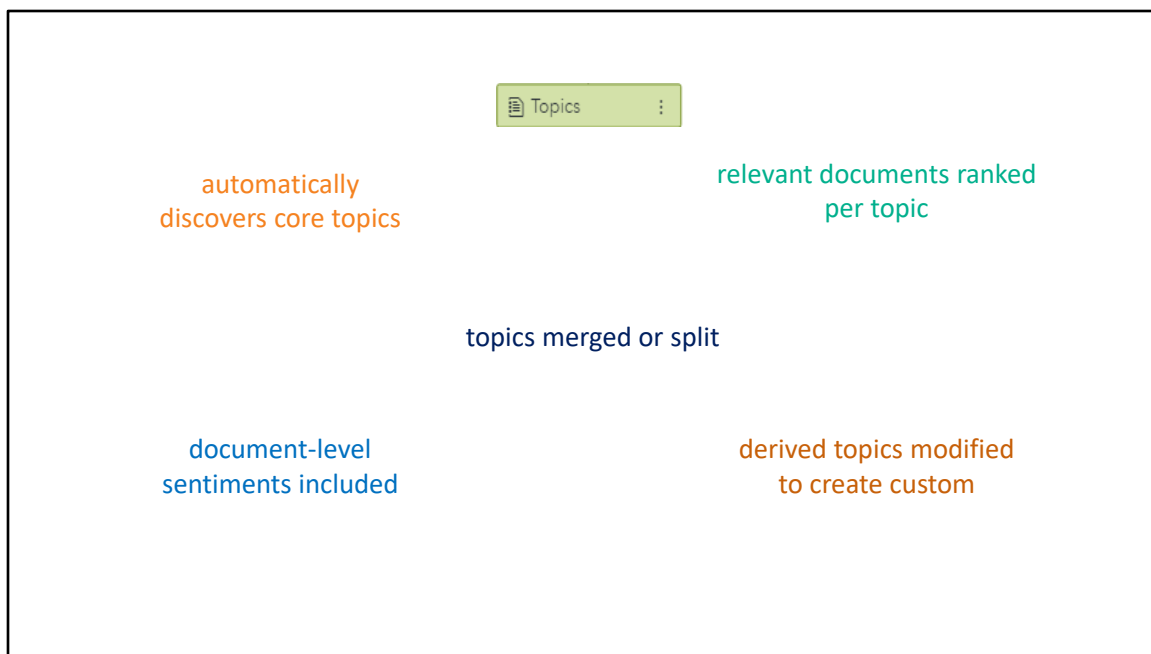
Documents

All (1414) Matched (224 of 1414) Search

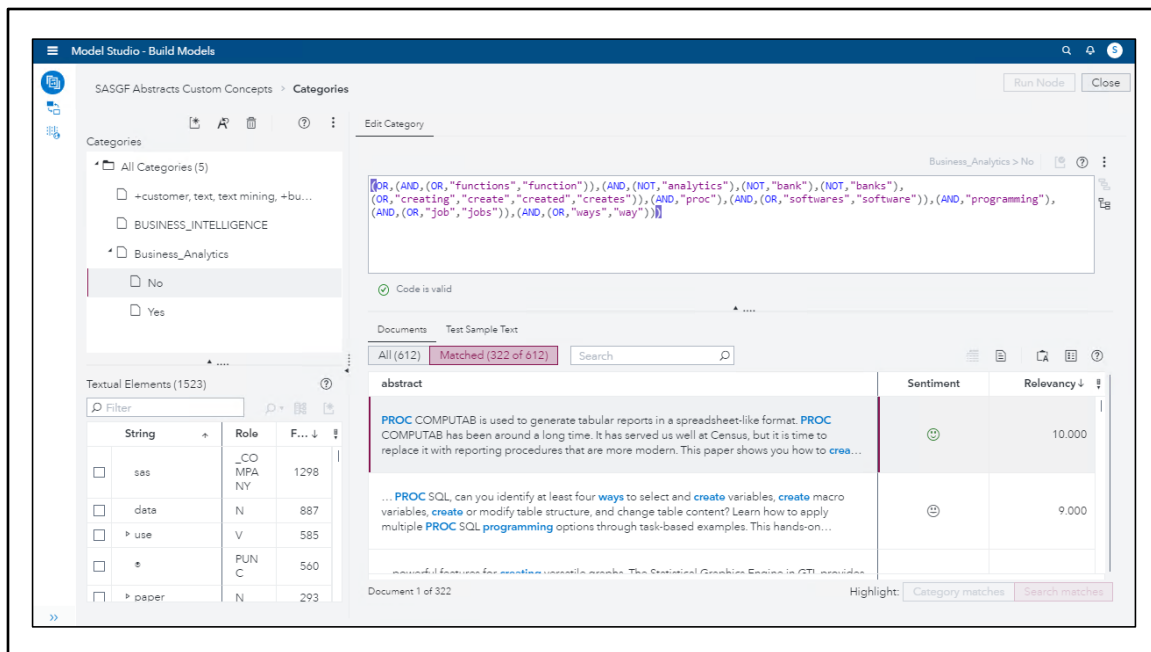
DrugReport	Relevancy ↓	Sentiment
... taken ecstapin xr regularly for over 5 years . the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently went off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheav...	0.334	☹
... Ecstapin for about 5 years . The withdrawal symptoms are horrible . I have brain zaps , nausea , vomiting , dizzy , can't concentrate, I fill like I am in a another world and vision problems I slowly took myself off of this deadly drug . I went from 150mg then to 37.5mg . I have been off it for about a week . My opinion with the withdrawals is like digging your own grave. I wouldn't recommend this drug to anybody.	0.330	☹

Document 1 of 224 Highlight: Topic matches Search matches

When you open the Topics node, you see a Topic pane, a Terms pane, and a Documents pane. You can select a topic, and then find matching terms and documents.



The Topics node automatically discovers core topics. The most relevant documents can be ranked with respect to a selected topic. Topics can be further merged or split. If a Sentiment node precedes the Topics node, document-level sentiments from the Sentiment node can be included. Derived topics can be modified to create custom topics.



When you open the Categories node, you see a Categories pane, an Edit window, a Textual Elements pane, and a Documents pane. If a variable in the input data has a role of category, the Categories pane shows the levels of the categorical variable, and if you select a level, you'll see the category rules associated with that level. You can also find the documents that match the categorical rules, and the terms that satisfy the category rules will be highlighted.

Categories

Model Studio - Build Models

SASGF Abstracts Custom Concepts > Categories

Categories

All Categories (5)

+customer, text, text mining, +bu...

BUSINESS_INTELLIGENCE

Business_Analytics

No

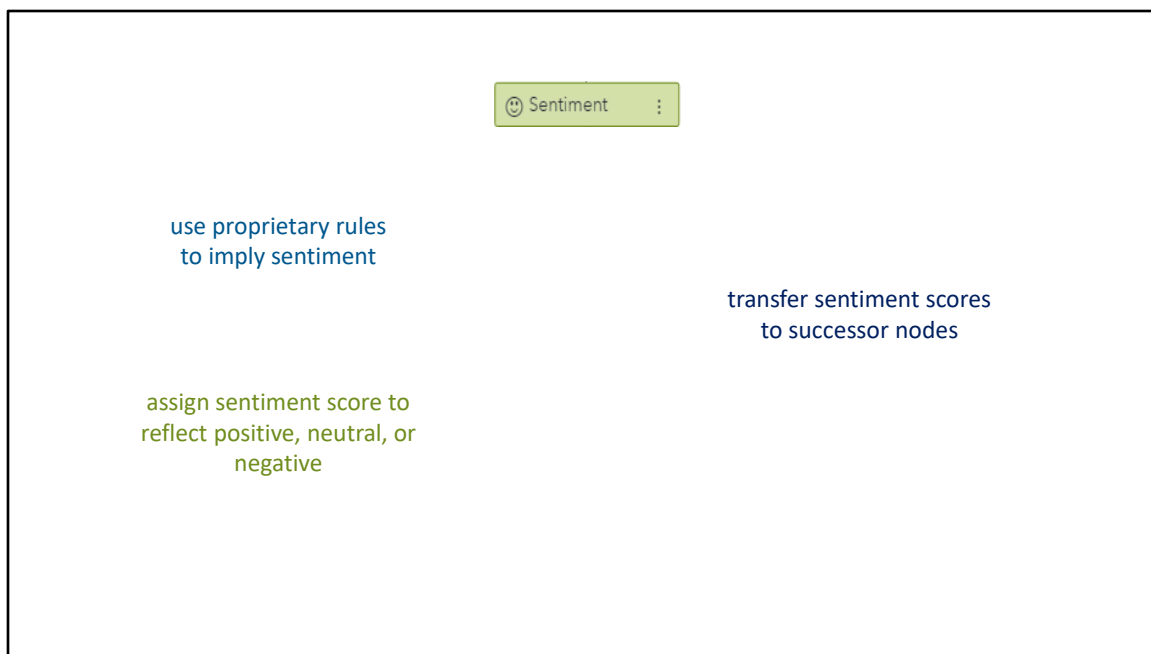
Yes

Boolean linguistic rules automatically derived

easy to understand

can be edited

When you supply a variable with the role of category, Boolean linguistic rules are automatically derived for that categorical variable. If you promote a topic to a category, then Boolean linguistic rules are also derived for the topic. The derived Boolean linguistic rules are easy to understand, and they can be edited. If you select results for the Categories node, you will see statistics on accuracy and specificity. Further customization or refinement by an analyst can help achieve the desired results.



The Sentiment analysis node uses proprietary rules that identify and analyze terms, phrases, and character strings to imply sentiment. Score code from the Sentiment node can be used to assign a sentiment score to each document to reflect positive, neutral, or negative sentiment. The sentiment scores can be transferred to successor nodes. Thus, the Topics node displays sentiment scores for each document if it is preceded by a Sentiment node.

Concepts, Terms, Topics, and Categories

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: SAS Visual Text Analytics Purpose



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Question: SAS Visual Text Analytics Purpose

What is the main purpose of using SAS Visual Text Analytics?
(Select all that apply.)

- a. concept extraction
- b. topic extraction
- c. content categorization
- d. sentiment extraction

Answer: SAS Visual Text Analytics Purpose

What is the main purpose of using SAS Visual Text Analytics?
(Select all that apply.)

- ☒ a. concept extraction
- ☒ b. topic extraction
- ☒ c. content categorization
- ☒ d. sentiment extraction

Correct answers: a, b, c, d

Question: SAS Visual Text Analytics Nodes



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Question: SAS Visual Text Analytics Nodes

Which of the following nodes can be used to extract information from within a document?

- a. Concepts
- b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

Answer: SAS Visual Text Analytics Nodes

Which of the following nodes can be used to extract information from within a document?

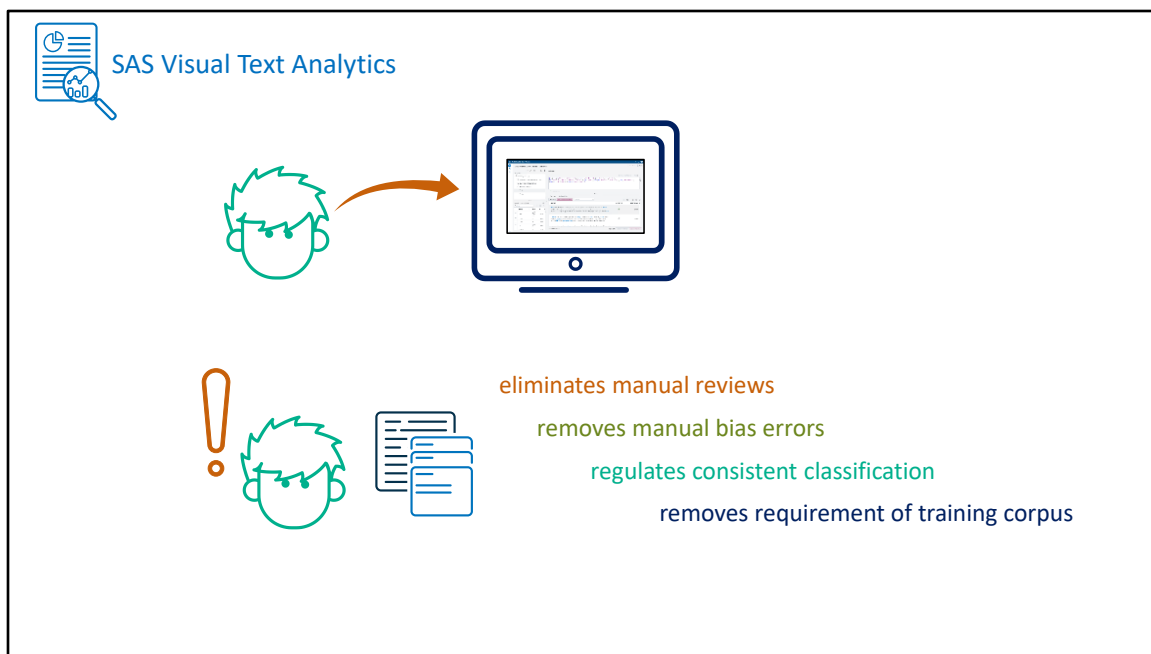
- ☒ a. Concepts
- b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

a.

Benefits of SAS Visual Text Analytics



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There are many reasons to use SAS Visual Text Analytics for text analytics.

SAS Visual Text Analytics makes it possible for analysts to examine all their content, interrogate it, and define classification schemes in a single, simple-to-use GUI.

It eliminates laborious manual document reviews, removes manual bias errors, and regulates consistent content classification.

SAS Visual Text Analytics removes the requirement to create a training corpus in advance of defining classification rules.

Resource

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



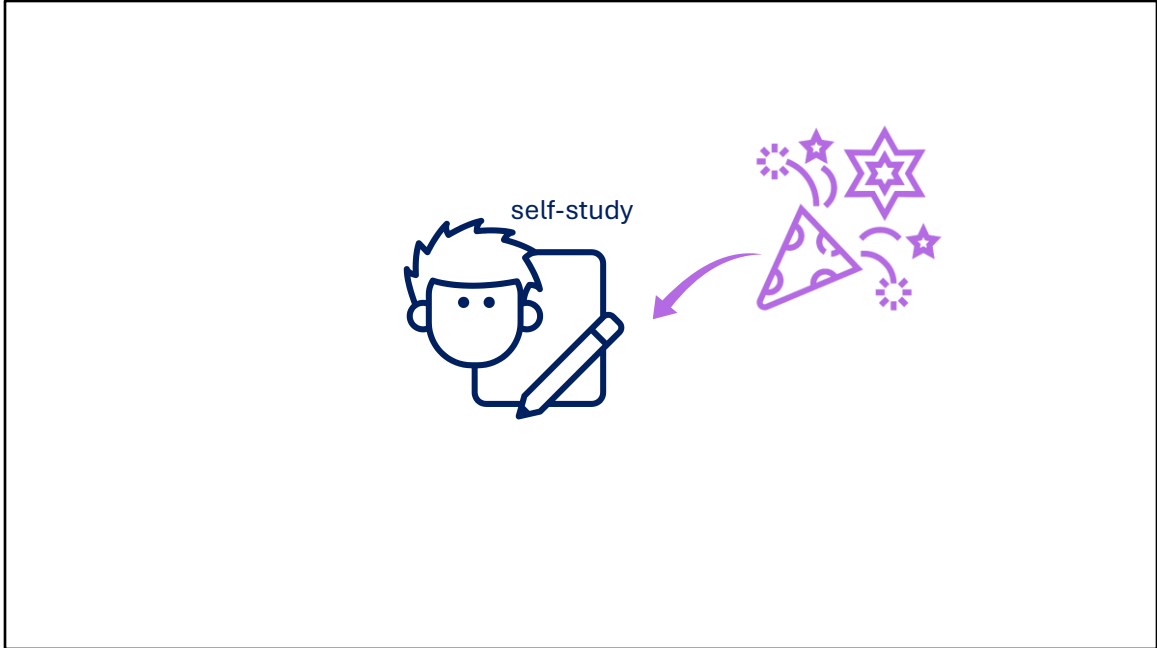
1.2 Language Challenges



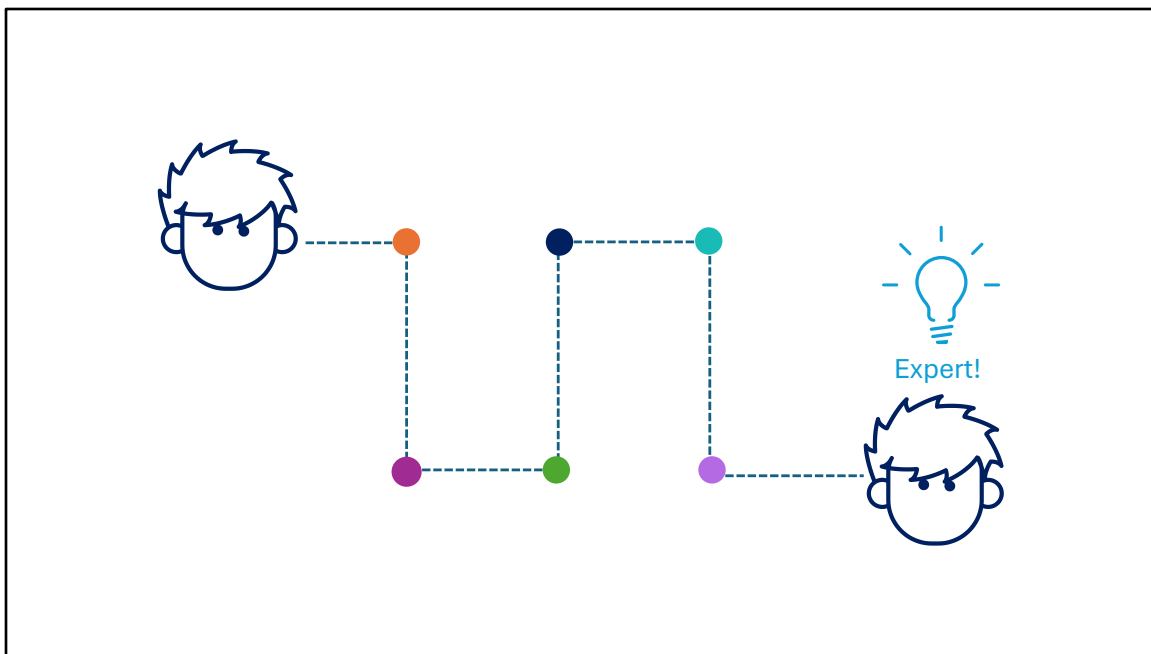
Self-Study Overview



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This section is classified as self-study because it is in the category of “fun facts that are not essential for success.” If you omit this section, it's unlikely that you will be less successful except for relatively rare cases.



However, if you are on a path to become a text analytics expert, some information in this section can equip you to tackle more serious document conversion and data transcoding problems.

Self-Study

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Lesson 2: SAS Visual Text Analytics Demonstrations



2.0 Lesson Overview



Lesson Overview

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



2.1 Importing Document Collections



Demo: Importing and Converting Document Files

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Document Formats

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Resource

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



2.2 Creating a Project with No Predefined Concepts



Demo: Creating a SAS Visual Text Analytics Project with No Predefined Concepts, Part 1

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Creating a SAS Visual Text Analytics Project with No Predefined Concepts, Part 2

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Creating a SAS Visual Text Analytics Project with No Predefined Concepts, Part 3

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Creating a SAS Visual Text Analytics Project with No Predefined Concepts, Part 4

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



The Document Text Window with the Term Map

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Predefined Concepts



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Question: Predefined Concepts

Select the true statements about predefined concepts in SAS Visual Text Analytics.

- a. SAS Visual Text Analytics provides nine predefined concepts, which are concepts whose rules are already written.
- b. You can choose to include predefined concepts or not during the project creation. You cannot add them in the interactive window after the node is run.
- c. If you include the predefined concepts, you can disable one or more predefined concepts by selecting the concepts and then clicking the Disable button.
- d. If you include the predefined concepts, you cannot add and edit custom concept rules.

Answer: Predefined Concepts

Select the true statements about predefined concepts in SAS Visual Text Analytics.

- ☒ a. SAS Visual Text Analytics provides nine predefined concepts, which are concepts whose rules are already written.
- ☒ b. You can choose to include predefined concepts or not during the project creation. You cannot add them in the interactive window after the node is run.
- ☐ c. If you include the predefined concepts, you can disable one or more predefined concepts by selecting the concepts and then clicking the Disable button.
- ☐ d. If you include the predefined concepts, you cannot add and edit custom concept rules.

Answer: a. and b.

Practice: Working with a SAS Visual Text Analytics Project

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



2.3 Creating a Project with Custom Concepts



Demo: Working with Custom Concepts, Part 1

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Working with Custom Concepts, Part 2

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Working with Custom Concepts, Part 3

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Custom Concepts



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Question: Custom Concepts

After an inspection, a New York City restaurant can receive grade values A, B, C, G, N, P, Z, and no grade. Only one grade designation is allowed. Which of the following is/are true? (More than one correct choice is possible.)

- a. For a category variable containing a restaurant grade value, the Categories node will derive Boolean rules for each grade value in the data set.
- b. The Boolean rule for two or more grade values can produce a TRUE result so that mutually exclusive text categorization is not possible using only the Boolean scores.
- c. Scored data produced using score code contains both a Boolean score and relevancy score for each category value.
- d. The relevancy score in scored data can be used to produce mutually exclusive category scores.

Answer: Custom Concepts

After an inspection, a New York City restaurant can receive grade values A, B, C, G, N, P, Z, and no grade. Only one grade designation is allowed. Which of the following is/are true? (More than one correct choice is possible.)

- ☒ a. For a category variable containing a restaurant grade value, the Categories node will derive Boolean rules for each grade value in the data set.
- ☒ b. The Boolean rule for two or more grade values can produce a TRUE result so that mutually exclusive text categorization is not possible using only the Boolean scores.
- ☒ c. Scored data produced using score code contains both a Boolean score and relevancy score for each category value.
- ☒ d. The relevancy score in scored data can be used to produce mutually exclusive category scores.

Answer: a., b., c., d.

Practice: Creating a Project

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Lesson 3: SAS Visual Text Analytics Nodes



3.0 Lesson Overview



Lesson Overview

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



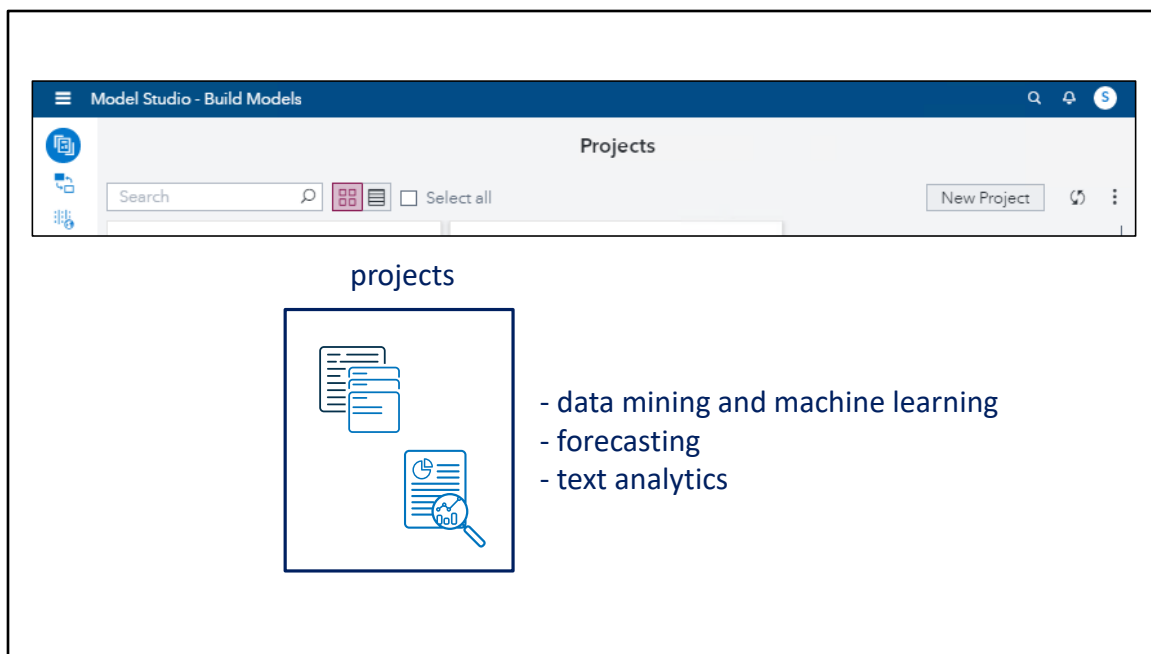
3.1 Projects



Creating Projects

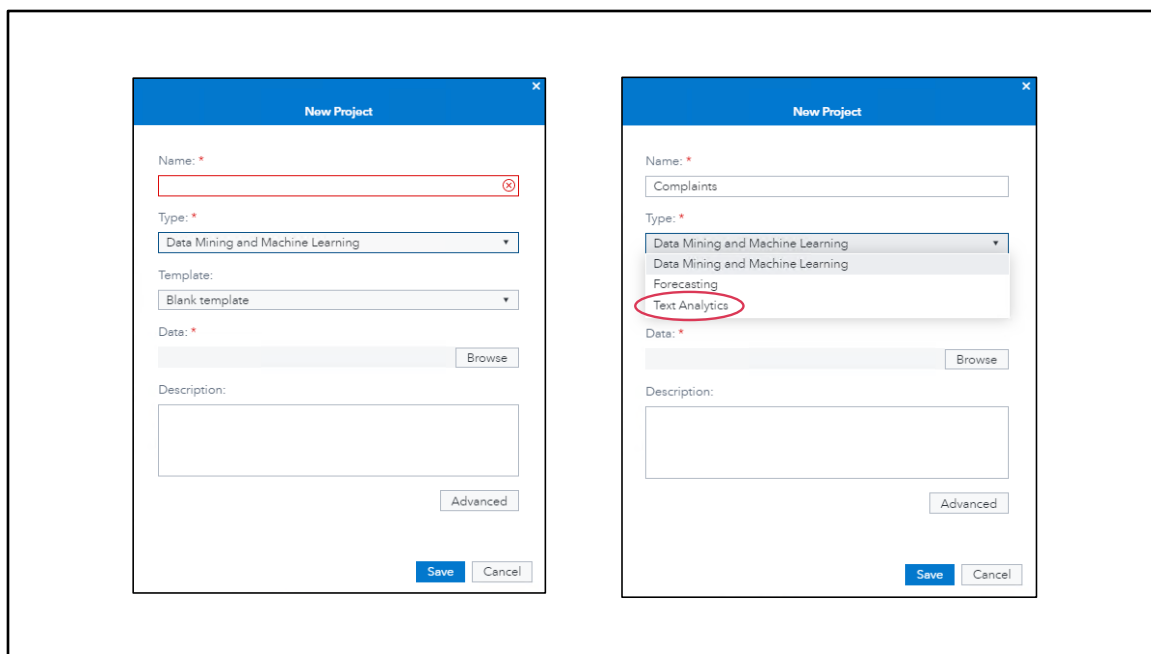


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You access SAS Visual Text Analytics through Model Studio. In Model Studio, you create *projects*, which are containers for your data and analysis. A project contains the input data, options, and analysis tasks. Three types of projects are supported in Model Studio and are related to three analysis areas: data mining and machine learning, forecasting, and text analytics. Model Studio is designed so that you can create and run multiple projects simultaneously.

To create a new project, click the **New Project** button in Model Studio.



SAS Visual Text Analytics supports the Text Analytics project type. When you select **Text Analytics**, you work with concepts, terms, sentiment, topics, and categories. Concepts, topics, and categories are built based on the terms in the project document collection.

The image shows a 'New Project' dialog box on the left and a vertical pipeline diagram on the right. The dialog box has fields for Name (filled with 'Complaints'), Type (set to 'Text Analytics'), Template (set to 'Text Analytics: Generate Concepts, Topics, a...'), Data (with a 'Browse' button), and Description. It includes 'Save' and 'Cancel' buttons at the bottom. The pipeline diagram on the right consists of six nodes connected by downward arrows: 'Data' (teal), 'Concepts' (orange), 'Text Parsing' (orange), 'Sentiment' (green), 'Topics' (green), and 'Categories' (blue). Each node has a small icon and a vertical ellipsis menu on its right side.

The default visual text analytics pipeline places the five nodes in sequence after a Data node. The default ordering is Concepts, Text Parsing, Sentiment, Topics, and Categories. If a Sentiment node appears before a Topics or Categories node, then sentiment information will be available in the document tables for those nodes.

You set the options for a node by selecting the node and changing the settings that are shown to the right of the pipeline. Each node is described in detail later.

You can also provide a project description.

DrugReport

Role:

Text

Language detection method:

Manual

Project Language:

English

Arabic

Chinese

Croatian

Czech

Danish

Dutch

English

Farsi

Finnish

French

German

Greek

Hebrew

Hindi

Hungarian

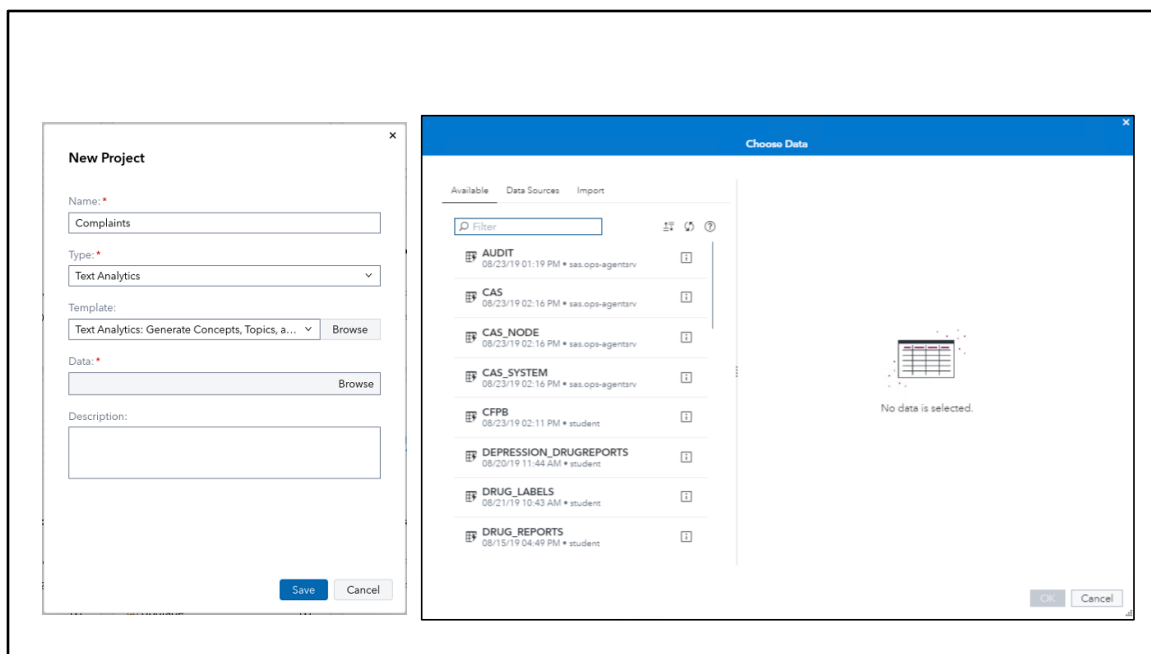
Indonesian

Italian

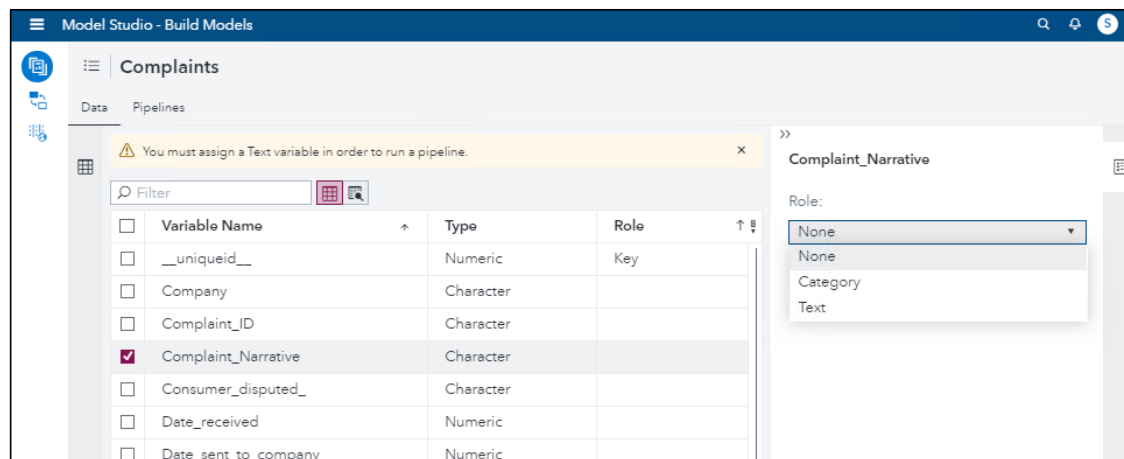
30+ Languages

Language for a Visual Text Analytics project is now set on the Data tab.

Over 30 languages are supported, and new languages are being added all the time. There are thousands of spoken languages, but many languages have a very small population of native speakers.

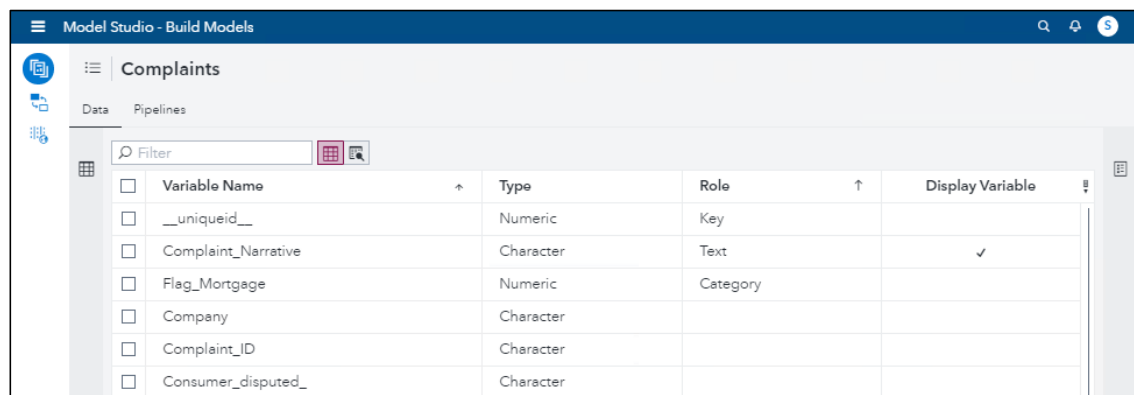


You can browse to find the analysis data set. The data set can already reside in memory, or you can import an existing SAS table to copy it into memory. You can also access a document collection and convert the documents into text format, then compile the documents into an analysis data set and copy it into memory.



The data set you select must have at least one variable with a role of Text. The variable with the role of Text contains the documents used for text analysis.

A categorical variable with the role Category is optional, but if one is identified, linguistic rules will be derived to classify each document into a level of the categorical variable.



Variable Name	Type	Role	Display Variable
__uniqueid__	Numeric	Key	
Complaint_Narrative	Character	Text	✓
Flag_Mortgage	Numeric	Category	
Company	Character		
Complaint_ID	Character		
Consumer_disputed_	Character		

In this example, **Complaint_Narrative** contains the documents used for the analysis. The **Flag_Mortgage** variable has value *one* for mortgage complaints, and the value *zero* otherwise. Linguistic rules will be derived to classify a document as *zero* or *one* with respect to the mortgage flag.

Variable Name	Type	Role	Display Variable
☐ __uniqueid__	Numeric	Key	
☐ Company	Character		
☐ Complaint_ID	Character		
☐ Complaint_Narrative	Character		
☐ Consumer_disputed_	Character		
☐ Date_received	Numeric		
☐ Date_sent_to_company	Numeric		
☐ DocLength	Numeric		
☐ Flag_Communication	Numeric		
☐ Flag_DataCollection	Numeric		

And, although a variable having a role of Key is required, one is supplied automatically by the software.

Resource

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



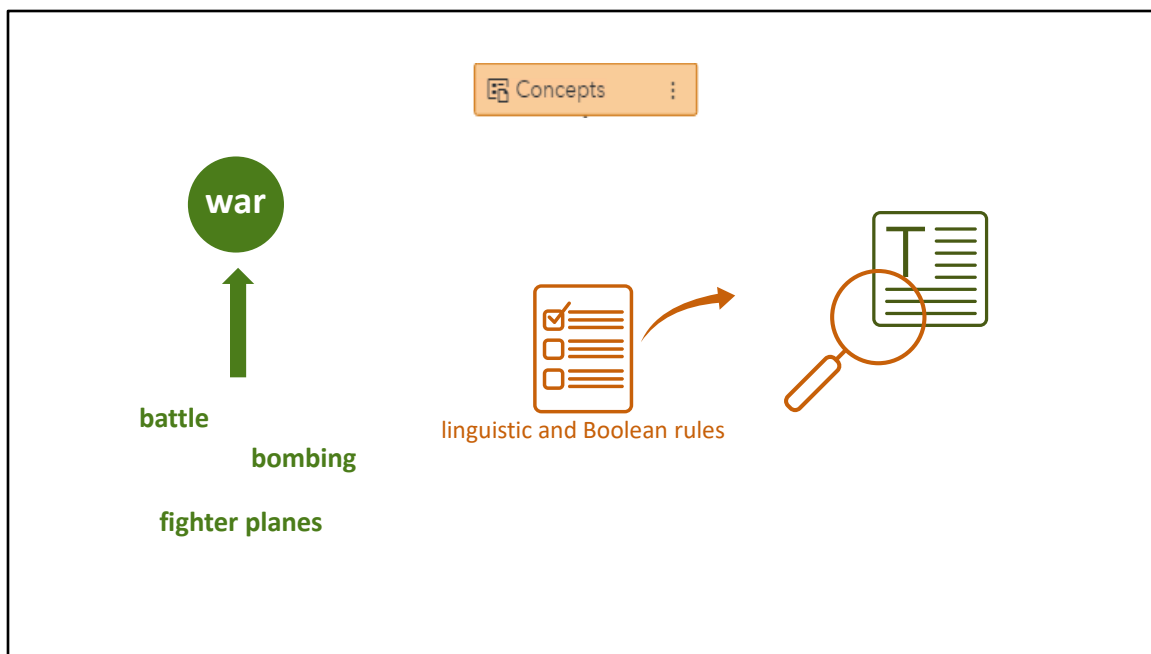
3.2 Concepts



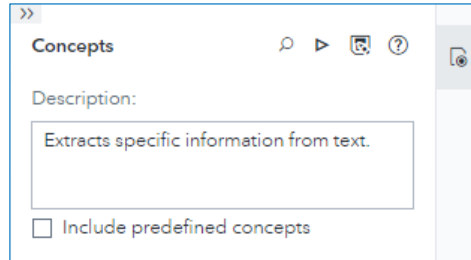
Exploring the Concepts Node



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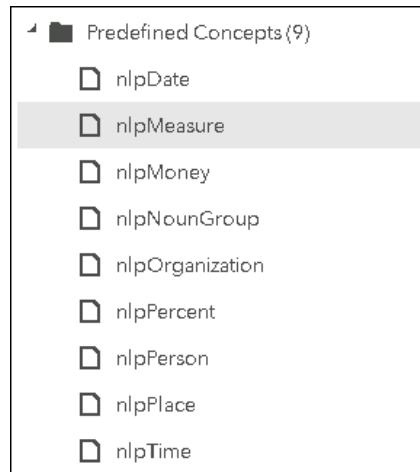
Concepts are useful for analyzing information in context. You can write rules for recognizing concepts that are important to you and thereby create custom concepts. For example, you can specify that the concept *war* is identified when the terms *battle*, *bombing*, and *fighter planes* are encountered in text.



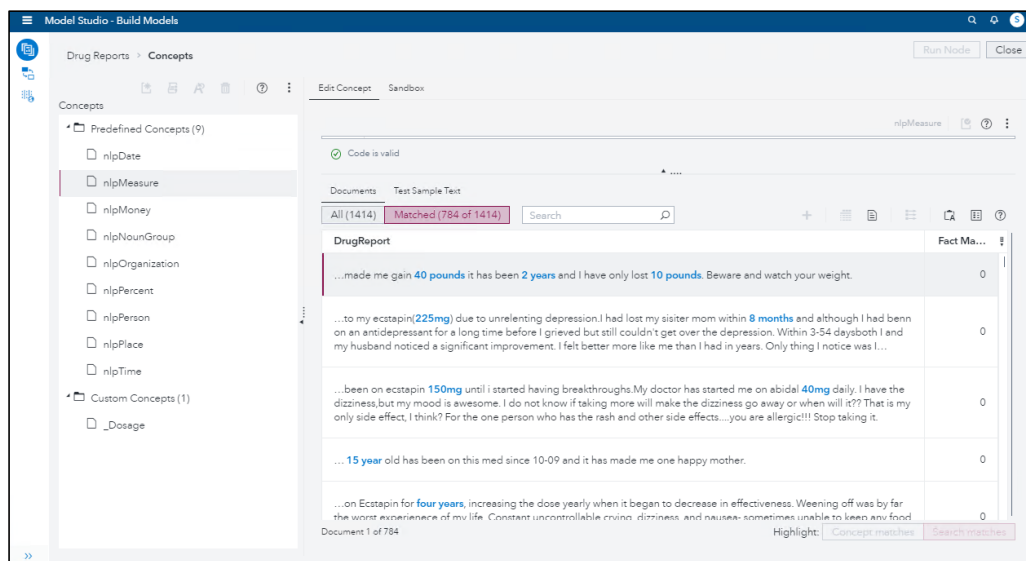
To examine the concepts settings, select the **Concepts** node.

The Concepts node has a single property: Include predefined concepts.

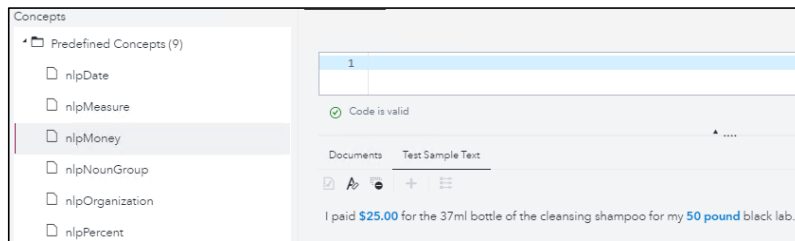
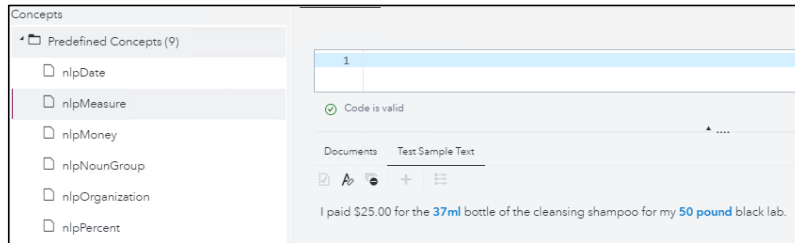
If you do not select the **Include predefined concepts** check box, and you do not provide custom concepts, then you can delete the Concepts node from the pipeline.



SAS Visual Text Analytics provides nine predefined concepts, which are concepts whose rules are already written. Predefined concepts save time by providing you with commonly used concepts and their definitions, such as *Measure* or *Organization*. You cannot edit predefined concepts or their rules, but you can append additional rules in the provided code editor. The predefined concepts supported by SAS Visual Text Analytics were created based on multiple language relevance.



When you click on a concept such as **nlpMeasure**, and select **Matched**, Model Studio presents the documents containing the concept, and highlights the terms that satisfy the concept. The concept rules that define a predefined concept are not shown. However, the definition of a custom concept can contain a reference to another concept, whether it's custom or predefined.



Although you can't see the rules that define a predefined concept, you can use the Test Sample Text feature to investigate what terms are highlighted in a given document related to the predefined concept rules.

3.3 Terms



Exploring the Text Parsing Node



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Start list

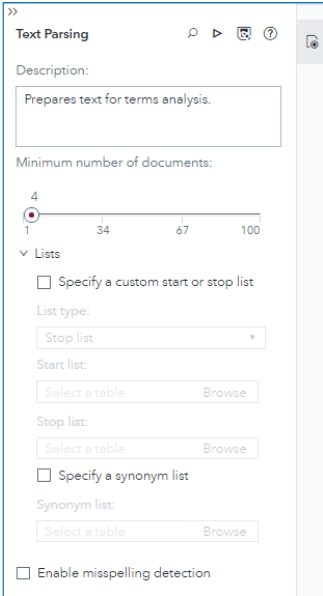
- terms to include
- inclusion dictionary

Stop list

- terms to exclude
- exclusion dictionary







Synonym list

- pairs of words to treat as single terms
- required format



The Text Parsing node has several properties. You use Start lists and Stop lists to control which terms are used and which are ignored.

A Start list is a data set that contains a list of terms that should be included in the analysis. Sentiment, topics, and categories use only the terms in the Start list. A Start list is also called an inclusion dictionary.

A Stop list is a data set that contains a list of terms that should be excluded from the analysis. You can use Stop lists to exclude terms that contain little information or that are extraneous to your Visual Text Analytics tasks. Sentiment, topics, and categories do not use the terms in the Stop list. A Stop list is also called an exclusion dictionary.

A default Stop list is provided for English (en_stoplist.sashdat) and for all supported languages.

Dictionary tables that are used as Start or Stop lists can be SAS tables. They do not need to be CAS tables.

The Minimum number of documents specifies how many documents a term must appear in to be allowed to be in a Start list. A term that is in a Start list, or is absent from a Stop list, will be placed in the Stop list if it appears in fewer than the minimum number of documents.

A synonym list is a SAS data set that identifies pairs of words that should be treated as single terms for the purpose of analysis. Synonym lists are stored in data sets and have a required format.

The Text Parsing node also has a spell-checking feature.

Text Parsing: Kept and Dropped Terms



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term

- label for group of characters, strings, or patterns that represent a single concept
- building blocks for topics, term maps, and category rules
- associated role is either blank, identifies term's part of speech, or identifies concept

token

- sequence of characters without separator

surface form

- variant of a term in a matched subset of text
- include inflected forms, synonyms, misspellings, and other ways to reference

stemming

- process of identifying inflected forms for nouns and verbs

Let's go over some vocabulary. A *term* is defined as a label for a group of characters, strings, or patterns that represent a single concept (an idea) as defined by underlying rules or algorithms. In SAS Visual Text Analytics, a term is the basic building block for topics, term maps, and category rules. Each term has an associated role that either is blank, identifies the term's part of speech, or identifies the concept that is associated with the term.

For alphabetic writing systems, a word is a *token* (a sequence of characters without a separator), and a term contains one or more words. There is a subtle difference between a term and a concept. When the Concept node appears before the Text Parsing node, a concept is eligible to be placed in the term table.

A *surface form* is a variant of a term that is in a matched subset of text. Surface forms can include inflected forms, synonyms, misspellings, and other ways of referring to a term.

Stemming is the process of identifying inflected forms for nouns and verbs. For example, *going* is an inflected form of *go*.

Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Kept Terms (1522)

☐	Term *	Role	Documents	Frequency ↓
☐	not	ADV	658	1174
☐	take	V	677	1105
☐	depression	N	492	616
☐	feel	V	371	518
☐	year	N	395	502
☐	drug	N	341	485
☐	day	N	344	479
☐	medication	N	304	446

Dropped Terms (6849)

☐	Term *	Role	Documents	Frequency ↓
☐	i	PRO	1081	5003
☐	be	V	1129	4183
☐	have	V	1069	2849
☐	and	CONJ	1028	2746
☐	the	DET	904	2560
☐	to	PPOS	897	2386
☐	it	PRO	882	2100
☐	a	DET	792	1655

Start List

Stop List

To examine the terms that are used in the analysis, right-click the Text Parsing node and select Open.

When you open the Text Parsing node, you see the Kept Terms and Dropped Terms tables. Kept Terms are terms that reside in the Start list, and Dropped Terms reside in the Stop list. By default, both lists are sorted in descending order based on the frequency that each term appears.

You can move a kept term to the Dropped Terms pane by selecting the term and clicking the **Drop terms** icon, and you can move a dropped term to the Kept Terms pane by selecting the term and clicking the **Keep terms** icon.

If you make changes to the terms, you must rerun the project to see the effects of your changes.

Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Kept Terms (1522) Filter

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☐ take	V	677	1105
☒ depression	N	492	616
☐ feel	V	371	518
☐ year	N	395	502
☐ drug	N	341	485
☐ day	N	344	479
☐ medication	N	304	446

Dropped Terms (6849) Filter

Term	Role	Documents	Frequency
☐ i	PRO	1081	5003
☐ be	V	1129	4183
☐ have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655

Documents

All (1414) Matched (492 of 1414) Search

DrugReport

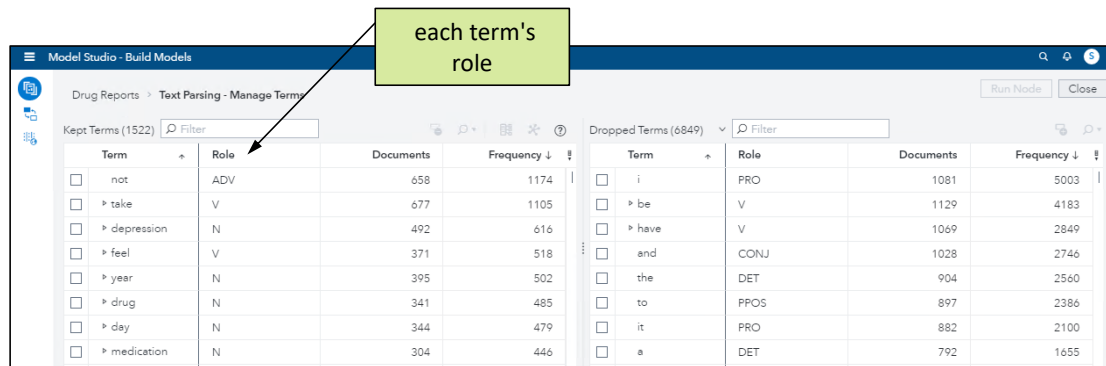
...due to unrelenting **depression**. I had lost my sister mom within 8 months and although I had been on an antidepressant for a long time before I grieved but still couldn't get over the **depression**. Within 3-5 days both I and my husband noticed a significant improvement. I felt better more like me than I had in years. Only thing I notice was I sometimes mix up words. Anybody else do that? Not bad enough for me to stop med tho.

...an abyss of **depression**, one which I have never experienced before, I had to go back on the prexifan and taper myself off for a month, even then I was having a continuous panic attack with heart pounding, no eating for 4-5 days, severe sense of loneliness, guilt, and poor self-worth. It was absolutely miserable that lasted for about 6-10 days. I have family support, but needed to see a psychiatrist (still waiting for visit). Be very cautious going on this med. It's great for a couple weeks, but very exhausting.

Document 1 of 492

Highlight: Term matches Search matches

You can view documents that contain a kept term by selecting the term and selecting **Matched** in the Documents pane.



Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Kept Terms (1522) | Filter

Term *	Role	Documents	Frequency ↓
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479
☐ > medication	N	304	446

Dropped Terms (6849) | Filter

Term *	Role	Documents	Frequency ↓
☐ i	PRO	1081	5003
☐ > be	V	1129	4183
☐ > have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655

The Role column displays each term's role, if one can be determined. A role could be a part of speech or a concept. To view the synonyms that were assigned to a term, click the triangle that appears next to that term.

Synonym list

- pairs of words to treat as single terms
- required format



TERM

TERMROLE

PARENT

PARENTROLE

Recall that a synonym list is a SAS data set that identifies pairs of words that should be treated as single terms for the purposes of analysis, and these lists have a required format.

You must include the following variables:

TERM, which contains a term to treat as a synonym of the parent, and PARENT, which contains the representative term to which the term should be assigned.

TERMROLE, which enables you to specify that the synonym is assigned only when the term occurs in the role that is specified in this variable. A term role is a function that is performed by a term in a particular context. Term roles include part-of-speech roles, entity roles, and user-defined roles.

And PARENTROLE, which enables you to specify the role of the parent.

Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Run Node Close

Kept Terms (1522) Filter

Term	Role	Documents	Frequency ↓
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479
☐ > medication	N	304	446

Dropped Terms (6849) Filter

Term	Role	Documents	Frequency ↓
☐ i	PRO	1081	5003
☐ > be	V	1129	4183
☐ > have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655



a

an

the

and

or

but

By default, words that provide little or no value are excluded from analysis. Examples of these words include the articles a, an, and the, and conjunctions such as and, or, and but. Other terms that are specific to your document collection, but provide little or no value, are also identified and excluded.

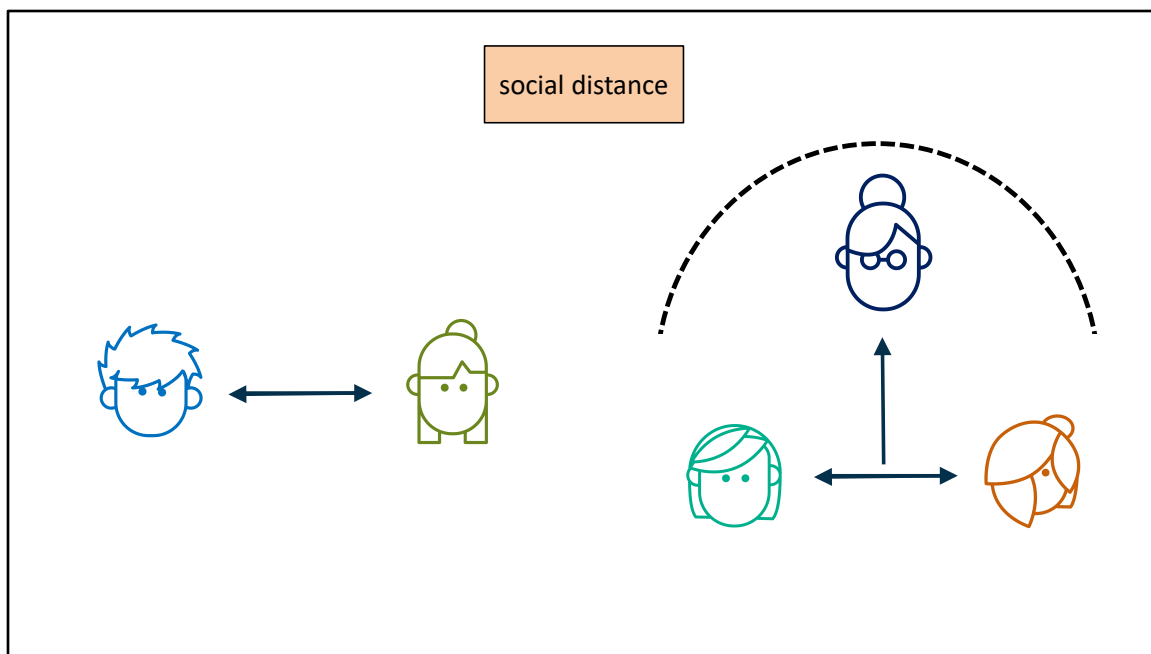
Similarity Scores and Terms Maps



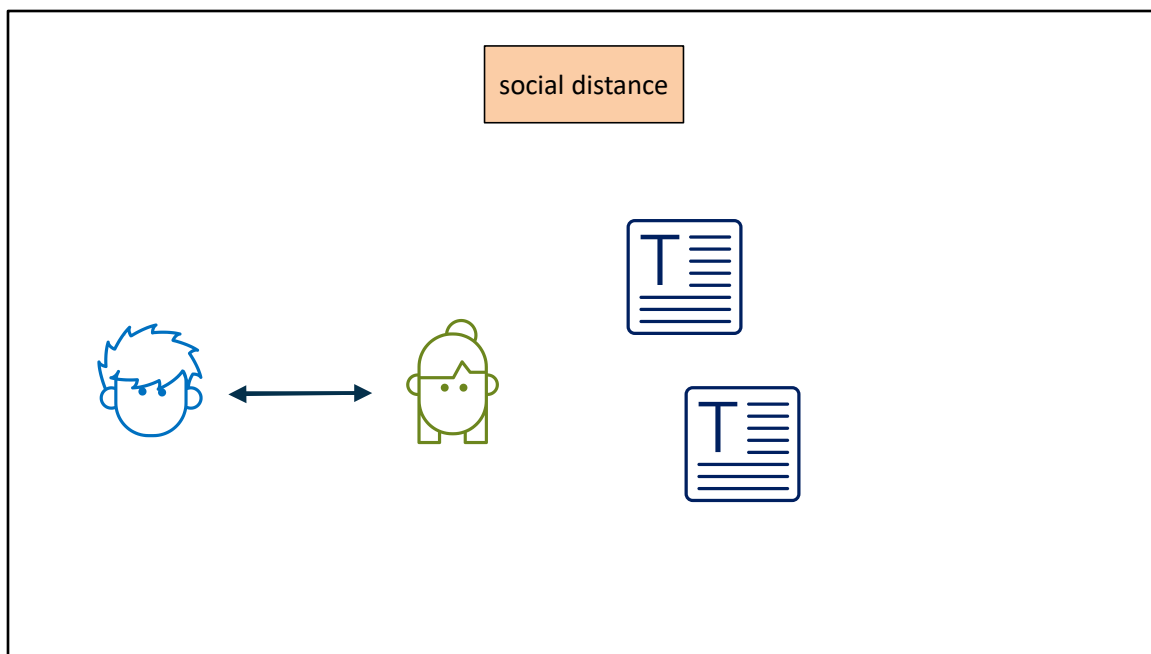
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	Term ↑	Similarity ↓	Role	Documents	Frequency
☒	▷ depression	1.000	N	492	616
☐	major	0.672	A	64	65
☐	anxiety	0.655	N	187	231
☐	major depression	0.640	nlpNounGroup	27	27
☐	immensely	0.614	ADV	4	4
☐	▷ satisfy	0.578	V	17	17
☐	chronic	0.576	A	22	23
☐	chronic pain	0.559	nlpNounGroup	5	6
☐	▷ combine	0.554	V	10	10
☐	depression	0.551	PN	9	10

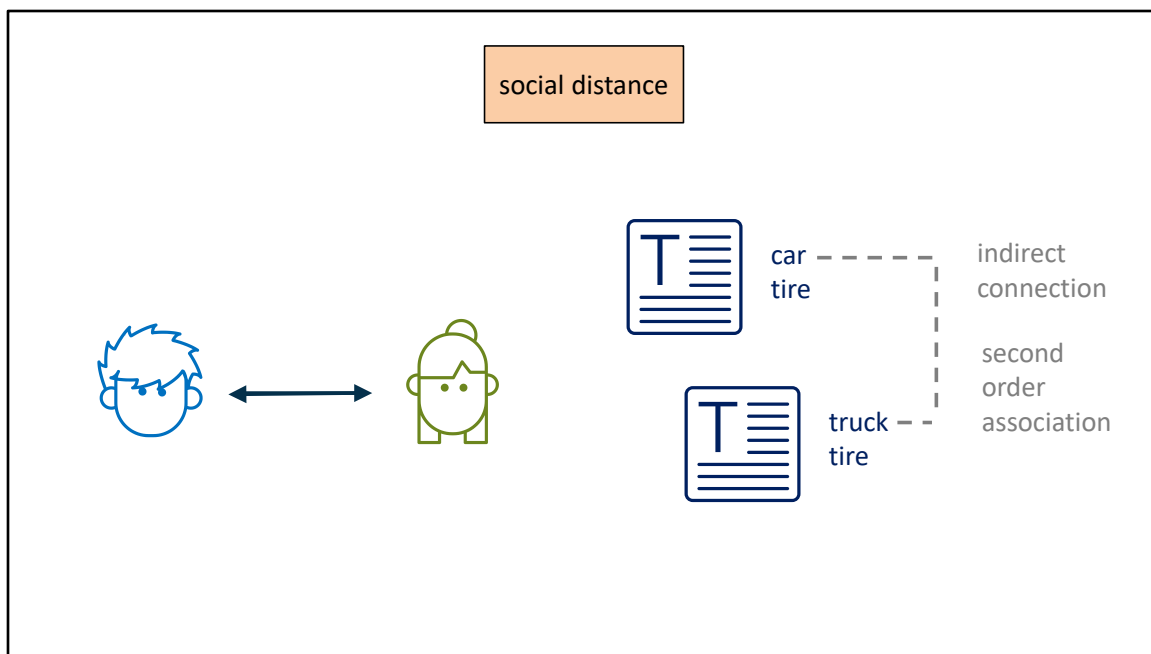
You can also view similarity scores for a term. Similarity scores are derived based on the linguistic distance between words as measured by co-occurrence frequencies. Social network metrics can be used to derive a social distance between two people based on community membership, and similar arguments can be used to derive the distance between words based on document membership (or co-occurrence). Thus, from a social network perspective, the personas *depression* and *major* have the most social interaction, while *anxiety*, *major depression*, and *immensely* are also in the social mix.



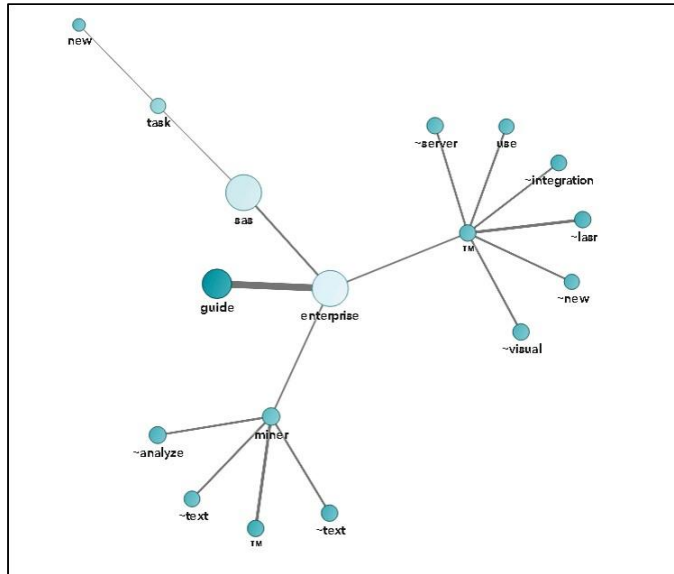
Social distance relates to two individuals interacting directly by speaking with each other or interacting indirectly by communicating through a third party.



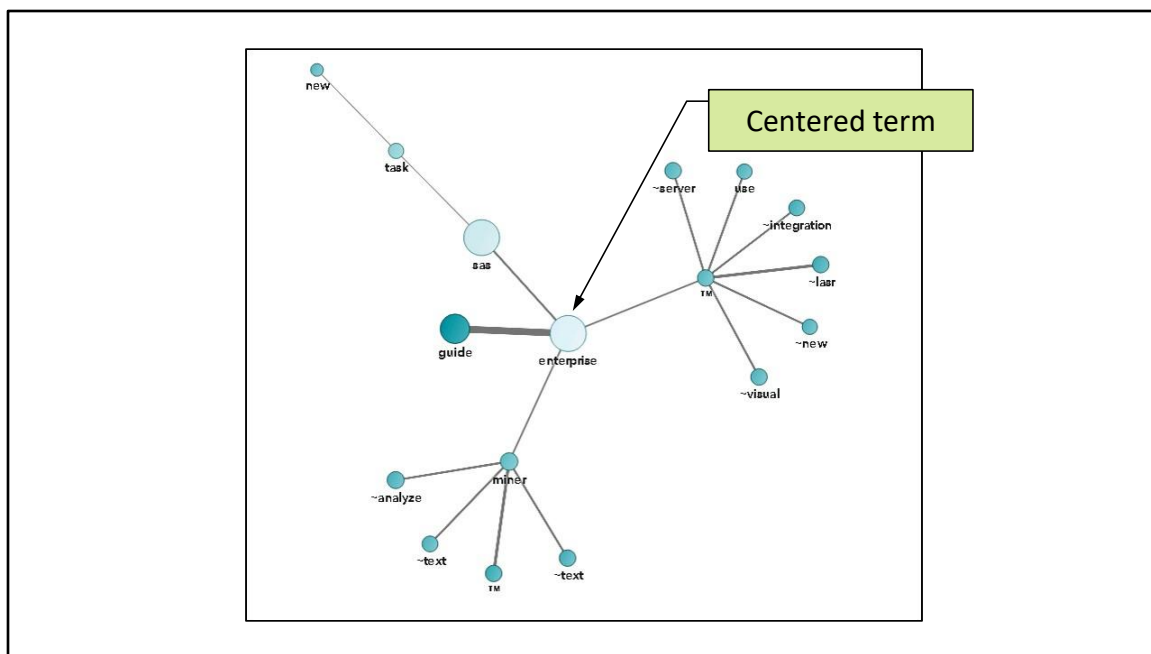
Words in the same document can be viewed as interacting directly. Words that do not appear together may be related because of shared direct relationships.



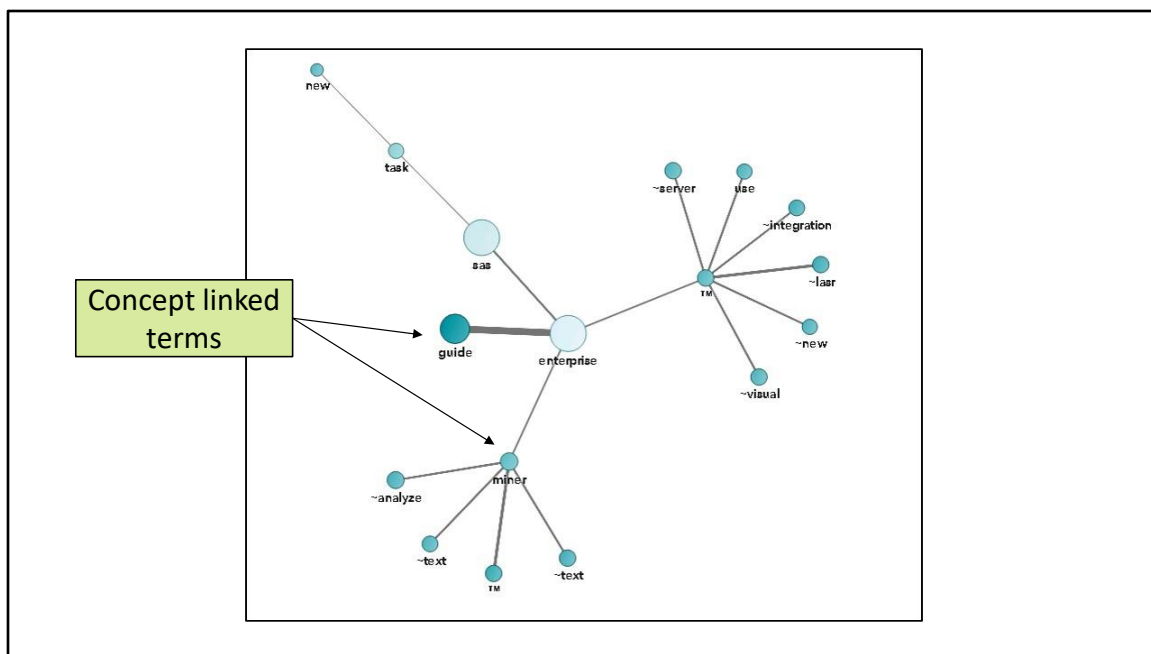
For example, one document may mention a car and a tire, while another mentions a truck and a tire, so car and truck are indirectly related through the word tire, and thus can be viewed as having an indirect social connection. This is called a *second order association*. The relationship depends on the context, and the word tire provides the context.



The Term Map window displays a term map for the selected term. The selected term,



called the centered term, is *enterprise*, and it is represented by the largest circle in the map. The term *guide* has the strongest association with *enterprise* based on the thickness of the line connecting the two terms.



Guide and *miner* are concept linked terms that co-occur with the centered term. For additional information about the term map display, click the **Help** icon in the term map display.

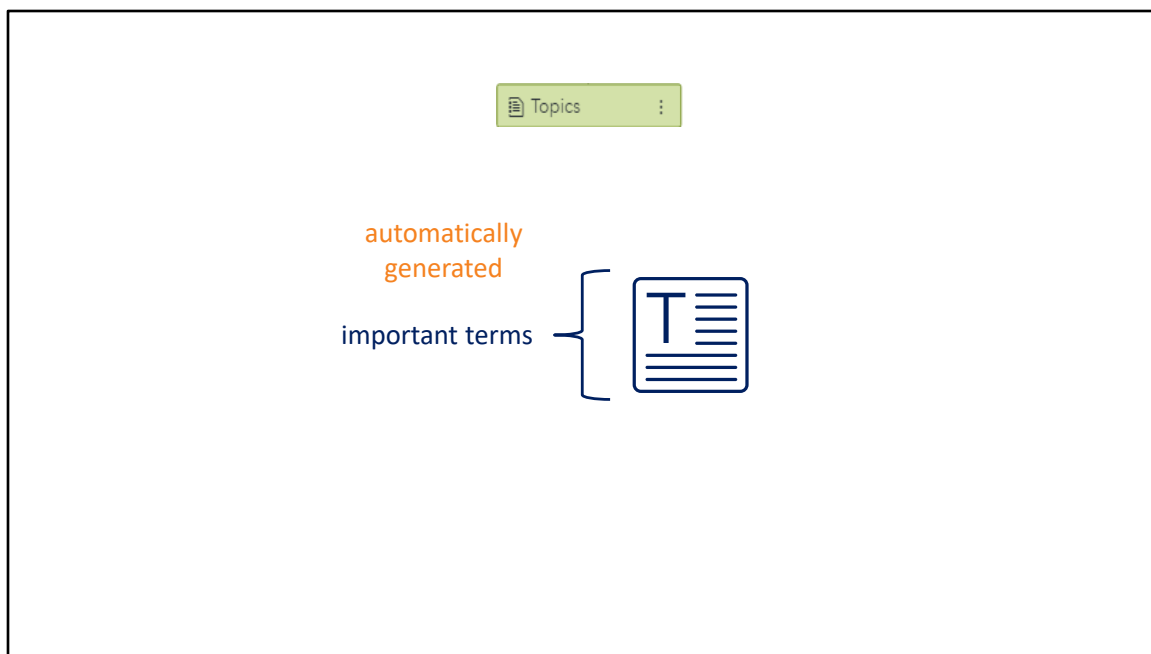
3.4 Machine-Generated Topics



Exploring Topics



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Topics are derived from natural groupings of important terms that occur in your documents. In SAS Visual Text Analytics, topics are automatically generated and assigned to documents. A single document can contain more than one topic.

Model Studio - Build Models

Drug Reports > Topics

Topics (10)

Topic	Created by	Documents
☒ +symptom, +withdrawal, off of, horrible, +dose	System	224
☐ +effect, +side effect, side, +side, sexual	System	223
☐ +work, ecstapin, best, +year, +try	System	223
☐ +day, +feel, mg, a day, +start	System	217
☐ abidal, +pain, just, +start, +work	System	208
☐ +medication, very, +good, +use, abidal	System	190
☐ +life, +save, +thought, +druo, more	System	184

Terms

All (1522) Matched Filter

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479

Documents

All (1414) Matched (224 of 1414) Search

DrugReport	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently went off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	⊖	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, I fill like I am in a another world and vision problems I slowly took myself off of this deadly drug . I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week . My opinion with the withdrawals is like diaing your own arve. I wouldn't recommend this druo to anybody.	⊖	0.330

Document 1 of 224

Highlight: Topic matches Search matches

The Topics page displays all the topics that SAS Visual Text Analytics identified. The default name of a topic is the top five terms based on relevancy scores. The terms table displays every term in the topic, its calculated relevancy, its assigned role, and the number of documents that contain that term. These terms are sorted in descending order based on their relevancy scores.

Document-level sentiments are also generated if they are preceded by a Sentiment node. The percentage of the documents in the topic that have a sentiment score of positive, negative, and neutral appears with each topic.

Model Studio - Build Models

Drug Reports > Topics

Topics (10)

Topic	Created by	Documents	Term	Role	Documents	Frequency
☒ +symptom, +withdrawal, off of, horrible, +dose	System	204				
☐ +effect, +side effect, side, +side, sexual	System	2		ADV	658	1174
☐ +work, ecstapin, best, +year, +try	System	2		V	677	1105
☐ +day, +feel, mg, a day, +start	System	217	☐ > feel	N	492	616
☐ abidal, +pain, just, +start, +work	System	208	☐ > year	V	371	518
☐ +medication, very, +good, +use, abidal	System	190	☐ > drug	N	395	502
☐ +life, +save, +thought, +drugs, more	System	184	☐ > day	N	341	485
					344	479

Split a topic into two topics

Documents

All (1414) Matched (224 of 1414) Search

DrugReport	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently want off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	⊖	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, i fill like i am in a another world and vision problems. I slowly took myself off of this deadly drug . I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week . My opinion with the withdrawals is like diazina your own grave. I wouldn't recommend this drugs to anybody.	⊖	0.330

Document 1 of 224

Highlight: Topic matches Search matches

If a topic seems to convey information about two or more themes, you might want to split the topic to see whether the two themes emerge.

The screenshot shows the SAS Model Studio - Build Models interface. The 'Topics' table is displayed with columns: Topic, Created by, Documents, Term, Role, Documents, and Frequency. A callout box with the text 'Merge two or more topics' points to the 'Merge Topic' icon in the toolbar. Below the Topics table, the 'Documents' section shows a list of documents with columns: DrugReport, Sentiment, and Relevancy. The first document is highlighted, showing a snippet of text about 'ecstapin xr' and 'side effects'.

Topic	Created by	Documents	Term	Role	Documents	Frequency
☒ +symptom, +withdrawal, off of, horrible, +dose	System	224				
☐ +effect, +side effect, side, +side, sexual	System	223				
☐ +work, ecstapin, best, +year, +try	System	217				
☐ +day, +feel, mg, a day, +start	System	208				
☐ abidal, +pain, just, +start, +work	System	190				
☐ +medication, very, +good, +use, abidal	System	184				
☐ +life, +save, +thought, +drugs, more	System					

DrugReport	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently want off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	😊	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, I fill like I am in a another world and vision problems I slowly took myself off of this deadly drug . I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week . My opinion with the withdrawals is like diazina your own arve. I wouldn't recommend this drugs to anybody.	😊	0.330

If you see two topics that seem related, you can merge them by selecting them and clicking **Merge Topic**. This action combines all the selected topics into the same topic.

The screenshot shows the SAS Model Studio - Build Models interface. The 'Topics' pane on the left lists 10 topics, each with a checkbox and a 'Documents' count. A green callout box with the text 'Add topics as categories.' points to a button in the 'Topics' pane. The 'Terms' pane on the right shows a table of terms with columns for Term, Role, Documents, and Frequency. The 'Documents' pane at the bottom shows a list of documents with columns for Document, Sentiment, and Relevancy.

Topic	Created by	Documents
☒ +symptom, +withdrawal, off of, horrible, +dose	System	224
☐ +effect, +side effect, side, +side, sexual	System	223
☐ +work, ecstapin, best, +year, +try	System	223
☐ +day, +feel, mg, a day, +start	System	217
☐ abidal, +pain, just, +start, +work	System	208
☐ +medication, very, +good, +use, abidal	System	190
☐ +life, +save, +thought, +drugs, more	System	184

Term	Role	Documents	Frequency
not	ADV	658	1174
		677	1105
		492	616
		371	518
> year	N	395	502
> drug	N	341	485
> day	N	344	479

Document	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently went off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	😊	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps, nausea, vomiting, dizzy, can't concentrate, I feel like I am in a another world and vision problems I slowly took myself off of this deadly drug. I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week. My opinion with the withdrawals is like diazina your own grave. I wouldn't recommend this drug to anybody.	😞	0.330

The next step in the analysis process is to identify which topics you want to promote to categories. To promote a topic to a category, select that topic in the Topics pane and click **Add topics as categories**. SAS Visual Text Analytics adds the selected topic to the Categories page. You can promote multiple topics to categories at one time.

>> Topics

Description:
Assigns documents to topics.

▼ Topic Discovery

☒ Automatically determine number of topics

Maximum topics:
25

Term density:
0 1 5 10

Document density:
0 1 5 10

You can edit the properties that affect all topics by changing the settings for the Topics node. Term density refers to how topics are populated with terms. It's defined by a number between 1 and 10. The default is 1. Increasing the term density number captures fewer documents.

This value affects the number of documents that belong to a topic. For example, having fewer terms in a topic captures fewer documents. Values that you enter are rounded to the nearest integer.

You can also designate a maximum number of topics that you want generated for the project. If you do not enter a value, the software uses default methods to generate topics from important terms.

Run the topics to see the results of your changes.

Topics (16)			Terms				
☐ Topic	Created by	Documents ↓	All (21918) Matched (13 of 21918) Filter				
☒ taxiway, runway, taxi, clearance, groundcontrol	User	16117	☐ Term	Relevancy ↓	Role	Documents	Frequency
☐ taxiway, runway, taxi, clearance, tower	User	4176	☐ crossrunway	1.000	_JNCURSION	546	844
☐ tower, takeoff, clearance, runway, hear	System	3335	☐ taxiway	1.000	_JNCURSION	2339	8705
☐ flap, knot, brake, +autopilot, gear	System	3222	☐ cross	1.000	_JNCURSION	2704	5113
☐ approach, runway, visual, visual approach, +locate	System	3004	☐ ramp	1.000	N	1012	1748
☐ +foot, descend, altitude, +autopilot, cross	System	2989	☐ taxi	1.000	_JNCURSION	3662	7700
☐ engine, emergency, declare, smoke, checklist	System	2869	☐ groundcontrol	1.000	_JNCURSION	1135	1982
☐ minimumequipmentlist, takeoff, flap, dispatch, flight	System	2867	☐ hear	1.000	V	2988	4509
			☐ control	1.000	N	7020	15306

You can select a subset of terms in a derived topic to create a custom topic. The custom topic assigns a term weight of 1 to each term. You can merge two topics to form a new topic. Then you can select terms from the merged topic to create a custom topic. This provides flexibility when you are creating custom topics. You can use subject-matter expertise to build customized dictionaries. Although a custom topic acts much like a custom concept that uses CLASSIFIER rules, the results can be different. This is because concepts apply Boolean operations, where topics apply numeric scoring.

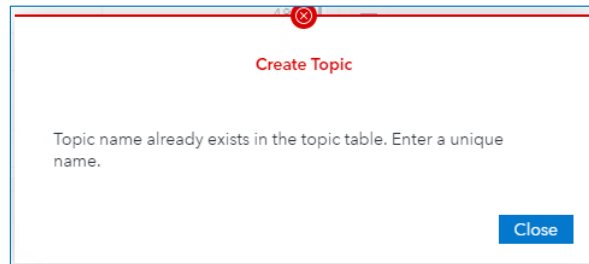
Select terms

Create a topic from terms

Term	Role	Documents	Frequency
☒ depression	N	492	616
☒ depressive	A	17	18
☒ depressive	PN	4	4
☒ depressive	N	4	4
☒ depressive episode	nlpNounGroup	4	5
☒ depressive symptom	nlpNounGroup	4	4
☐ describe	V	16	16
☐ desire	N	12	12

Term	Relevancy ↓	Role	Documents	Frequency ↓
☐ depressive	1.000	N	4	4
☐ depression	1.000	N	492	616
☐ depressive	1.000	A	17	18
☐ depressive symptom	1.000	nlpNounGroup	4	4
☐ depressive episode	1.000	nlpNounGroup	4	5
☐ depressive	1.000	PN	4	4

To create a custom topic, select all the terms that define the topic by clicking the square to the left of each term in the Terms pane. After you have selected all the terms that apply, click the **Create a topic from terms** icon in the upper right of the Terms pane. A new topic will be added to the Topics pane having the name of five of the selected terms. All terms in the custom topic will have a relevancy score of one.



Unfortunately, if you select the first five terms that define an existing topic, you will not be allowed create a custom topic. The following error message appears: Topic name already exists in the topic table. Enter a unique name.

To overcome this unfortunate limitation, you can try to emulate the desired topic by creating separate topics, and then joining them. A new set of relevancy values will be derived for the joined topic, so you will not get relevancy weights equal to one. Since you cannot delete topics, this approach often leads to a surplus of topics that will increase the run time of a pipeline.

Term and Document Density

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



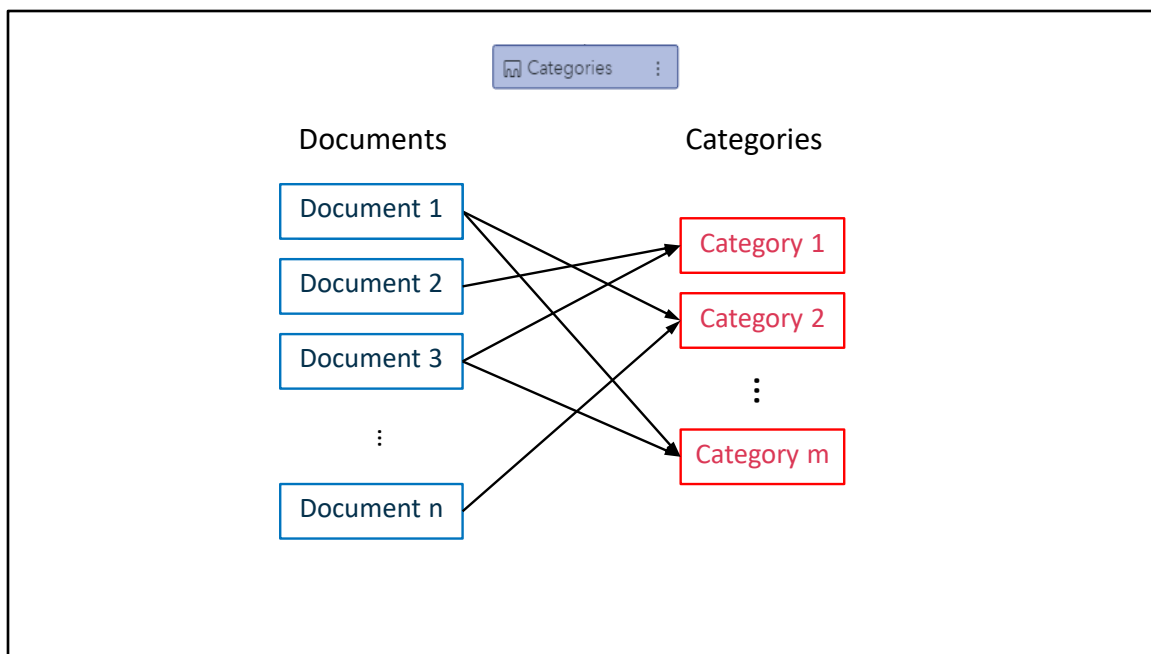
3.5 Categories



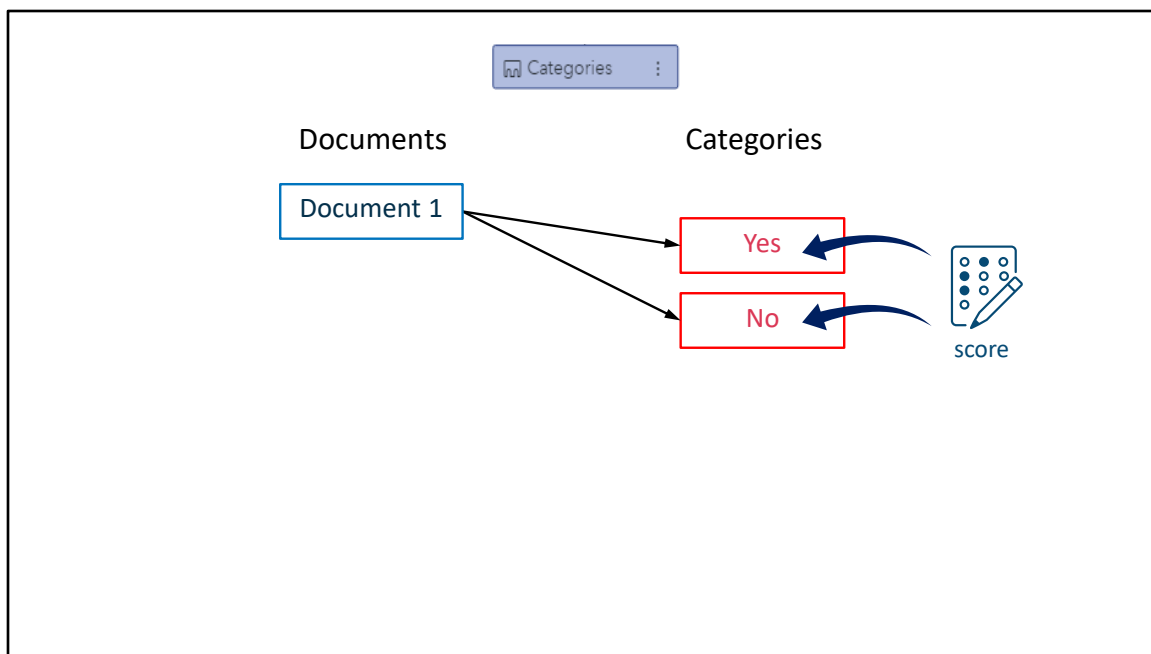
Reviewing Category Rules and Results



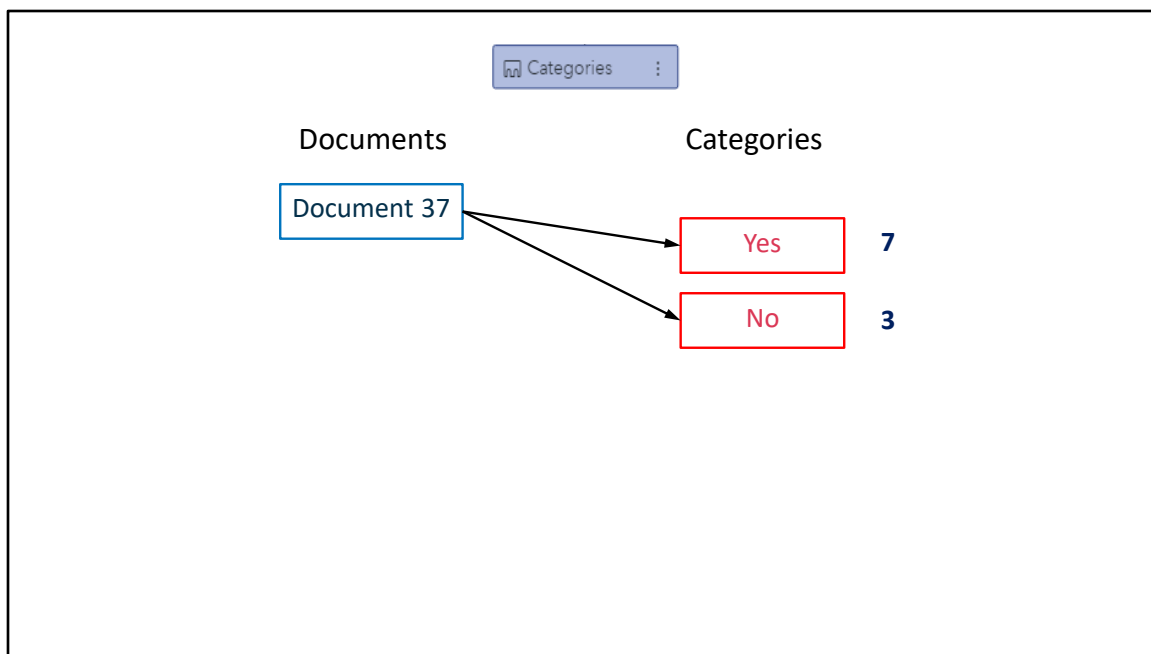
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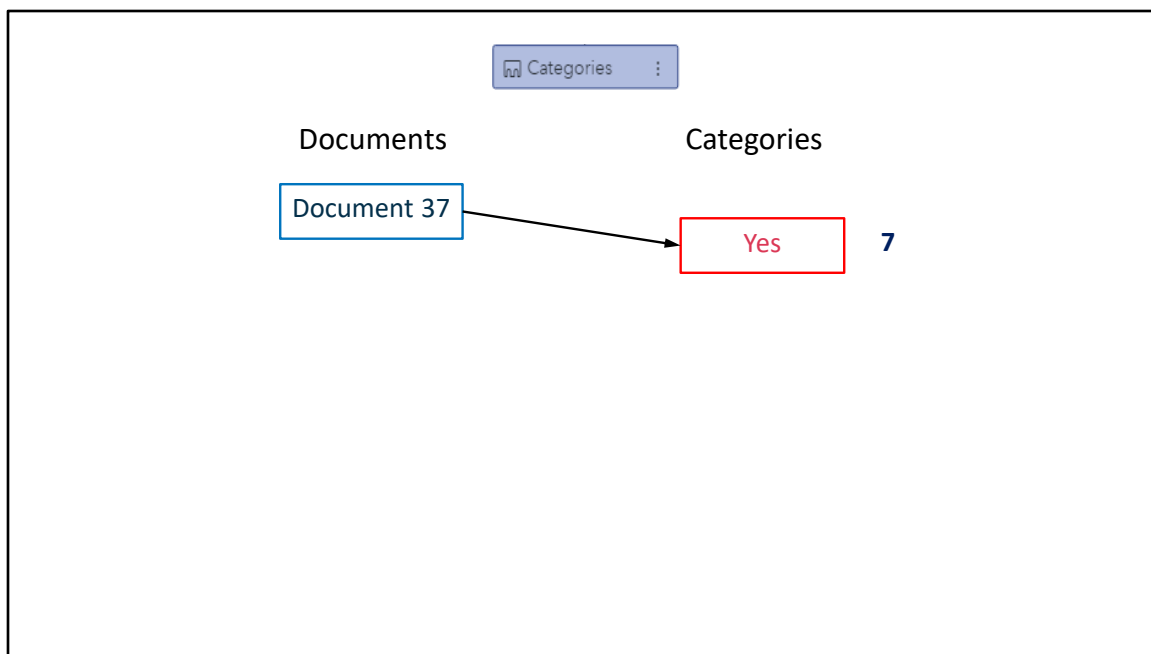
The Categories node allows a document to fall into more than one category.



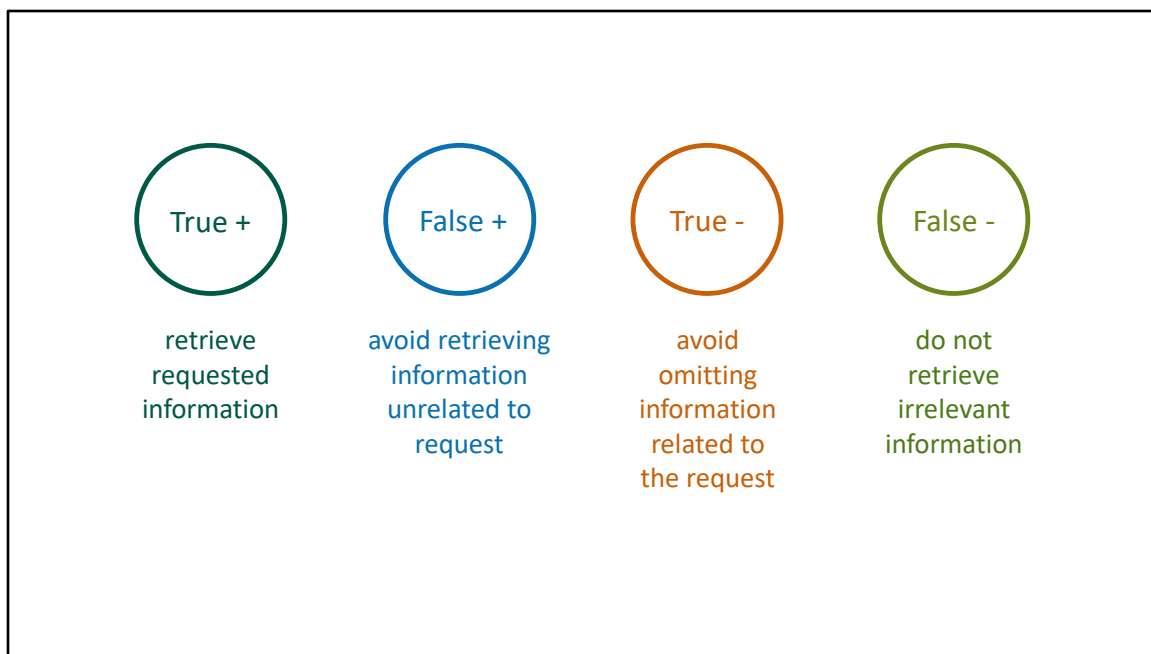
If a document is classified as belonging to both a Yes and No category for a binary category variable, a score is assigned to each category so that ambiguities can be resolved.



Suppose document 37 is assigned to the Yes category with a score of 7 and to the No category with a score of 3. Since the Yes category score is higher,



you can choose to assign document 37 to the Yes category.



When a category rule is based on a category target variable, you can determine the accuracy of the classification. There are four possible outcomes from a classified document: true positive, false positive, true negative, and false negative. For a true positive, you want to retrieve the information that was requested. For a false positive, you want to avoid retrieving information that is unrelated to the request. For a true negative, you want to avoid omitting information that is related to the request. And for false negatives, you do not want to retrieve information that is irrelevant.

Notation

ND = Number of documents

NS = Number of selected documents (retrieved)

TP = Number of true positives

FP = Number of false positives

TN = Number of true negatives

FN = Number of false negatives

		Action	
		Retrieved	Omitted
Actual	Contain Info	TP	FN
	Absent Info	FP	TN
Column Totals		NS	ND-NS

Category rules can be assessed based on summary statistics derived using the four possible outcomes.

Misclassification Rate = Fraction
misclassified

$$\text{_MISC_} = (FP + FN)/ND$$

Precision = Fraction of retrieved
documents that are relevant

$$\text{Precision} = TP/(TP + FP)$$

Recall = Fraction of relevant
documents that are retrieved

$$\text{Recall} = TP/(TP + FN)$$

		Action	
		Retrieved	Omitted
Actual	Contain Info	TP	FN
	Absent Info	FP	TN
Column Totals		NS	ND-NS

The misclassification rate, often presented as a percentage, is the fraction of documents that are incorrectly classified. Precision is the fraction of documents that are classified as positives that are true positives. Recall is the fraction of all positive documents that are classified as positive.

F1 Statistic = Harmonic mean of precision and recall

$$F1 = 2 / [(1/Precision) + (1/Recall)] = (2 * Precision * Recall) / (Precision + Recall)$$

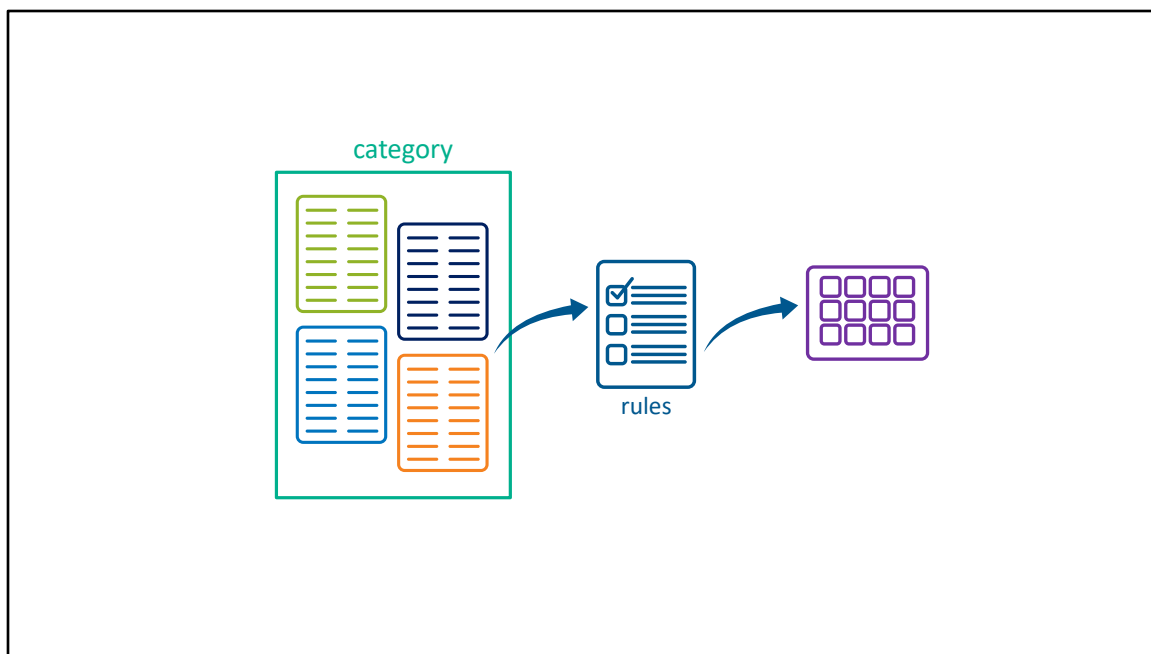
(F1 is also called the *F-Measure*.)

The F measure, or F1 statistic, is the harmonic mean of precision and recall and is one way to measure the tradeoff between accurate information retrieval and comprehensive information retrieval.

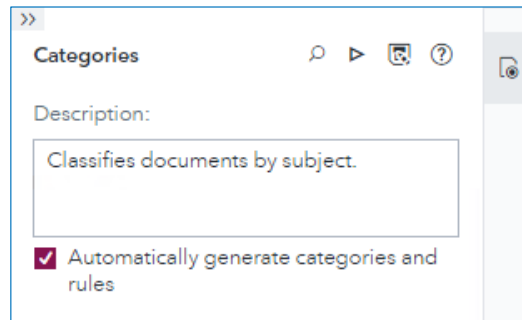
Exploring the Category Node



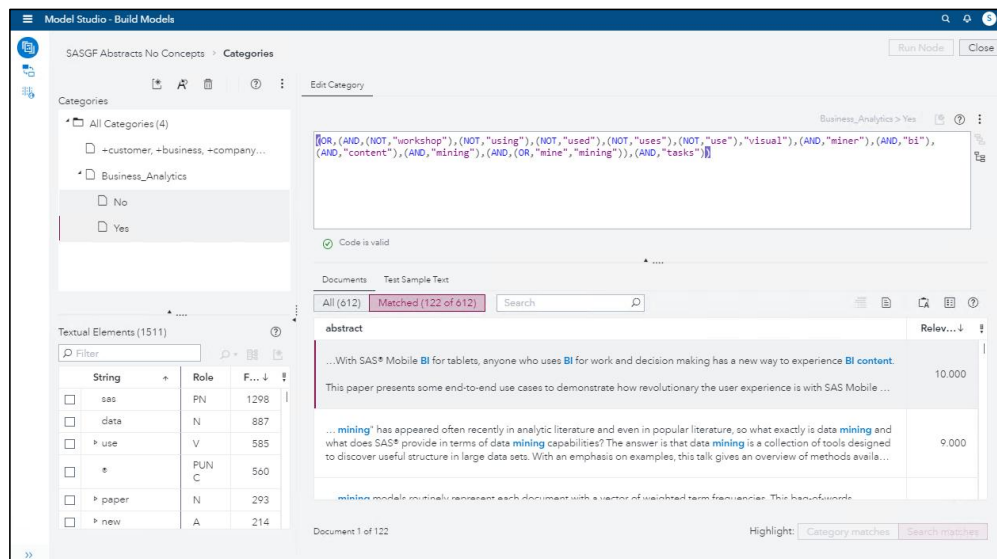
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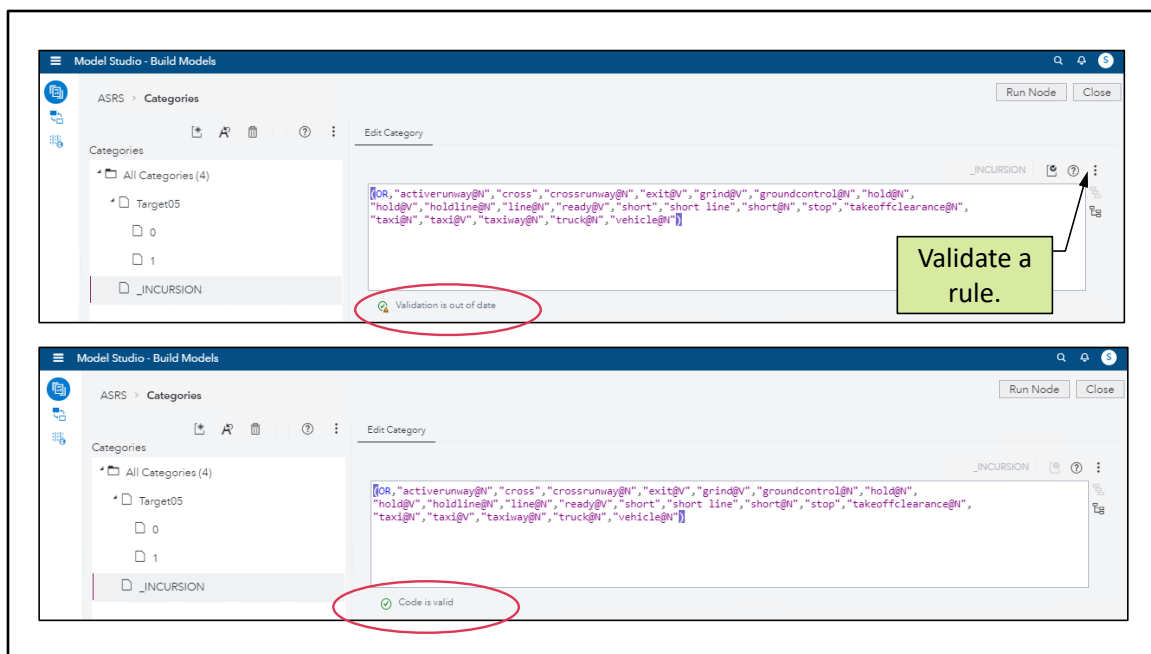
A *category* identifies a group of documents that share a common characteristic. The rules for each category are executed against the input data set.



The Categories node has a single property that determines if linguistic rules will automatically be derived for all category variables.



The Categories node uses machine learning and natural language processing to derive category rules for each category variable. The rules can be understood without any prerequisite knowledge of NLP algorithms. The rules can also be edited to incorporate domain knowledge that is not available to the software.



You can validate manually edited or created rules to ensure that there are no syntax errors.

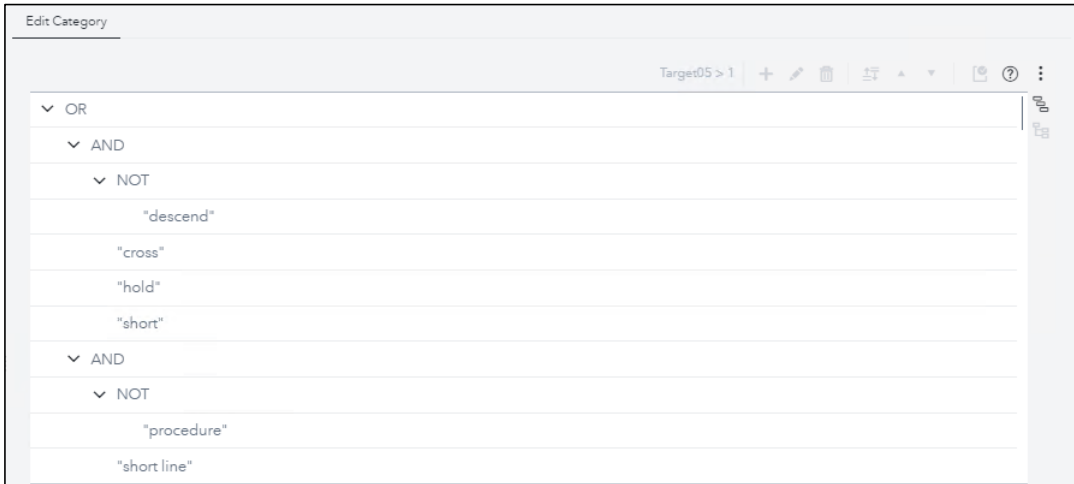
The screenshot shows a SAS interface for testing a rule. At the top, there are tabs for 'Documents' and 'Test Sample Text'. Below the tabs, there are filters for 'All (21519)' and 'Matched (11658 of 21519)', along with a search bar. A yellow box labeled 'Test a rule.' points to the 'Test Sample Text' tab. The main area displays a table with three columns: 'Text', 'Sentiment', and 'Relevancy'. The table contains three rows of text, each with a negative sentiment icon (a sad face) and a relevancy score. A pop-up window shows the text 'Groundcontrol cleared me to cross runway two right, but I had to stop for a fuel truck.' with terms highlighted in blue.

Text	Sentiment	Relevancy ↓
taxi JUST bare PAST THE instrumentlandingsystem HOLD SHORT LINE AT approach END OF runway AT sdf airport._ aircraft WERE SENT AROUND.THIS occur FOR sever REASON _ THE control say TAXI TO runway HOLD SHORT OF THE approach.NOT HOLD SHORT OF...	☹	130.000
cross OF HOLD SHORT LINE FOR runway _WE WERE clear TO THE hold PAD FOR runway _ AT dca airport TO WAIT OUT A ground STOP FOR ORD.AFTER approximate _ hour WE WERE give A reroute AND TOLD TO START engine AND TAXI TO HOLD SHORT OF runway _I THOUGHT...	☹	107.000
...WAS clear TO TAXI bywayof taxiway P taxiway left FOR A runway depart.AS I near taxiway south ON taxiway left I WAS ask IF I WAS FAMILIAR WITH ORANGE transit I state I WAS UNFAMILIAR AND WOULD LIKE A progressive	☹	104.000

Document 1 of 11658

Groundcontrol cleared me to cross runway two right, but I had to stop for a fuel truck.

You can test a rule to ensure it's capturing the information as intended. The highlighted terms show how the rule is invoked for a specific test case.



The tree view makes it easier to decipher the logic behind a rule. The hierarchy of the Boolean rules is made clear by the tree formatting. For the rule given, a NOT rule is within each AND rule, and the two AND rules are within the parent OR rule.

3.6 Document Scoring



Scoring New Documents

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Resource

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Lesson 4: Concept and Category Rule Definitions



4.0 Lesson Overview



Lesson Overview

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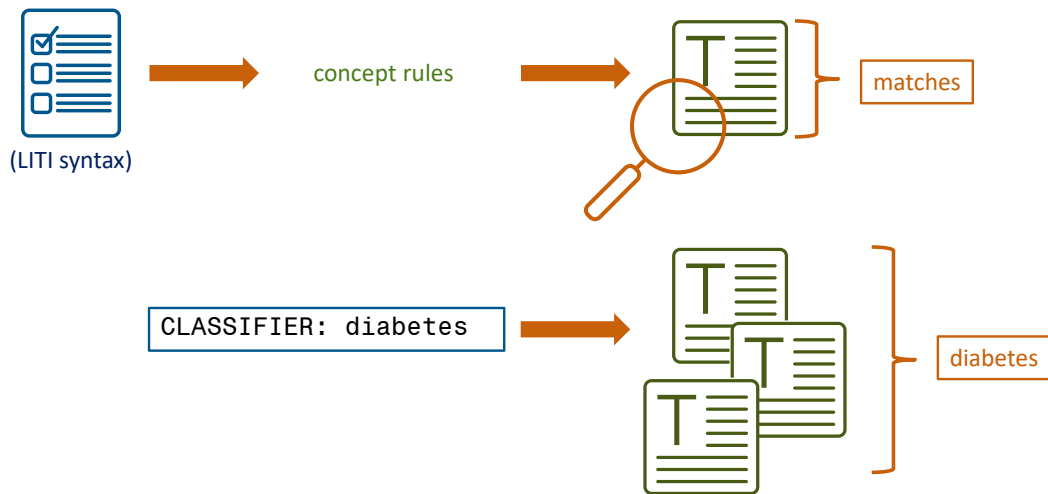
4.1 SAS Visual Text Analytics Rules



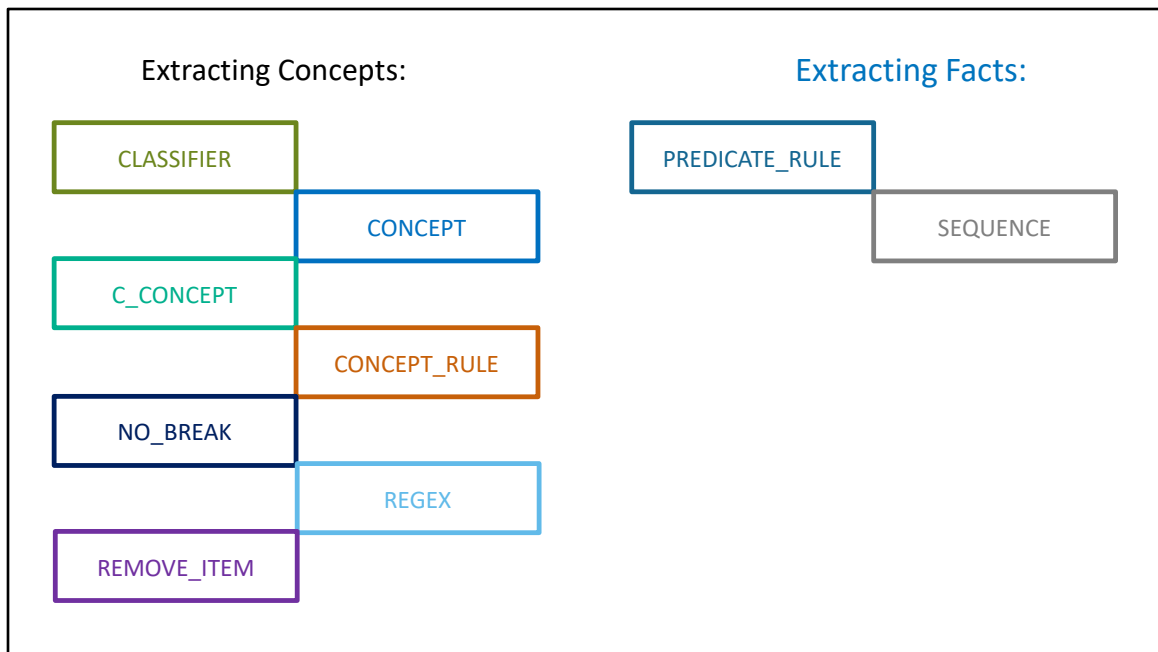
Overview of Concept Rules



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Concept rules are written using LITI (language interpretation and text interpretation) syntax. You write concept rules that recognize items in context so that you can extract only the pieces of the document that match the rule. For example, in the rule **CLASSIFIER: diabetes**, **CLASSIFIER** is the rule type and is written in uppercase letters, **diabetes** is the argument, and they are separated by a colon. The rule extracts all documents in your document collection that contain the word *diabetes*. Rule modifiers can be used to further refine the set of matches, because the rule syntax varies greatly depending on the rule type. Also, a single concept can reference one or more additional concepts.

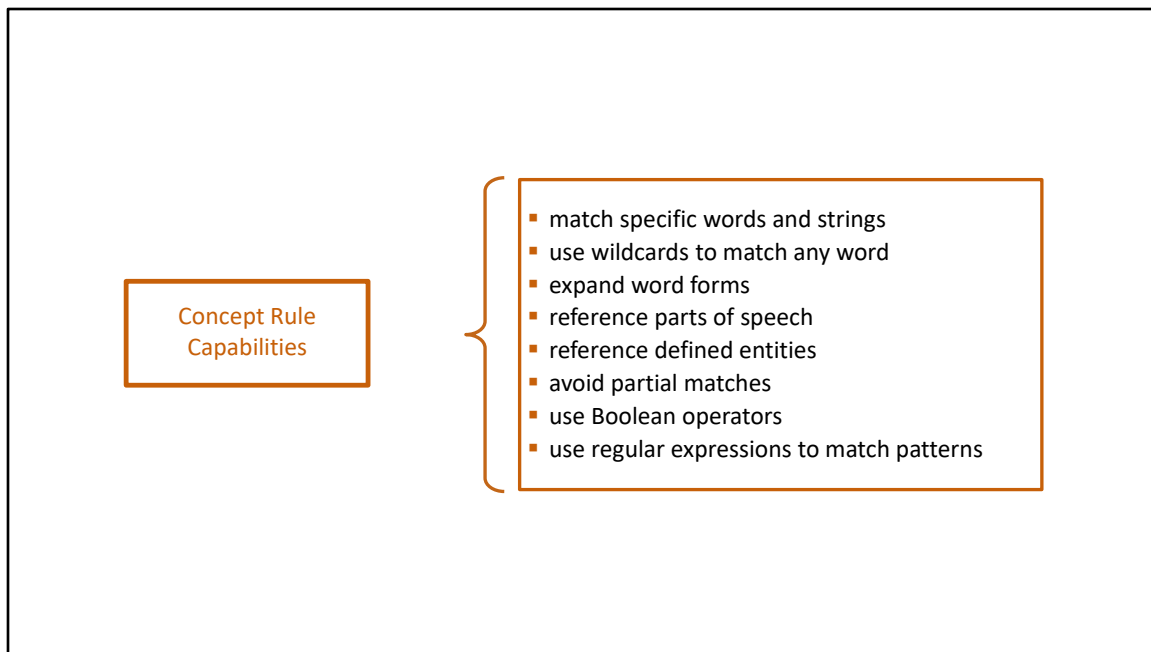


It's important to understand the rule types so that you can select those that efficiently generate the most matches for your purposes. The following rule types are used for extracting concepts: CLASSIFIER, CONCEPT, C_CONCEPT, CONCEPT_RULE, NO_BREAK, REGEX, and REMOVE_ITEM.

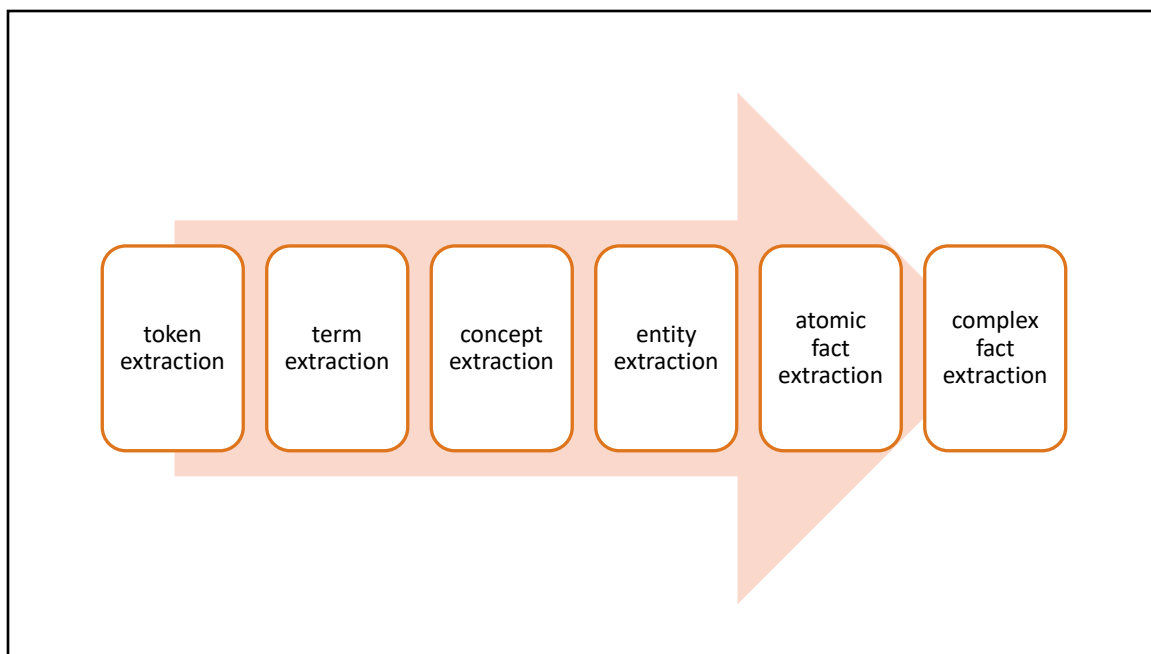
Facts (also called *predicates*) are related pieces of information that are located and matched together. For extracting facts, you can use the PREDICATE_RULE and SEQUENCE rule types.

A concept can be defined using more than one rule type.

For the purposes of this course, we aren't covering REMOVE_ITEM or SEQUENCE, but you can use the documentation to learn more about these rule types.



Each of the concept rule types provides one or more of the following capabilities: match specific words and strings, use wildcards to match any word, expand word forms, reference parts of speech, reference defined entities, avoid partial matches, use Boolean operators, and use regular expressions to match patterns.



As you move from one level of complexity to the next level, you engage additional natural language features of the software. Transitioning from terms to concepts and entities, you can benefit from predefined concepts, or you can exploit your domain expertise to add custom concepts.

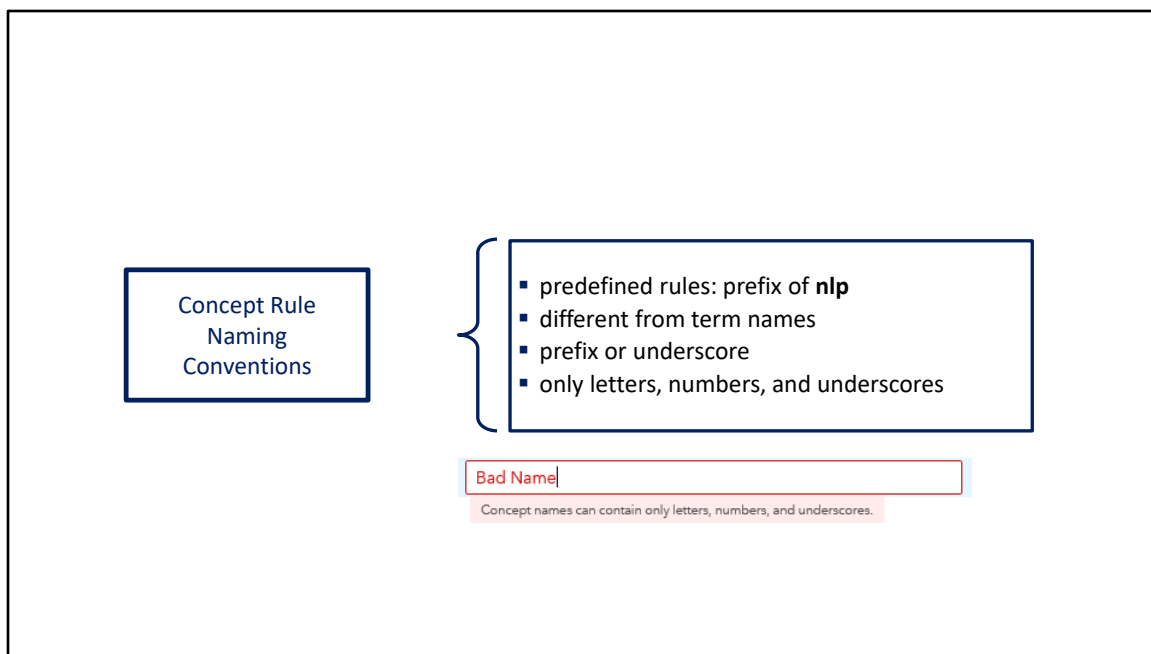
Parsing a document begins with token extraction. Tokens are just contiguous sets of characters separated by white space or punctuation. When you add language features, such as parts-of-speech, you increase the complexity from token extraction to term extraction. With more sophisticated natural language processing to identify concepts, such as noun phrases, measurements, or places, you can identify and extract more complex information from a document. Finally, by associating two or more concepts, you can extract facts or predicates, such as information that combines a university concept with a presidential concept to identify universities named after U.S. presidents. This is called atomic fact extraction.

Complex fact extraction is the goal of artificial intelligence, namely having computers that communicate directly with humans. Text analytics focuses on a specific document collection and avoids cumbersome overhead that would be needed to accommodate ad hoc queries spanning all human knowledge.

Concept Rule Naming Conventions



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Predefined concept rules use the prefix **nlp** for natural language processing. Concept names should not be the same as term names. You can use a prefix or an underscore to avoid ambiguities with terms. Concept names can contain only letters, numbers, and underscores. The example concept names we'll use all have at least one underscore to distinguish them from actual terms.

Concept Rule Coding Conventions

- **Punctuation to qualify matches:**
 - ~~CLASSIFIER, CONCEPT~~
- **Colon:**
 - separates rule types and tags: **CLASSIFIER:**
 - between arguments in PREDICATE_RULE
 - before part-of-speech tag: **:Prep**
- **Comma:**
 - separates arguments in rule definitions
 - separates logical operators in PREDICATE_RULE

You use punctuation to qualify the matches for all rule types except CLASSIFIER and CONCEPT.

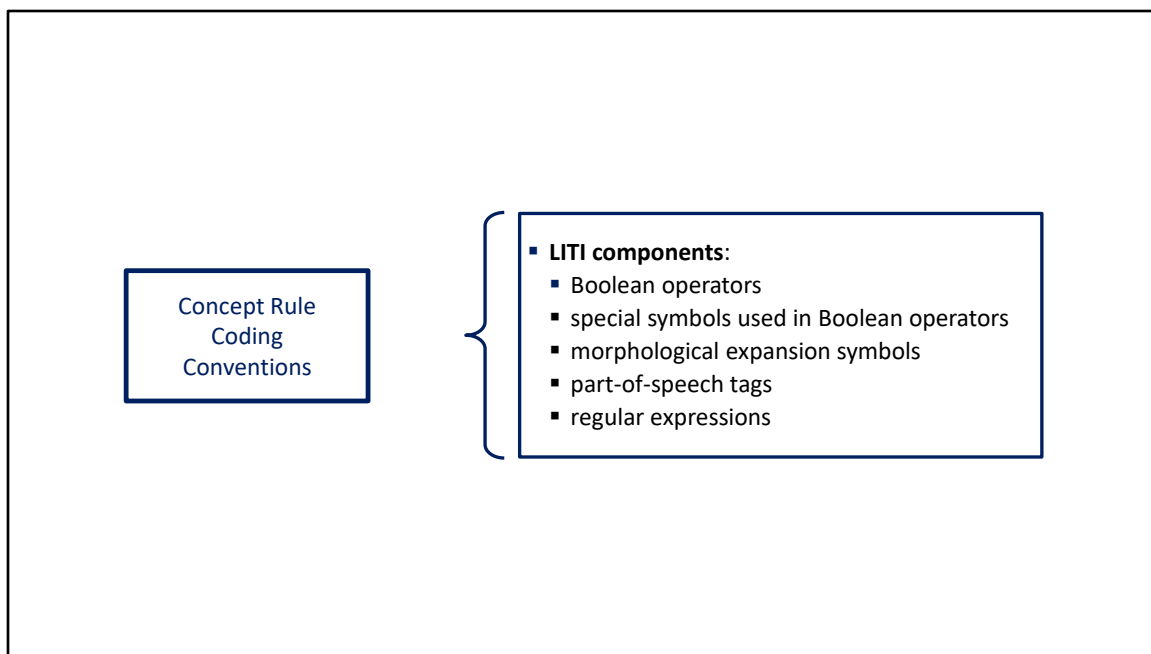
A colon separates rule types and tags after a concept rule type, for example CLASSIFIER:. You use a colon between arguments in a PREDICATE_RULE definition, and before a part-of-speech tag, for example, :Prep.

A comma separates elements such as arguments in a rule definition. You add a space after the comma and before the next element. Also, you can use a comma to separate logical operators in a PREDICATE_RULE definition.

Concept Rule Coding Conventions

- **Single space:**
 - separate strings, concept names, part-of-speech tags, rule modifiers
 - CONCEPT, CONCEPT_RULE, and C_CONCEPT
- **Quotation marks:**
 - enclose concept names and strings
 - CONCEPT_RULE, REMOVE_ITEM, PREDICATE_RULE
- **Parentheses:**
 - group elements
 - CONCEPT_RULE, REMOVE_ITEM, SEQUENCE, PREDICATE_RULE
 - Boolean operators (AND, OR, SENT, DIST_n, and ORDDIST_n)
- **Square braces:**
 - group elements
 - REGEX

You use a single space to separate strings, concept names, part-of-speech tags, and rule modifiers in CONCEPT, CONCEPT_RULE, and C_CONCEPT definitions. You use quotation marks to enclose concept names and strings in CONCEPT_RULE, REMOVE_ITEM, and PREDICATE_RULE definitions. You use parentheses to group the elements in CONCEPT_RULE, REMOVE_ITEM, SEQUENCE, and PREDICATE_RULE rule types, and within certain Boolean operators (AND, OR, SENT, DIST_n, and ORDDIST_n). And you use square braces to group elements in the REGEX rule type.



Other LITI components include Boolean operators, special symbols used in Boolean operators, morphological expansion symbols, part-of-speech tags, and regular expressions.

Additional Concept Rule Guidelines

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Boolean Operators for Extracting Concept Rules and Facts

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Special Symbols Used in Boolean Rules

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Morphological Expansion Symbols in Concept Rules

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Part-of-Speech Tags (for English)

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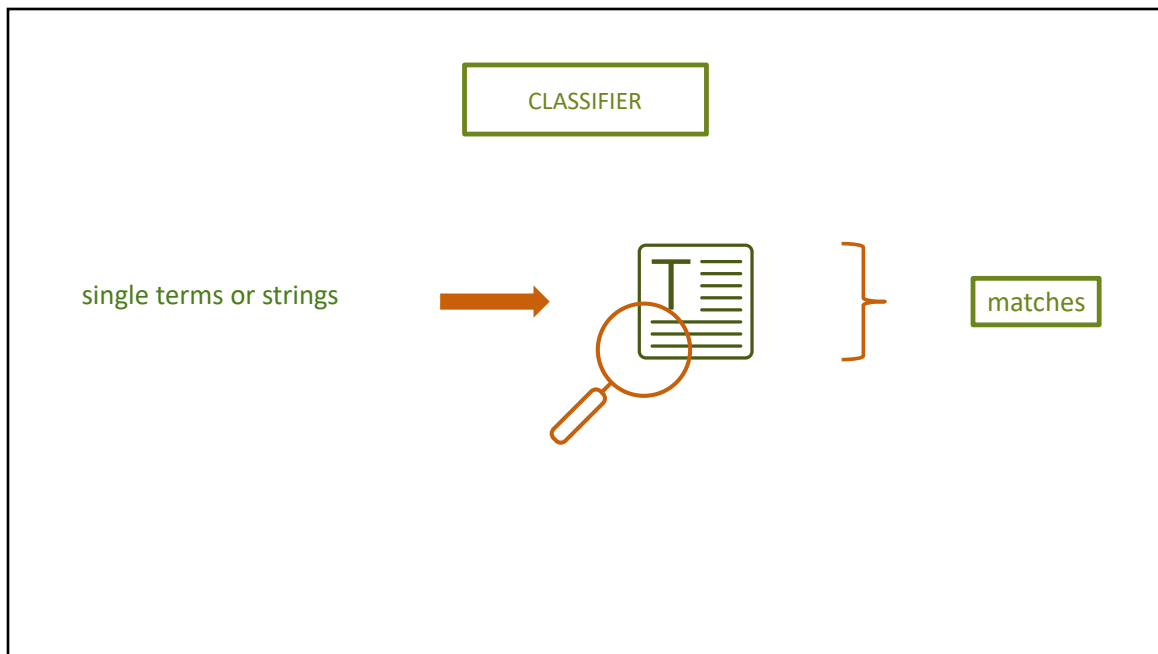
4.2 SAS Visual Text Analytics Concept Rules



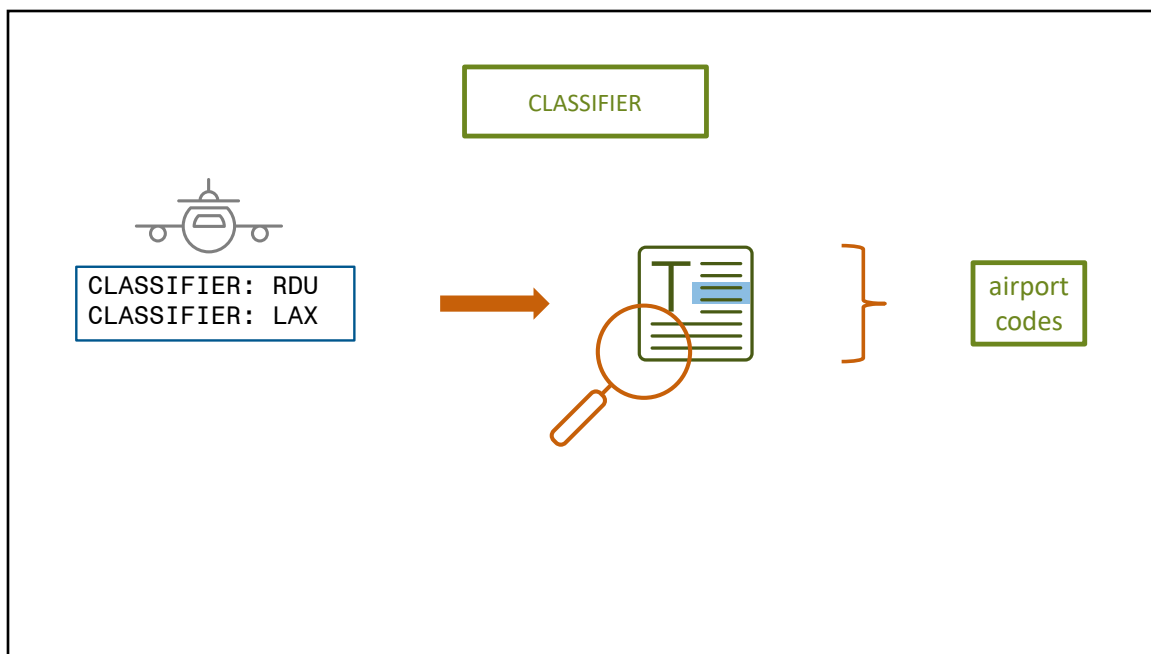
Exploring the CLASSIFIER Rule



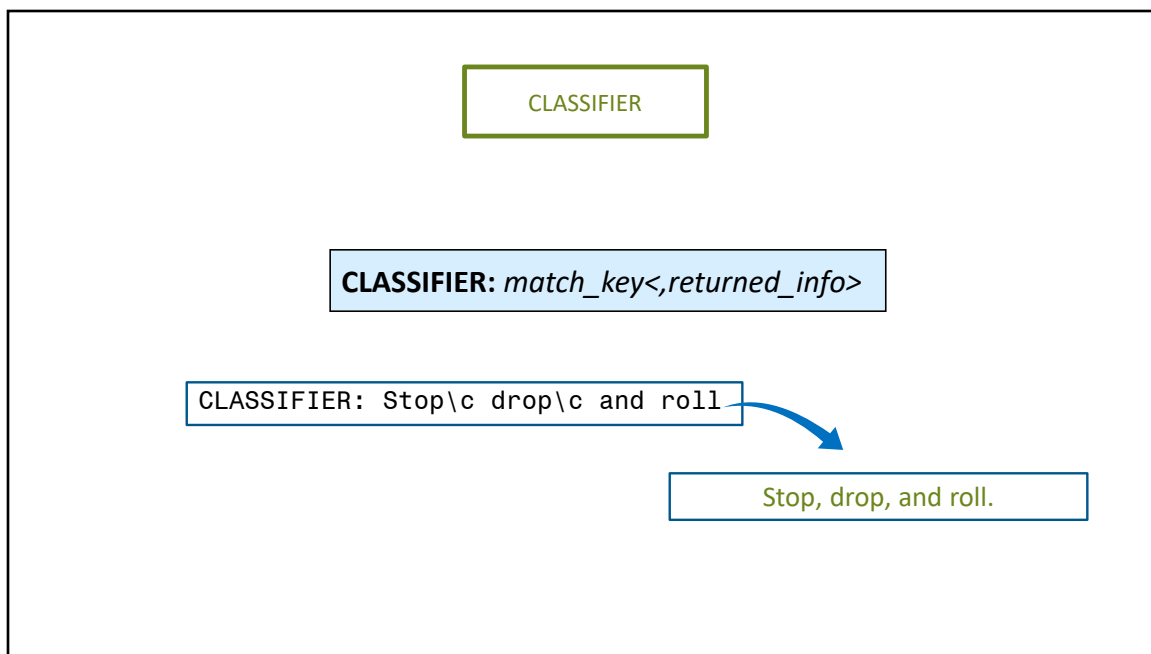
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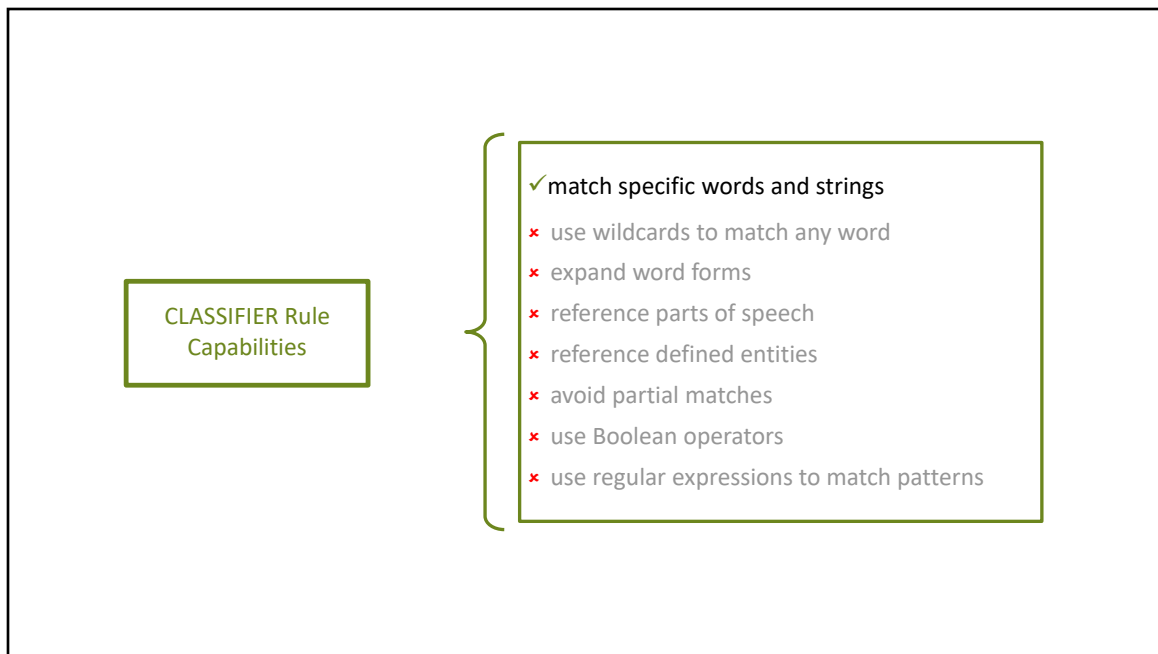
Let's start with the CLASSIFIER rule type. This rule identifies single terms or strings that you want to match in context.



For example, in a concept definition, you can create CLASSIFIER rules that contain specific airport codes. The portions of text that contain the airport codes are considered matches to the CLASSIFIER rules.



In the syntax, *match_key* is the literal string to match in the document. *Returned_info* is an optional string to return for the matched concept. You can omit the comma if you don't specify *returned_info*. You can use the character combination \c to match a comma within the *match_key*. For example, the sentence *Stop, drop, and roll.* is returned as a match for the rule **CLASSIFIER: Stop\c drop\c and roll.**



So out of the available concept rule type capabilities, the CLASSIFIER rule matches specific words and strings.

CLASSIFIER Rules for an Aviation Concept

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CLASSIFIER Rule Examples



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CLASSIFIER

Rule	Example Matches
CLASSIFIER: Orion Glimmer C20	<i>Orion Glimmer C20</i> <i>orion glimmer c20</i> <i>ORION GLIMMER C20</i>
CLASSIFIER: very good value	<i>Very good value</i> <i>VERY good value</i> <i>very GOOD value</i>
CLASSIFIER: disappointing	<i>Disappointing</i> <i>disappointing</i>

Here are some examples of CLASSIFIER rules and their matches. Note that the matching is not case sensitive, and each CLASSIFIER rule corresponds to one term, or dictionary entry.

CLASSIFIER

Rule	Example Result
CLASSIFIER: [coref=drug]:Elevex	If the term <i>Elevex</i> appears at least once in the document, the term <i>drug</i> also returns matches for <i>Elevex</i> .
CLASSIFIER: [coref=author,writer,Larsson]: Stieg Larsson	If the string <i>Stieg Larsson</i> appears at least once in the document, the terms <i>author</i> , <i>writer</i> , and <i>Larsson</i> also return matches for <i>Stieg Larsson</i> .

Coreference, represented with the operator `coref=`, is a relationship between two words or phrases in which both refer to the same person or thing. So, in the first CLASSIFIER LITI definition, if the term *Elevex* appears at least once in the documents, the term *drug* also returns a match. In the example, three matches are identified: *Elevex*, *drug*, and *Elevex*. If the term *Elevex* did not appear, then no matches would be identified, and no information would be extracted.

In the second definition, if the string *Stieg Larsson* appears at least once in the document, the terms *author*, *writer*, and *Larsson* also return matches. A total of five matches are identified, so this example returns five rows of matches in the scored data set related to this concept, and possible other rows for other concepts.

CLASSIFIER

Rule	Example Result
CLASSIFIER: [export=AR:accounts receivable]:Sokolov	If a document matches <i>Sokolov</i> , then the CLIENT concept is flagged. If the document also contains <i>accounts receivable</i> , then the AR concept is flagged.

Additional modifiers are accommodated (for example, the EXPORT feature can be used to invoke a separate concept when a classifier term is in a document). Suppose the CLIENT concept is defined by this statement: **CLASSIFIER:[export=AR:accounts receivable]:Sokolov**. CLASSIFIER colon square bracket export equal AR colon accounts receivable close bracket Sokolov

If a document matches *Sokolov*, then the CLIENT concept is flagged. If the document also contains *accounts receivable*, then the AR concept is flagged.

Demo: Using the CLASSIFIER Rule

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Question: CLASSIFIER Rule



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Question: CLASSIFIER Rule

Which CLASSIFIER rule has the correct syntax to extract the text *SAS Visual Text Analytics*?

- a. CLASSIFIER: SAS Visual Text Analytics_C
- b. CLASSIFIER: SAS Visual Text Analytics
- c. CLASSIFIER: "SAS Visual Text Analytics"
- d. CLASSIFIER: ('SAS Visual Text Analytics')

Answer: CLASSIFIER Rule

Which CLASSIFIER rule has the correct syntax to extract the text *SAS Visual Text Analytics*?

- a. CLASSIFIER: SAS Visual Text Analytics _C
- ☒ b. CLASSIFIER: SAS Visual Text Analytics
- c. CLASSIFIER: "SAS Visual Text Analytics"
- d. CLASSIFIER: ('SAS Visual Text Analytics')

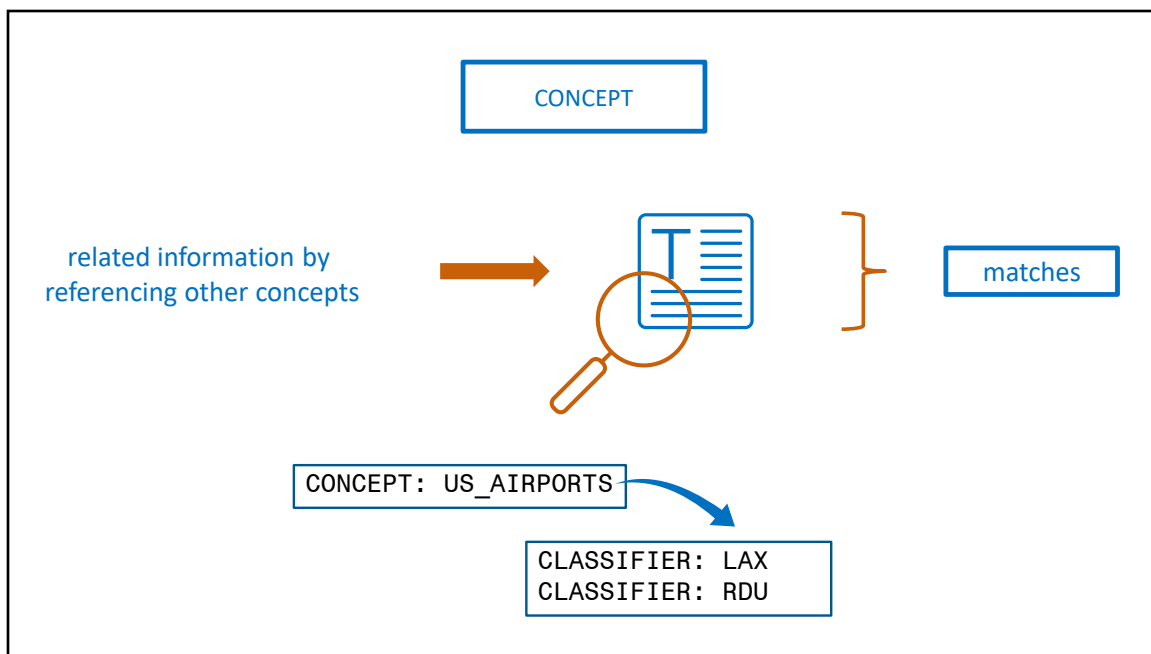
Answer: b

You can eliminate choice A because the CLASSIFIER rule does not support an _C marker. Choices C and D are eliminated because they would require matches on double quotes, parentheses, and single quotes. Choice B supplies the exact word sequence with no extraneous characters that would lead to a match for SAS Visual Text Analytics.

Exploring the CONCEPT Rule

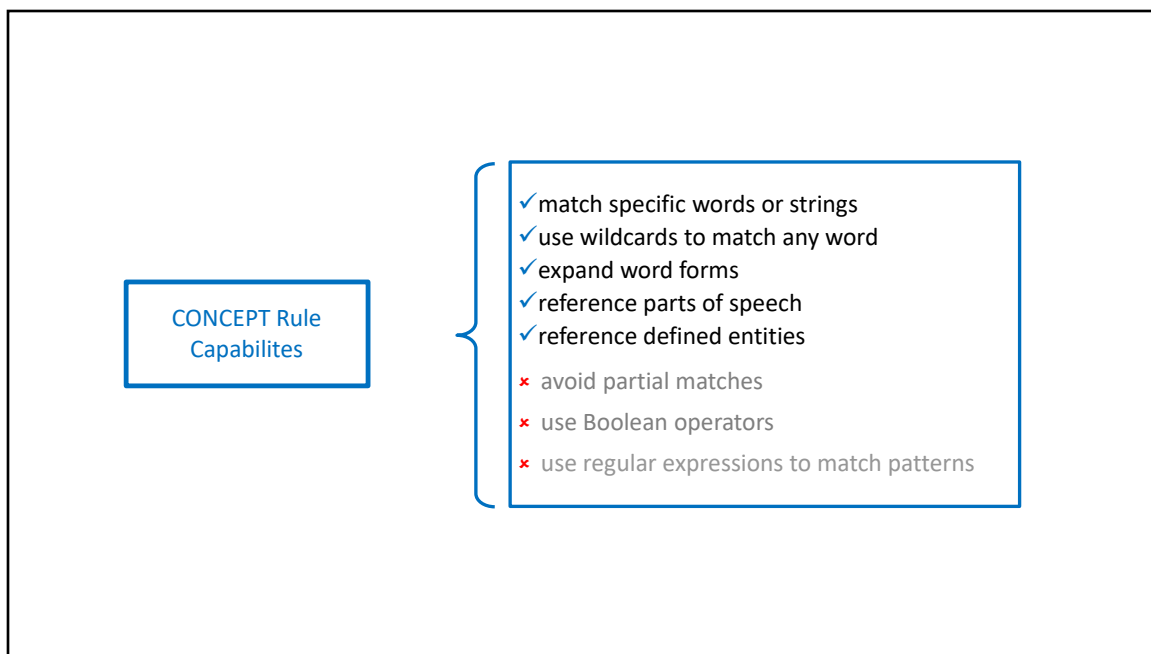


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Next we'll explore the CONCEPT rule. The CONCEPT rule identifies related information by referencing other concepts. For example, to capture documents that contain certain US airport names and locations, you can create a CONCEPT rule type in the definition. The CONCEPT rule type can reference any other concept. For example, it can reference a concept that contains a list of CLASSIFIER rules defining airport codes, thereby accessing a list of airport codes.

CONCEPT is a rule type. It's not to be confused with a "concept" in the general sense.



The capabilities of the CONCEPT rule include match specific words or strings, use wildcards to match any word, expand word forms, reference parts of speech, and reference defined entities.

CONCEPT

CONCEPT: *argument-1*<*argument-n*>

one space before:

- tokens
- concept names
- part-of-speech tags
- *_w* and *_cap* terms
- *_c*

In the syntax, *argument* can be a concept name, a rule modifier, or string. When you write CONCEPT definitions, enter at least one space before each of the following items: tokens (or literal strings), concept names, part-of-speech tags, *_w* and *_cap* terms, which are wildcards, and the context marker, *_c*, when preceded by a token, comma, or the name of a concept.

CONCEPT

Rule and Match Key	Example Matches
CONCEPT : SAS _w	SAS <i>employees</i> SAS <i>software</i>
CONCEPT : _w SAS	<i>Base</i> SAS <i>foundation</i> SAS
CONCEPT : SAS _cap	SAS <i>Institute</i> SAS <i>Analytics</i>

When you use the wildcard **_w**, the results match any word. So for example, if you specify **SAS _w**, the example matches include *SAS employees* and *SAS software* because they include any word that follows the word SAS. Or if you specify **_w SAS**, the example matches are any word followed by the word SAS, so *Base SAS* and *foundation SAS*.

When you use the wildcard **_cap**, the result matches any word that begins with an uppercase letter. So for example, if you specify **SAS _cap**, the example matches include *SAS Institute* and *SAS Analytics* because *Institute* and *Analytics* are both uppercase.

You specify both wildcards in lowercase.

CONCEPT

Symbol	Position	Result
@	Suffix	Stems the word that precedes the @ to include all forms of the word
@N	Suffix	Stems the word that precedes the @N to include noun forms of the word
@V	Suffix	Stems the word that precedes the @V to include verb forms of the word
_C	Suffix	Makes the rule case-sensitive

Here are some examples of additional arguments for CONCEPT rules. Three of these relate to expanding word forms. These three use morphological expansion symbols. Using the @ symbol stems the word that precedes the symbol to include **all forms** of the word. Using @N stems the word that precedes the @N to include **noun forms** of the word. Using @V stems the word that precedes the @V to include **verb forms** of the word, and using _C makes the rule **case-sensitive**.

CONCEPT

Rule	Example Matches
CONCEPT : go @	I will <i>go</i> . She is <i>going</i> . They <i>went</i> . He is <i>gone</i> .
CONCEPT : break @	It was <i>broken</i> . The camera <i>broke</i> . It kept <i>breaking</i> down. Give me a <i>break</i> !

So for example, if you specify **go @**, the example matches include all forms of the word *go*: *go*, *going*, *went*, and *gone*. And if you specify **break @**, the example matches includes all forms of the word *break*: *broken*, *broke*, *breaking*, and *break*.

CONCEPT

Part of Speech	Tag	Grammar Definition	Example Matches
Noun	N	SAS :N	SAS <i>employee</i> SAS <i>software</i>
Preposition	Prep	:Prep the drug	<i>of</i> the drug <i>from</i> the drug
Possessive Determinant	PossDet	:PossDet symptoms	<i>my</i> symptoms <i>her</i> symptoms <i>their</i> symptoms

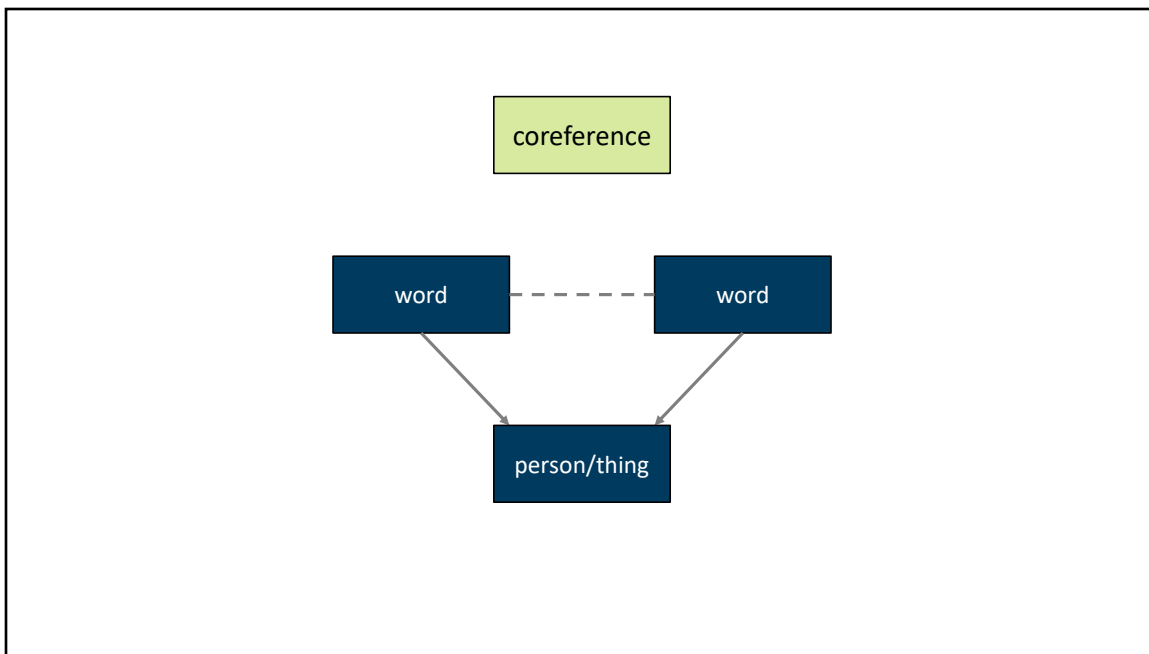
You precede part-of-speech tags with a colon when writing the definition. As an example, if you specify **:Prep the drug**, example matches include *of the drug* and *from the drug* because *of* and *from* are both prepositions.

Part-of-speech tags are case-sensitive, and you must type them in the case specified in the User's Guide.

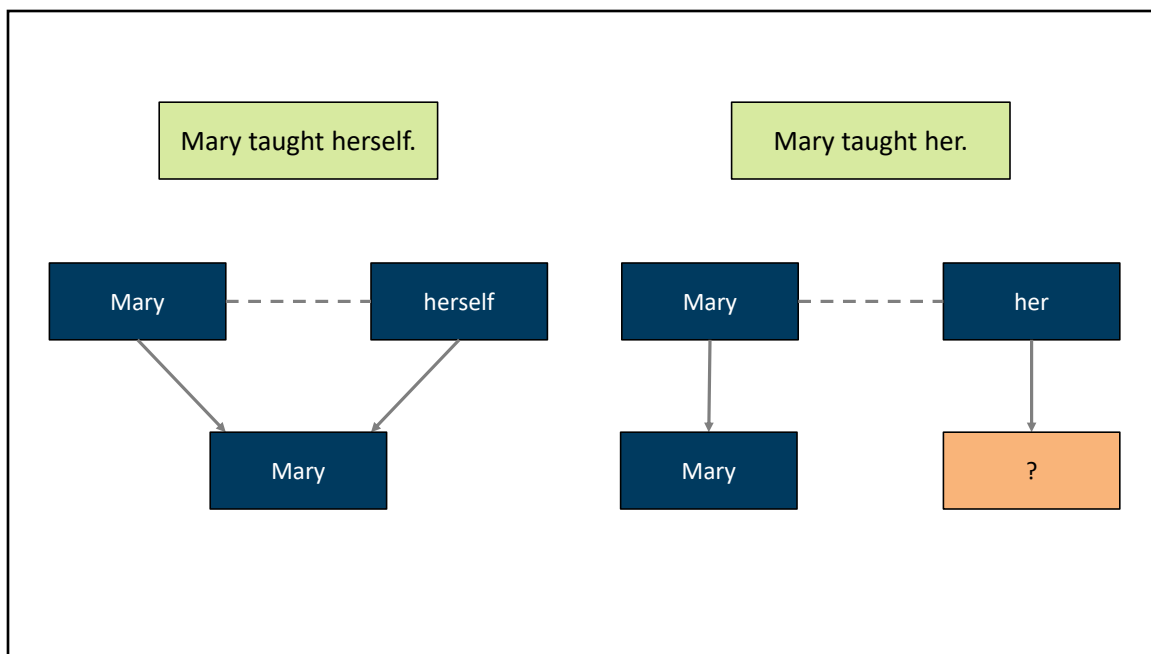
Coreference in the CONCEPT Rule



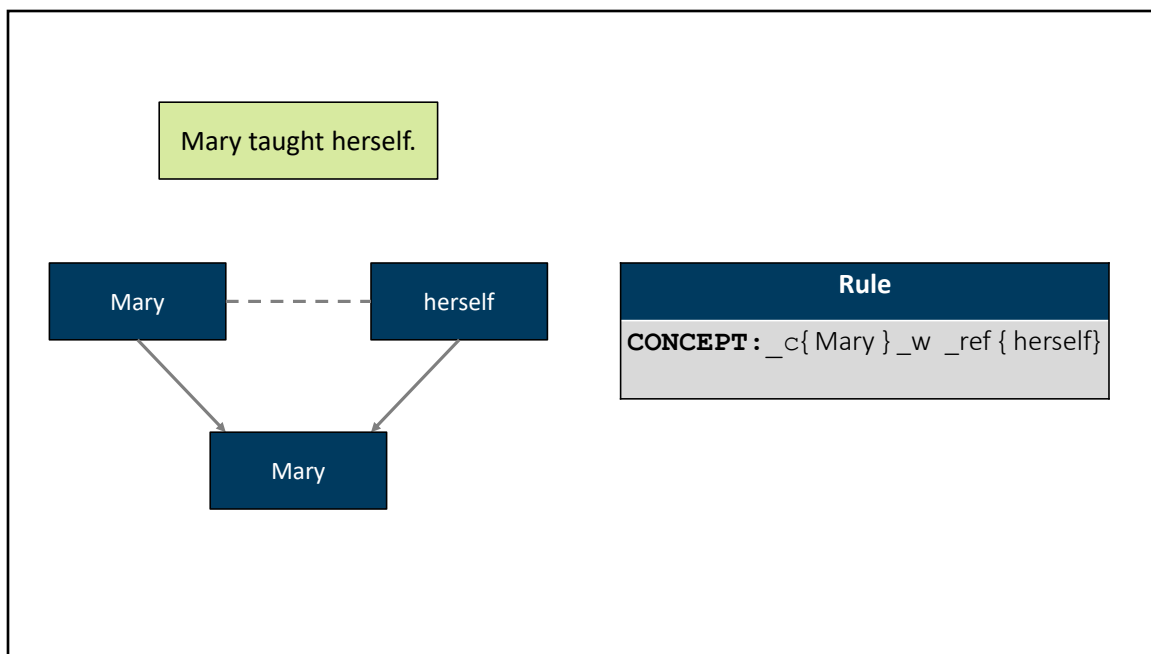
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Recall that coreference is a relationship between two words or phrases in which both refer to the same person or thing.



For example, in the phrase *Mary taught herself*, both the terms *Mary* and *herself* refer to the same person. However, in the phrase *Mary taught her*, the terms *Mary* and *her* refer to different people.



You can specify a canonical form and a referring term in a CONCEPT definition in a way that's similar to that in a CLASSIFIER definition. In the sentence *Mary taught herself*, *Mary* is the canonical form and *herself* is the referring term.

The canonical form is expressed in braces following the `_c` marker, and the referring term is expressed in braces following the `_ref` operator. You specify the `_c` and `_ref` operators in lowercase.

The definition can also include other items that are valid for the given definition type, such as literal text, part-of-speech tags, and references to other concepts.

Symbol	Result
>	Match all occurrences of the referring term
_F	Match all occurrences of the referring term that follow the canonical form
_P	Match all occurrences of the referring term that precede the canonical form

You can add symbols after the `_ref` operator to match additional occurrences of the referring term. The greater than symbol matches all occurrences of the referring term. `_F` matches all occurrences of the referring term that follow the canonical form. And `_P` matches all occurrences of the referring term that precede the canonical form.

Rule	Example Result
CONCEPT: _c{ doctor } :sep _ref { she }>	When I called my <i>doctor</i> , <i>she</i> said I should come in as soon as possible. At my next appointment, <i>she</i> changed my prescription.

Also, the text :sep is the part-of-speech tag for separator characters, including the period, comma, colon, and semicolon. In this example, the results match the term *doctor*, and both occurrences of *she*, the referring term. The second occurrence of *she* is matched because of the greater than (>) modifier.

Demo: Using the CONCEPT Rule

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Question: CONCEPT Rule



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Question: CONCEPT Rule

Which CONCEPT rule has the incorrect syntax to extract the text *SAS Visual Text Analytics*?

- a. CONCEPT: SAS Visual Text Analytics
- b. CONCEPT: SAS Visual _cap _cap
- c. CONCEPT: "SAS Visual Text Analytics"
- d. CONCEPT: _w Visual Text Analytics

Answer: CONCEPT Rule

Which CONCEPT rule has the incorrect syntax to extract the text *SAS Visual Text Analytics*?

- a. CONCEPT: SAS Visual Text Analytics
- b. CONCEPT: SAS Visual _cap _cap
- ☒ c. CONCEPT: "SAS Visual Text Analytics"
- d. CONCEPT: _w Visual Text Analytics

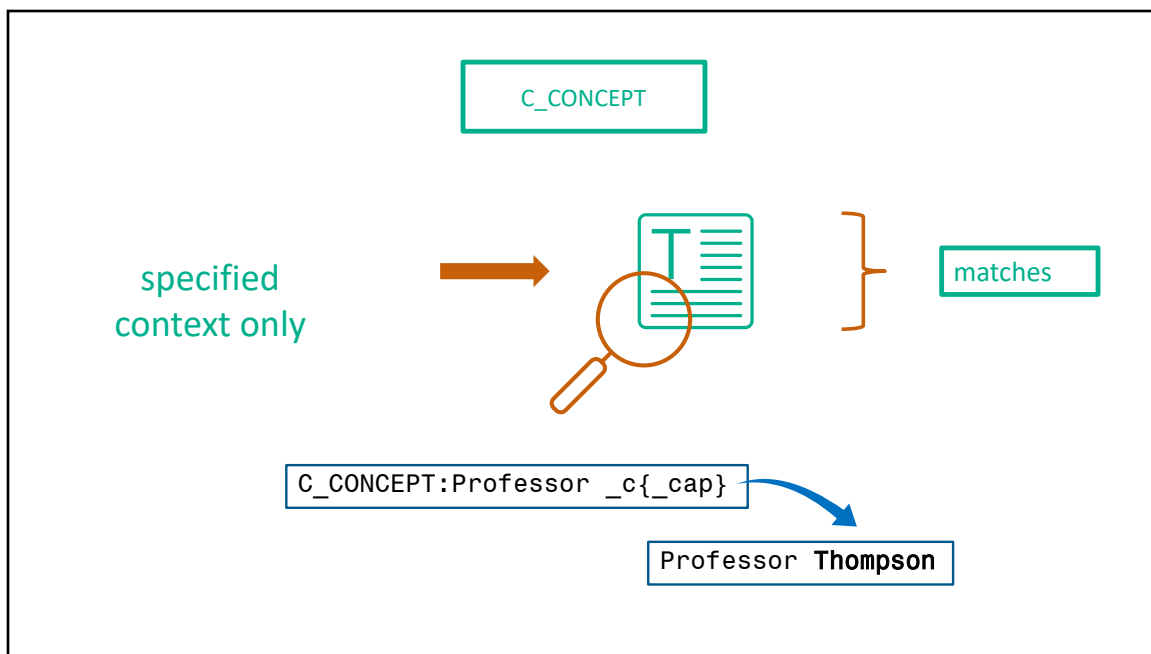
Answer: c

A CONCEPT rule without modifiers and special keywords behaves like a CLASSIFIER rule. Thus, the quotes in the rule would require quotes to appear in the matching string. All other options lead to a match. Option A is identical to a CLASSIFIER rule. Option B expands _cap to be a word that begins with an uppercase letter. Option D expands _w to match any word. If you were trying to match **SAS visual text analytics**, then options A and D would provide a match, but B would not because in "text analytics", neither word begins with an uppercase letter.

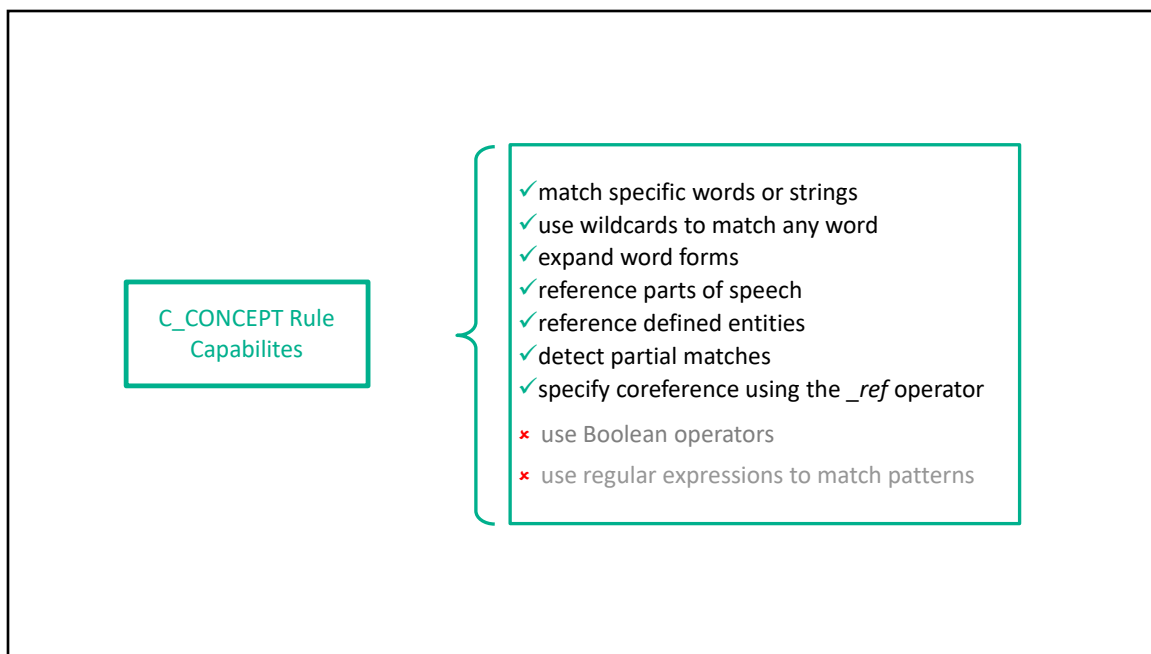
Exploring the C_CONCEPT Rule



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The `C_CONCEPT` rule returns matches that occur only in the specified context. For example, to extract matches that include names of university professors, you could create a `C_CONCEPT` rule that identifies matches on a concept (previously defined) that identifies last names only when the matched names are preceded by the word *Professor*.



The C_CONCEPT rule has all the capabilities of the CONCEPT rule plus the ability to match concepts with a context and detect partial matches. You specify coreference using the *_ref* operator.

C_CONCEPT

C_CONCEPT: <argument> _c{argument}<argument>

In the syntax, the context marker `_c` is required, and you can only specify one. It's used to locate related terms within a specific context. You specify the marker in lowercase, and the text or item to be matched is enclosed in braces. *Argument* can be a concept name, a rule modifier, or a string.

C_CONCEPT

Rule	Example Results
C_CONCEPT : New York _c { _cap _cap }	New York <i>Stock Exchange</i> New York <i>Public Library</i> New York <i>Daily News</i> New York theater district
C_CONCEPT : _c { _cap } Sea	<i>Bering</i> Sea <i>North</i> Sea <i>Baltic</i> Sea Bering Straight

Here are some examples of C_CONCEPT LITI rules and their matches. The first rule matches any two-word proper nouns that follow *New York* if the first letter of each word is uppercase. The second concept matches any word that begins with an uppercase letter followed by the word *Sea*.

C_CONCEPT

Rule Body	Location Specified	Example Results
Orion _c{Glimmer} C20	Definition of the product camera	Orion <i>Glimmer</i> C20
_c{I love} my camera	Tonal keyword, positive node	<i>I love</i> my camera
Auto Focus is _c{not fast enough}	Product definition, negative node for the focus feature	Auto Focus is <i>not fast enough</i>

The Rule Body shows the context in which information is identified and extracted. For the first rule, when *Glimmer* appears between *Orion* and *C20*, it is matched, and the term *glimmer* is extracted when scoring the document.

The second rule looks for a positive judgment associated with the specific keywords *my* and *camera*, and if found, returns the positive expression.

The third example looks for a negative assessment in context and extracts the negative expression when the context occurs.

C_CONCEPT

Rule	Example Results
C_CONCEPT : New York _c {Daily News }>	The <i>Daily News</i> of New York City is the fourth most widely circulated daily newspaper in the United States with a daily circulation of 605,677, as of November 1, 2011. The <i>Daily News</i> was founded by Joseph Medill Patterson in 1919. It was not connected to an earlier <i>New York Daily News</i> , which had been founded in the 1850s.

You can use the greater than symbol with the _c marker to specify a partial match. Every occurrence of the bracketed term is a match if the entire rule appears at least once in the input text. The example rule causes a match of *Daily News* in the context of being preceded by *New York*, and if *New York Daily News* appears at least one time in the document, then all occurrences of *Daily News* are matched, whether they appear before or after the initial match of *New York Daily News*.

Demo: Using the C_CONCEPT Rule

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Question: C_CONCEPT Rule



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Question: C_CONCEPT Rule

The concept rule C_CONCEPT: SAS _c{enterprise miner} is validated using the following test text: SAS Enterprise Miner is a solution to create accurate predictive and descriptive models on large volumes of data across different sources in the organization. Identify the text string that should be highlighted when you validate the rule.

- a. SAS Enterprise Miner
- b. Enterprise Miner
- c. SAS
- d. none of the above

Answer: C_CONCEPT Rule

The concept rule C_CONCEPT: SAS _c{enterprise miner} is validated using the following test text: SAS Enterprise Miner is a solution to create accurate predictive and descriptive models on large volumes of data across different sources in the organization. Identify the text string that should be highlighted when you validate the rule.

- a. SAS Enterprise Miner
- ☒ b. Enterprise Miner
- c. SAS
- d. none of the above

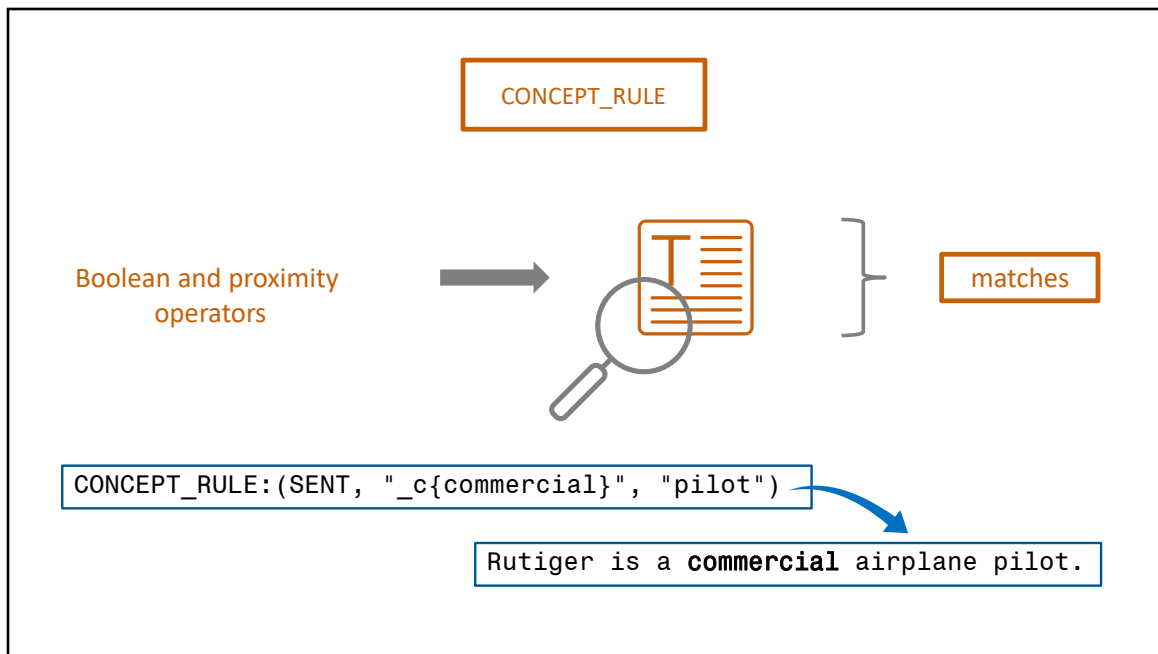
Answer: b

Answers A and C are incorrect because only the arguments in the _c marker will be matched, and the word SAS does not appear in the _c marker argument. B is correct because the _c marker contains the words Enterprise Miner. D is incorrect because B is correct.

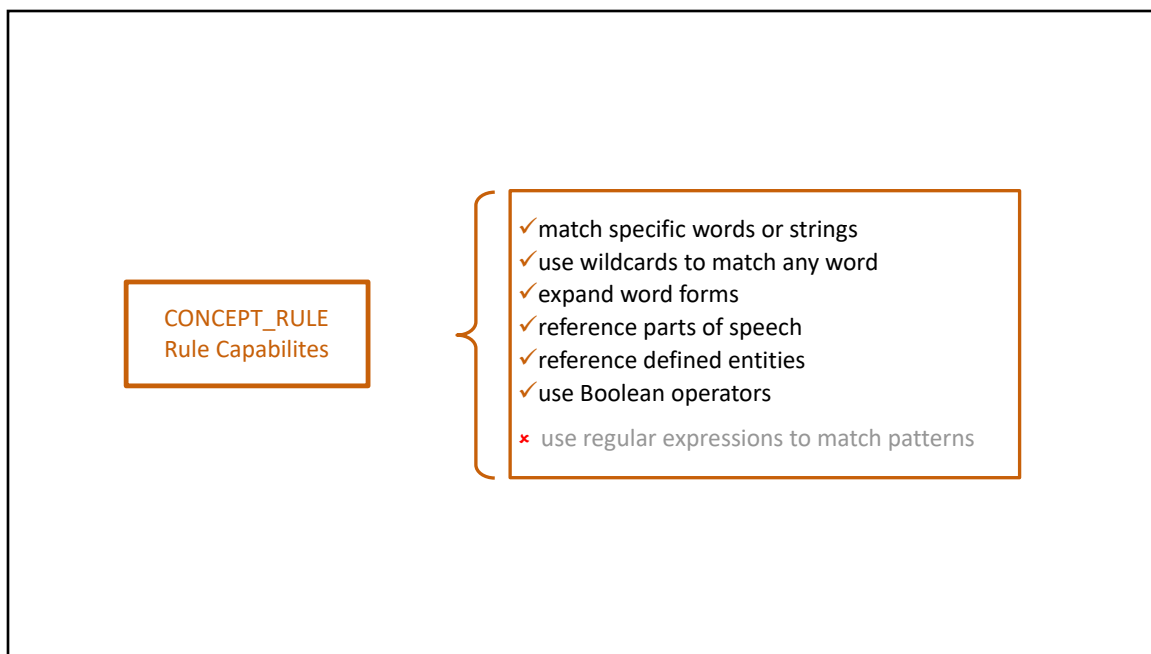
Exploring the CONCEPT_RULE Rule



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The CONCEPT_RULE rule uses Boolean and proximity operators to determine matches. In the example, if the word *commercial* appears in the same sentence as *pilot*, *commercial* is extracted from the document as information about a type of pilot.



The CONCEPT_RULE rule type has all the features of a C_CONCEPT rule, plus the ability to use Boolean operators to determine context matches.

CONCEPT_RULE

CONCEPT_RULE: (<Boolean-rule-1>...<Boolean-rule-n>)

(Boolean-operator, "_c{argument-1}",<"argument-2">...<"argument-n">)

The syntax is Boolean-rule-1, and then optional nested Boolean rules 2 thru n. At least one Boolean operator is required. A Boolean-operator is applied to arguments written as "argument-1", "argument-2", and so on. Arguments must be enclosed in double quotation marks, and there must be a space or blank before each argument. Exactly one argument must be an _c marker argument, unless a logical OR condition is specified, then more than one _c marker is allowed. The _c marker does NOT have to be the first argument, but it must be positioned so that when the rule resolves to TRUE, the _c marker term will exist in the document and will be extracted. Consequently, an _c marker should never appear in a NOT condition.

The other arguments that appear in quotation marks provide context for the match and must be present or absent as specified for a match to occur.

CONCEPT_RULE

Concept Name	Rule
<i>YearCpt</i>	REGEX : \d\d\d\d
<i>BirthYear</i>	CONCEPT_RULE : (SENT, "born", "_c{YearCpt}")

Example Results

Charles Dickens was *born* in a modest home in Portsmouth, England, in **1812**.
J.K. Rowling, *born* **1965**, is best known as the author of the Harry Potter series.

In this example, the **BirthYear** concept uses a **CONCEPT_RULE** definition that references the **YearCpt** concept. **YearCpt** matches exactly four contiguous numeric digits, so unusual years are possible, such as 2525. The Boolean operator **SENT** causes the **BirthYear** concept to extract the **YearCpt** year when it appears in the same sentence as the word *born*. Although the word *born* is highlighted in the Example Results to illustrate how the rule works, only the year is matched because it is the argument to the **_c** marker.

Demo: Using the `CONCEPT_RULE` Rule

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Question: CONCEPT_RULE



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Question: CONCEPT_RULE

Which CONCEPT_RULE body has the correct syntax?

- a. Orion _cap _c{is wonderful}
- b. (AND, Orion _cap, _c{is wonderful})
- c. (SENT, "OrionGlimmerprice", "_c{affordable}")
- d. (SENT, "OrionGlimmerprice", "affordable")

Answer: CONCEPT_RULE

Which CONCEPT_RULE body has the correct syntax?

- a. Orion _cap _c{is wonderful}
- b. (AND, Orion _cap, _c{is wonderful})
- ☒ c. (SENT, "OrionGlimmerprice", "_c{affordable}")
- d. (SENT, "OrionGlimmerprice", "affordable")

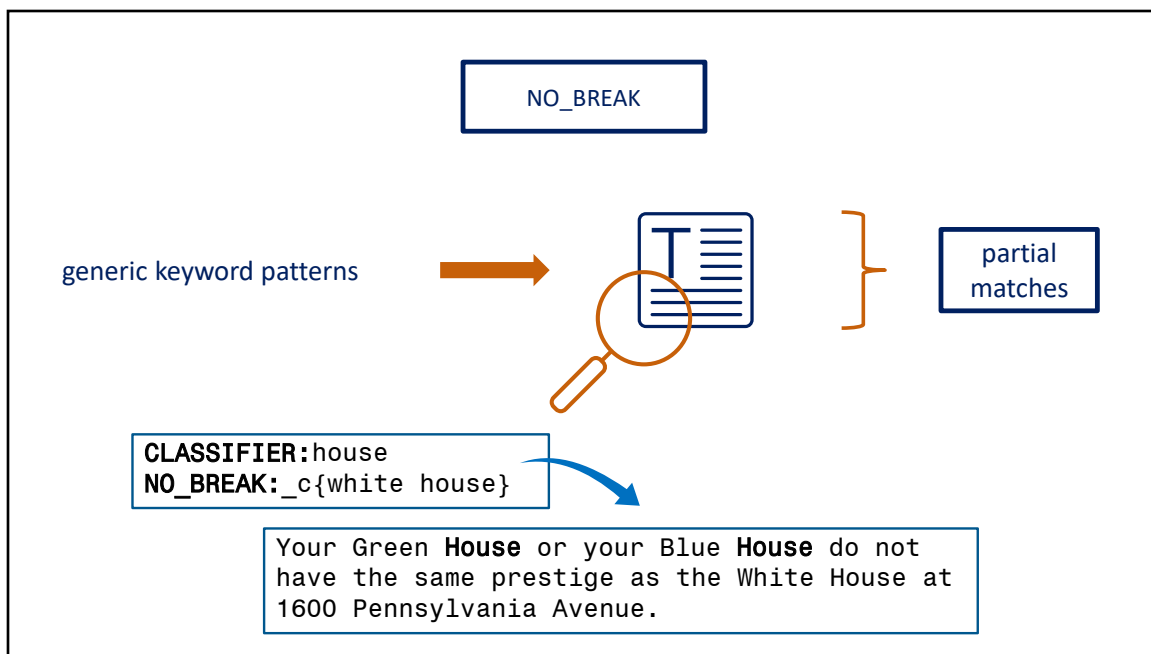
CONCEPT_RULE rules must include at least one Boolean operator and one _c marker. Arguments to Boolean operators are specified in double quotation marks.

Correct answer: c

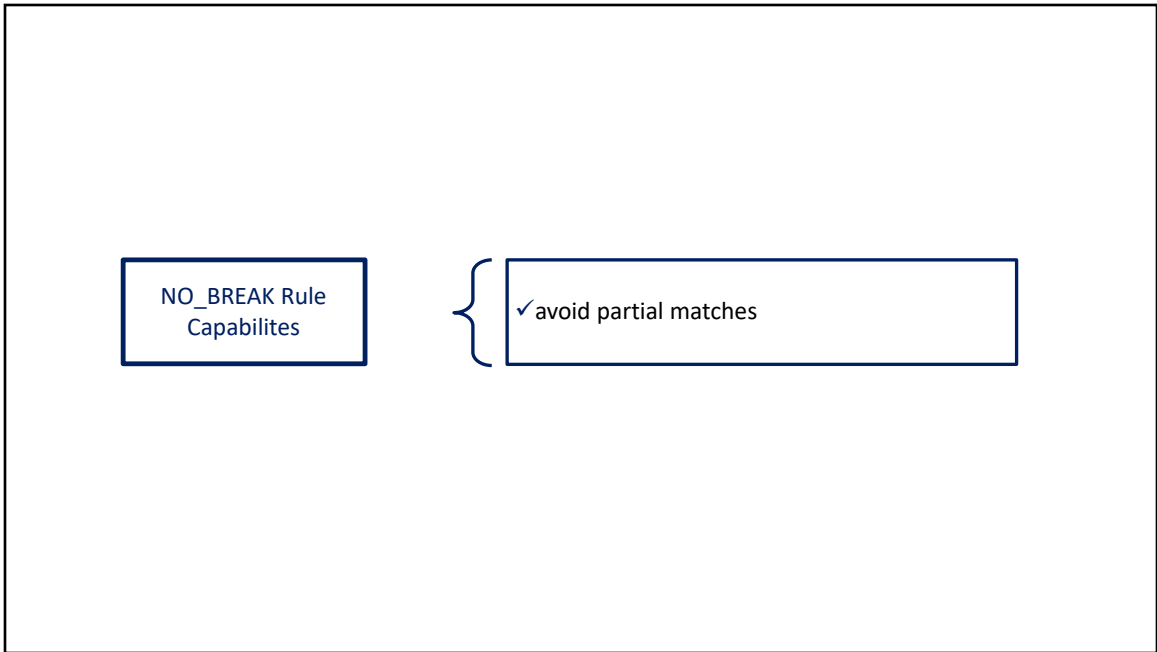
Exploring the NO_BREAK Rule



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The NO_BREAK rule prevents partial matches for generic keyword patterns. In this example, references to *house* are matched and extracted unless the reference is to the presidential residence, the *White House*.



The only capability of the NO_BREAK rule, therefore, is to avoid partial matches.

NO_BREAK

NO_BREAK: *_c{argument}*

- applies across entire taxonomy
- insert all in one concept
- globally implemented rules:
 - NO_BREAK
 - REMOVE_ITEM
- troubleshoot matching behavior

In the syntax, the *_c* marker is required and is lowercase. It can surround only one argument, and the argument can be a concept or a string.

There are some special considerations for the NO_BREAK rule. NO_BREAK applies across the entire taxonomy, regardless of where the rule appears or whether the rule is enabled or disabled.

You should not insert NO_BREAK rules just anywhere. It's helpful to insert them all in one concept. That is, create a concept that contains globally implemented rules only (NO_BREAK or REMOVE_ITEM). Having such rules all in one place aids in troubleshooting the matching behavior across your taxonomy.

NO_BREAK

Concept Name	Rule
MAX_LIKE	CLASSIFIER: maximum likelihood
OTHER_LIKE	CLASSIFIER: likelihood NO_BREAK: _c{MAX_LIKE}

Example Results

Ordinary gamma models are fit using *maximum likelihood* estimation.
Survival models obtain estimates using a partial *likelihood* formulation.

In this example, the **OTHER_LIKE** concept uses a NO_BREAK definition that references the **MAX_LIKE** concept. The term *likelihood* is matched unless it satisfies the rules for the **MAX_LIKE** concept. The example features two documents. The first document matches the **MAX_LIKE** concept. The second document has terms to match the **OTHER_LIKE** concept.

Demo: Using the NO_BREAK Rule

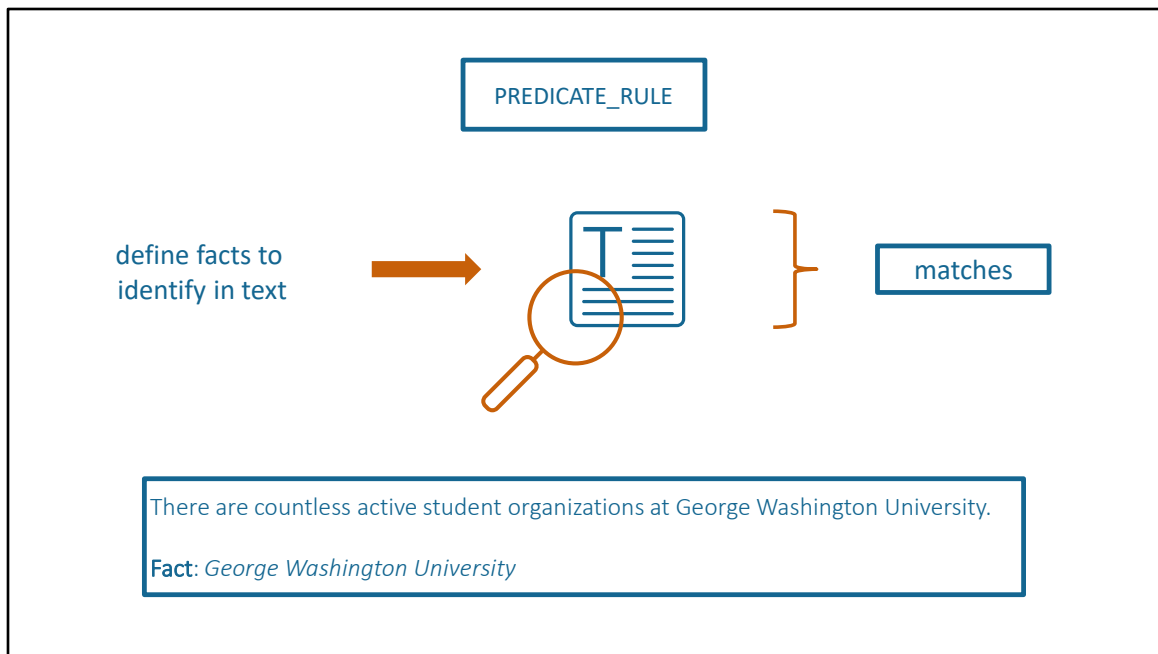
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Exploring the PREDICATE_RULE Rule

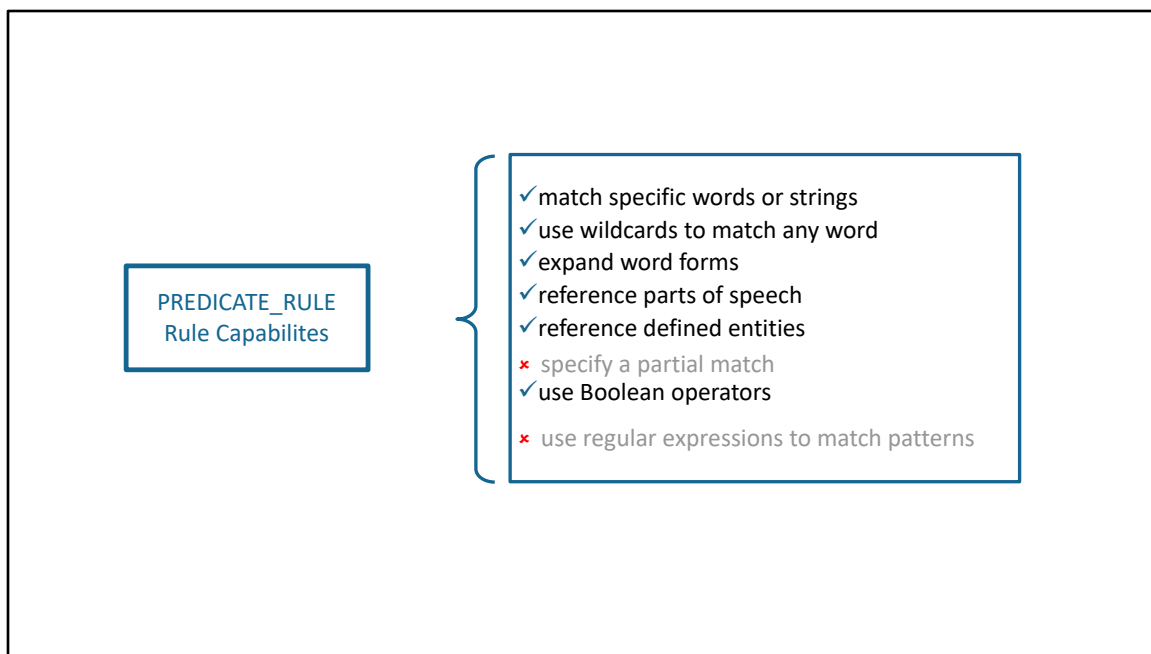


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You can use the **PREDICATE_RULE** rule to define facts that you want identified in text. Facts, which are also called *predicates*, are related pieces of information that are located and matched together.

For example, *George Washington University* can be considered as a fact in the following sentence: *There are countless active student organizations at George Washington University.*



The PREDICATE_RULE rule can match specific words or strings, use wildcards to match any word, expand word forms, reference parts of speech and defined entities, and use Boolean operators.

PREDICATE_RULE

```
PREDICATE_RULE: (arg1<,arg2,arg3,...>):  
                (Boolean_Op, "_arg1{match_key}",  
                <,"_arg2{match_key}", "_arg3{match_key}"...>)
```

In the syntax, at least one Boolean operator is required, and you must specify at least one argument, such as **arg1**. Arguments must be named using lowercase letters and digits and begin with a lowercase letter. The `_c` marker is not allowed in the `PREDICATE_RULE` type, but you can specify multiple markers. This enables you to define arguments that locate relationships between two or more concepts. When multiple markers are included in the rule, the match is highlighted from the first marker to the last marker.

Match_key can include Boolean operators, literals, part-of-speech tags, wildcards, and references to other concepts.

PREDICATE_RULE

Concept Name	Rule
<i>CountryNames</i>	CLASSIFIER: France CLASSIFIER: French CLASSIFIER: India CLASSIFIER: Indian CLASSIFIER: Canada ...
<i>OfficialTitles</i>	CLASSIFIER: Prime Minister CLASSIFIER: President
<i>WorldLeaders</i>	PREDICATE_RULE: (name,title,co): (SENT, "_name{_cap _cap}", "_title{OfficialTitles}", "_co{CountryNames}")

In this example, the **WorldLeaders** concept uses a **PREDICATE_RULE** definition that references the concept **OfficialTitles** and the concept **CountryNames**. An additional argument **name** references two words beginning with uppercase letters. Every instance of these three arguments is obtained, and all words in between the first and last occurrence of any argument, inclusive, is matched if all three argument types appear within the same sentence.

PREDICATE_RULE

Rule	Example Matches
PREDICATE_RULE: (aa,bb): (SENT, "_aa{good}", "_bb{quality}")	The <i>quality is good</i> . It's very <i>good quality</i> .
PREDICATE_RULE: (one,two): (AND, "_one{value}", "_two{price}")	It's a good <i>value for the price</i> . Couldn't believe the <i>price</i> . <i>What a value!</i>
PREDICATE_RULE: (xx,yy): (DIST_3, "_xx{not}", "_yy{disappointed}")	You will <i>not be disappointed</i> . I was <i>not at all disappointed</i> .

In these examples, the first PREDICATE_RULE looks for the words *good* and *quality* in the same sentence and matches these words and all words in between when the Boolean SENT operator is satisfied.

The second PREDICATE_RULE finds the words *value* and *price* when both occur in the document and matches the two specified words and all words in between.

The third PREDICATE_RULE uses a Boolean DIST_3 operator looking for *not* and *disappointed* within three words of each other.

PREDICATE_RULE in SAS Visual Text Analytics

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Demo: Using the PREDICATE_RULE Rule

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Question: PREDICATE_RULE



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Question: PREDICATE_RULE

Which PREDICATE_RULE body has the correct syntax?

- a. (aa,bb): (DIST_5, '_aa{cameraprice}', '_bb{expensive}')
- b. (yy,zz): (DIST_5, _yy{"cameraprice"}, _zz{"expensive"})
- c. (xx,yy): (DIST_5, "_xx{cameraprice}", "_yy{expensive}")
- d. (aa,bb): (DIST_5,_aa{cameraprice},_bb{expensive})

Answer: PREDICATE_RULE

Which PREDICATE_RULE body has the correct syntax?

- a. (aa,bb): (DIST_5, '_aa{cameraprice}', '_bb{expensive}')
- b. (yy,zz): (DIST_5, _yy{"cameraprice"}, _zz{"expensive"})
- ☒ c. (xx,yy): (DIST_5, "_xx{cameraprice}", "_yy{expensive}")
- d. (aa,bb): (DIST_5,_aa{cameraprice},_bb{expensive})

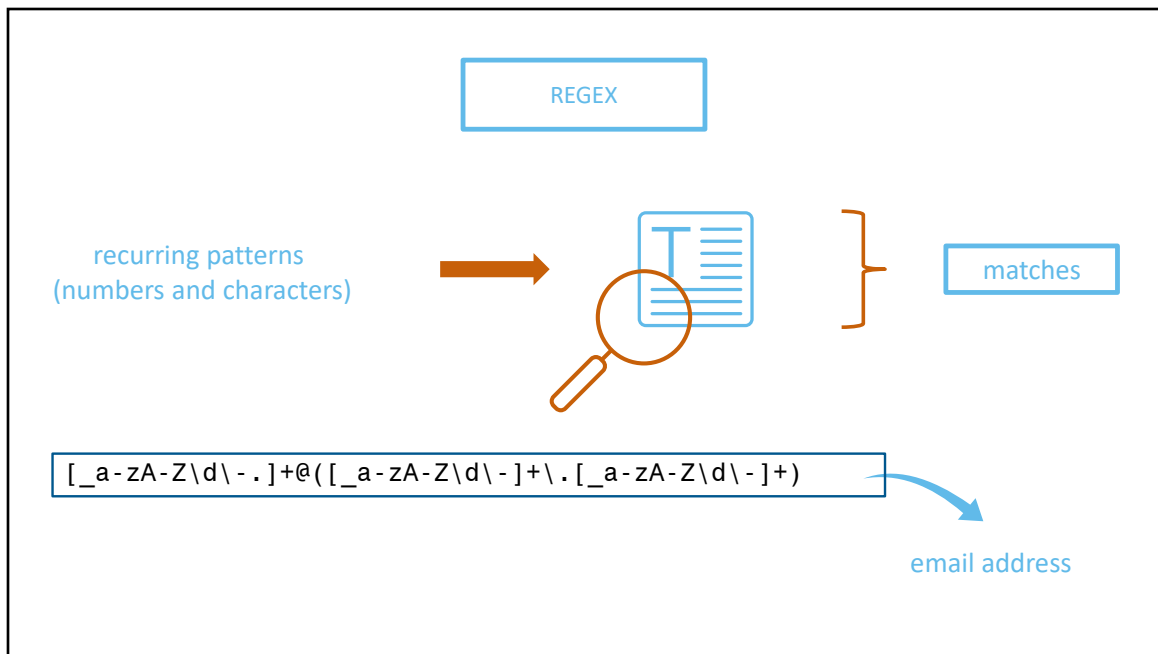
Boolean operators are followed by a comma and a space.
Arguments are specified in double quotation marks and
are separated by a comma and a space.

Correct answer: c

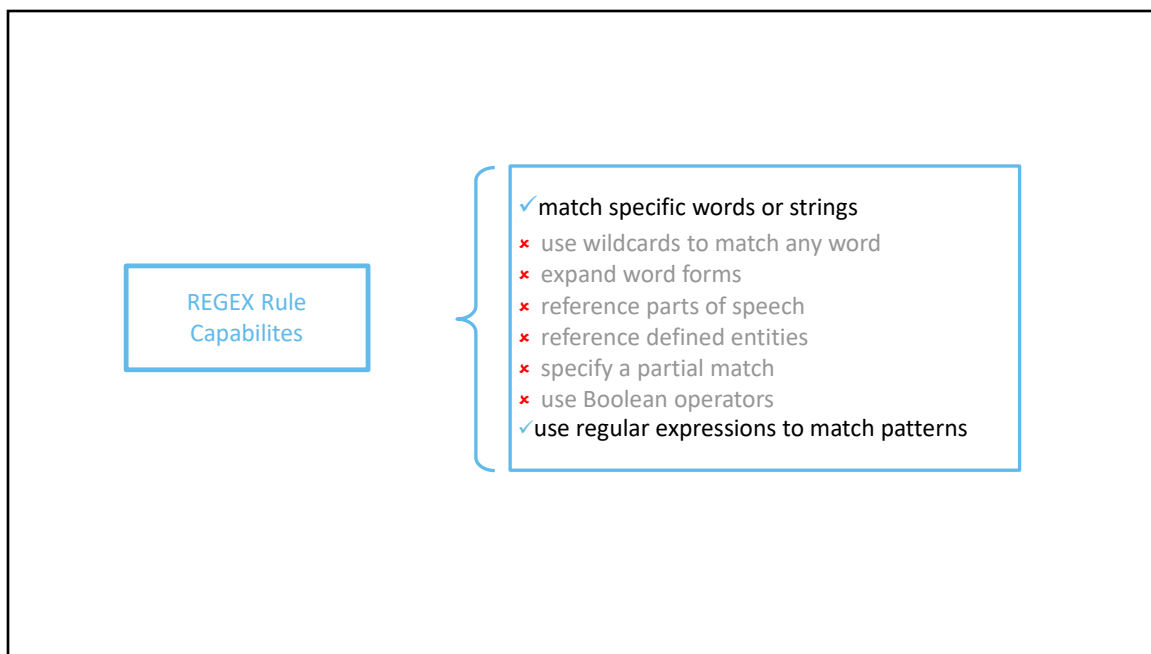
Exploring the REGEX Rule



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The REGEX rule identifies recurring patterns of information in text that can be expressed in numbers and characters, such as telephone numbers, license plate numbers, part numbers for manufacturing components, hyphenated words, and so on. Use regular expressions when you want to locate matches on text that has recognizable patterns, such as email and street addresses.



The REGEX rule capabilities include match specific words or strings, and use regular expressions to match patterns.

REGEX

- elements grouped using []
- meta characters, [], (,), ?, *, +, ., \, |, and \$, must be preceded with a backslash
 - [\?] returns ?
- cannot refer to concepts
- match a number with decimal notation (e.g., 392.55)
 - **REGEX:** [0-9]+[0-9\.\,][0-9]+

The REGEX rule elements are grouped using square brackets. Meta characters such as commas, question marks, and dollar signs must be preceded by a backslash to indicate that they have a literal meaning. For example, [\?] matches a question mark in text. You cannot refer to concepts in a REGEX expression. You can match one or more consecutive digits without decimals, such as matching a number with decimal notation (for example, 392.55).

This rule matches any digit 0 to 9, followed by either a digit, a comma, or a period, and then followed by any number of digits.

REGEX Matching, Guidelines, and Examples

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Regular Expressions: A Brief History

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Demo: Using the REGEX Rule

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Question: REGEX Rule



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Question: REGEX Rule

Which REGEX rule has the correct syntax to extract a part number that starts with XYZ- and any five-digit numbers (for example, XYZ-12345)?

- a. REGEX:[XYZ][\-][1234567890]
- b. REGEX:XYZ[\-][1234567890]+
- c. REGEX:[XYZ][\-][\d\d\d\d\d]
- d. REGEX:XYZ\-\d\d\d\d\d

Answer: REGEX Rule

Which REGEX rule has the correct syntax to extract a part number that starts with XYZ- and any five-digit numbers (for example, XYZ-12345)?

- a. REGEX:[XYZ][\-][1234567890]
- b. REGEX:XYZ[\-][1234567890]+
- c. REGEX:[XYZ][\-][\d\d\d\d\d]
- ☒ d. REGEX:XYZ\-\d\d\d\d\d

Answer: d

The square brackets around a character string such as XYZ will only match one of the characters in the string. Option A will match X-1, but not XYZ-12345. A similar argument applies to option C. Option B will match XYZ-1234567890123456789, in other words, there is no limit on the number of digits. Note that the square brackets around a single character has no effect. Thus, [\ -] will work, but is not preferred.

Resource

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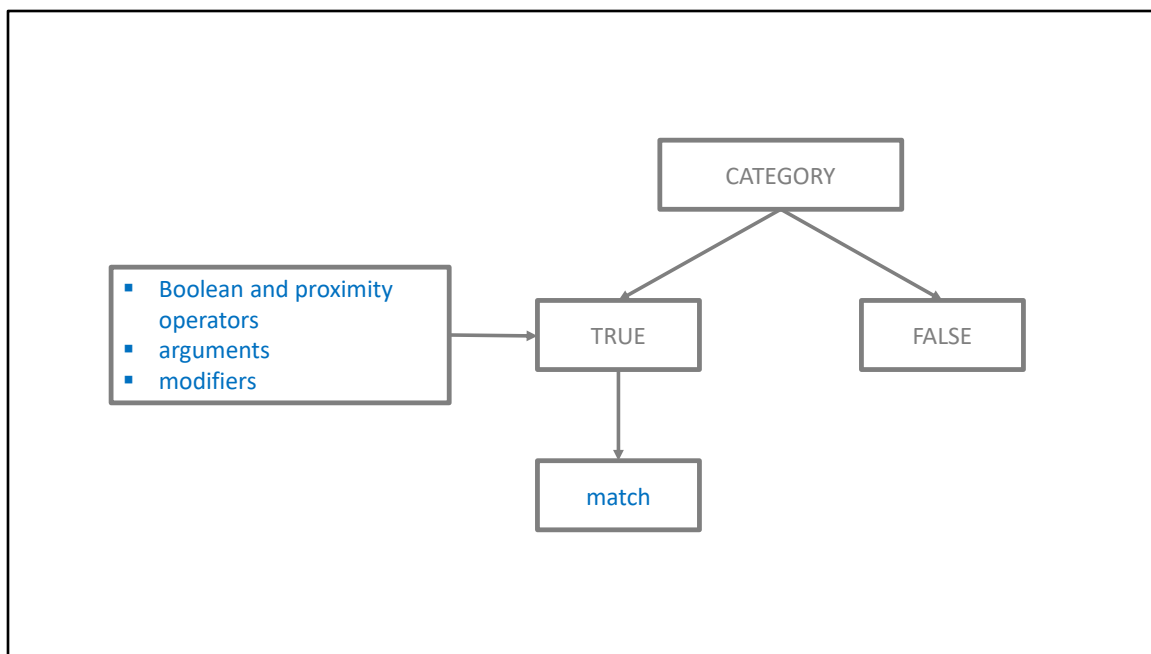
4.3 SAS Visual Text Analytics Category Rules



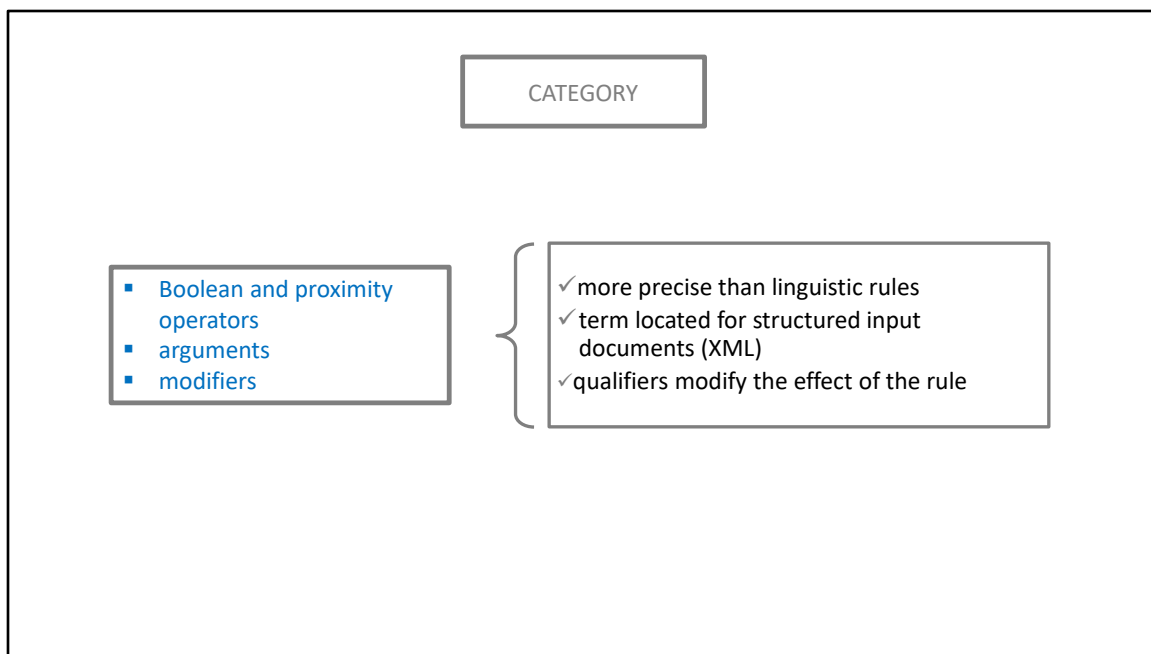
Introduction to CATEGORY Rules



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CATEGORY rules resolve to True or False, and True results in a match. CATEGORY rules use Boolean and proximity operators, arguments, and modifiers to define the conditions that are necessary for category matches. These rules are simpler to write than LITI rules and are recommended when there is no need to extract specific information from the data.



You can use Boolean operators to write more precise and unambiguous rules. Boolean rules can also be used when a term is located for structured input documents such as XML in order to return a match. As with linguistic rules, you can add qualifiers to modify the effect of the rule.

CATEGORY

(**OPERATOR**, *argument1*, *argument2*,...)

(AND, "AMAZON", (OR, "online retailer", "e-commerce"))

For a CATEGORY rule, use the syntax **OPERATOR**, **argument1**, **argument2**, where arguments can be terms, strings, or nested rules. Boolean and proximity operators and their arguments are enclosed in parentheses and separated with commas. The arguments are included in quotation marks. Rules can be nested.

In the example, the rule is true when the term *Amazon* appears with either the terms *online retailer* or *e-commerce*.

CATEGORY

Symbol	Position	Result
@	Suffix	Stems the word that precedes the @ to include <i>all forms</i> of the word.
@N	Suffix	Stems the word that precedes the @N to include <i>noun forms</i> of the word.
_C	Suffix	Makes the rule case-sensitive.
@V	Suffix	Stems the word that precedes the @V to include <i>verb forms</i> of the word.
*		Wildcard matching – Matches any characters that occur at the beginning or end of the word. For example, the argument "travel*" returns the matches <i>travels</i>, <i>traveled</i>, <i>traveler</i>, and <i>traveling</i>. The argument "*room" matches <i>bedroom</i>, <i>cloakroom</i>, <i>ballroom</i>, <i>room</i>, and so on.

You can add special symbols to terms that are specified in Boolean rules to expand word forms, perform case-sensitive matching, and specify a wildcard. A wildcard matches any characters that occur at the beginning or end of the word. For example, the argument "travel*" returns the matches *travels*, *traveled*, *traveler*, *traveling*, and so on. The argument "*room" matches *bedroom*, *cloakroom*, *ballroom*, *room*, and so on.

CATEGORY Boolean Rule Syntax and Examples

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Demo: Using the CATEGORY Rule

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Question: CATEGORY Rule



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Question: CATEGORY Rule

Which sub-category Boolean rule has the incorrect syntax?

- a. (AND, (NOT, (OR, "Enterprise Guide", "DDE", "Drug")), (OR, "Business Intelligence", "Enterprise Miner"))
- b. (SENT, (NOT, (OR, "Text Miner", "REG", "GLM")), (OR, "Business Intelligence", "Enterprise Miner"))
- c. (SENT, ["WAR_CONCEPT"], "europe")
- d. (ORD,_tmac:"@NAME/FIRST",_tmac:"@NAME/LAST")

Answer: CATEGORY Rule

Which sub-category Boolean rule has the incorrect syntax?

- a. (AND, (NOT, (OR, "Enterprise Guide", "DDE", "Drug")), (OR, "Business Intelligence", "Enterprise Miner"))
- b. (SENT, (NOT, (OR, "Text Miner", "REG", "GLM")), (OR, "Business Intelligence", "Enterprise Miner"))
- ☒ c. (SENT, ["WAR_CONCEPT"], "europe")
- d. (ORD,_tmac:"@NAME/FIRST",_tmac:"@NAME/LAST")

Answer: c

Square brackets [] should be used to reference concepts instead of curly braces {}.

Practice: Creating Appropriate Concept or Category Rules

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Lesson 5: Scoring Visual Text Analytics Models



5.0 Lesson Overview



Lesson Overview

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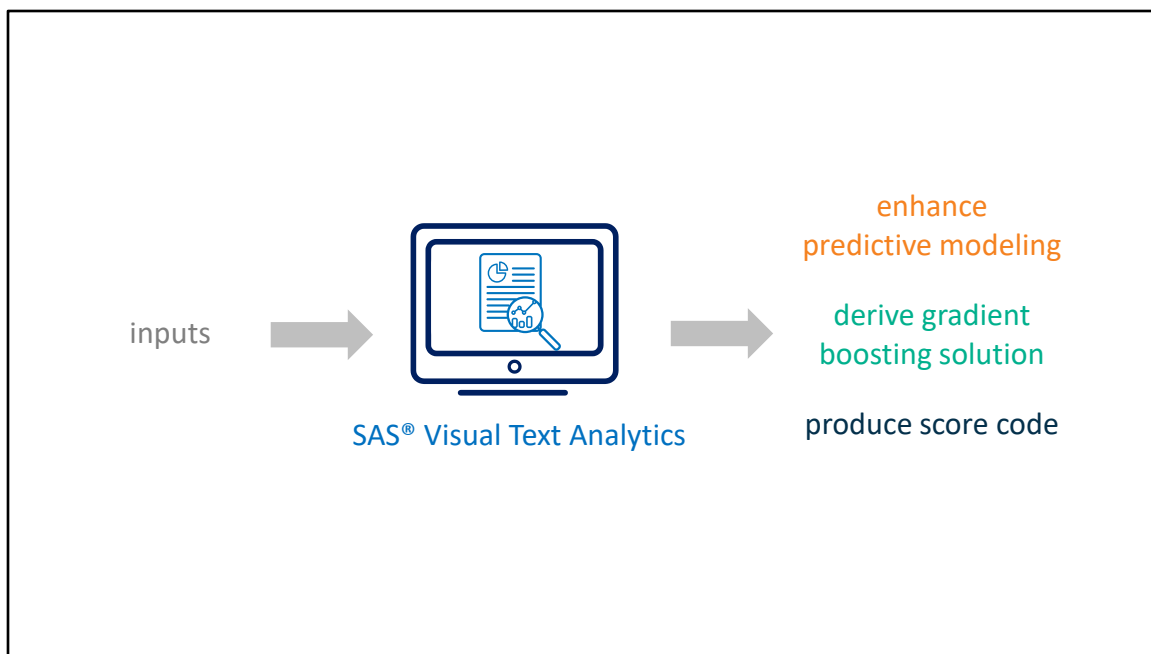
5.1 Introduction to Text Analytics Scoring



Integrating SAS Viya Projects and Software



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Although SAS Visual Text Analytics provides a powerful software environment for text processing and analysis, there are situations where text analytics is one of several components of a larger project. For example, text analytics inputs can be used to enhance predictive modeling, where additional transactional and operational inputs can be combined to derive a gradient boosting solution supported by SAS Visual Data Mining and Machine Learning.

SAS Visual Text Analytics nodes produce score code that can generate data for analysis using other SAS products.



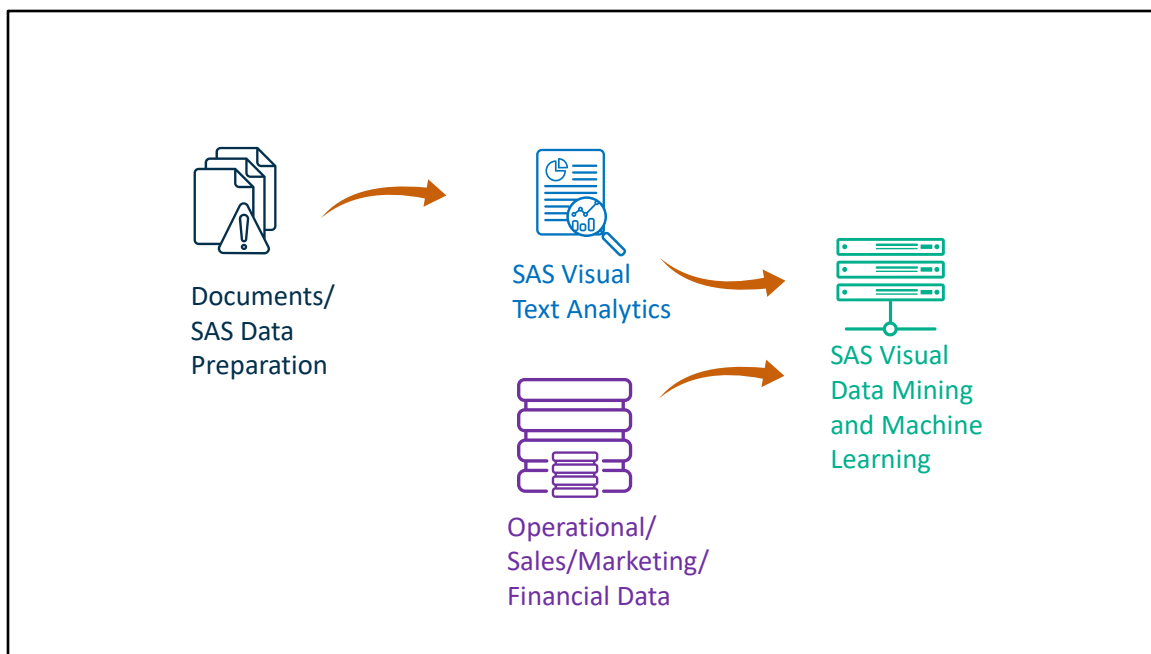
SAS® Visual Text Analytics

- Expand assessment of models by using SAS Visual Analytics to explore scored data.
- Enhance predictive modeling in SAS Data Mining and Machine Learning by using inputs from SAS Visual Text Analytics.
- Add alternative algorithms to categorize documents by exporting SAS Visual Text Analytics results to SAS Visual Data Mining and Machine Learning.

You can expand assessment of SAS Visual Text Analytics models by using SAS Visual Analytics to explore scored data.

You can enhance predictive modeling in SAS Data Mining and Machine Learning by using inputs from SAS Visual Text Analytics.

And you can add alternative algorithms, such as gradient boosting and neural networks, to categorize documents by exporting SAS Visual Text Analytics results to SAS Visual Data Mining and Machine Learning.



Document processing and conversion is supported by SAS Data Preparation, which is accessed through the Explore and Visualize data task in the show list of applications menu.

SAS Visual Text Analytics can derive concepts, topics, and categories for documents. Scored documents produce additional variables derived from the concepts, topics, and categories score code. The scored data can be combined with additional data, such as operational, sales, marketing, and financial data. The combined data can be processed by SAS Visual Data Mining and Machine Learning to produce solutions for supervised and unsupervised learning.

5.2 Scoring Concepts



Overview of the ASRS Project



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ASRS Data



- Voluntary safety reporting system
 - NASA
 - FAA

In this section, we're going to work with ASRS (Aviation Safety Reporting System) data. The ASRS is a voluntary safety reporting system. It's supported by the U.S. National Aeronautics and Space Administration, with assistance from other U.S. government agencies, like the U.S. Federal Aviation Administration.

ASRS Data



- Identify airports mentioned
- Determine most frequently occurring airports

The ASRS project seeks to identify all airports mentioned in the ASRS safety reports. One deliverable from the project is a list of the most frequently occurring airport references.

Demo: Scoring Airport Concepts, Part 1

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Demo: Scoring Airport Concepts, Part 2

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Demo: Scoring Airport Concepts, Part 3

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Practice: Extracting Numeric Data from Text (Self-Study)

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5.3 Scoring Topics



Overview of ASRS Topics



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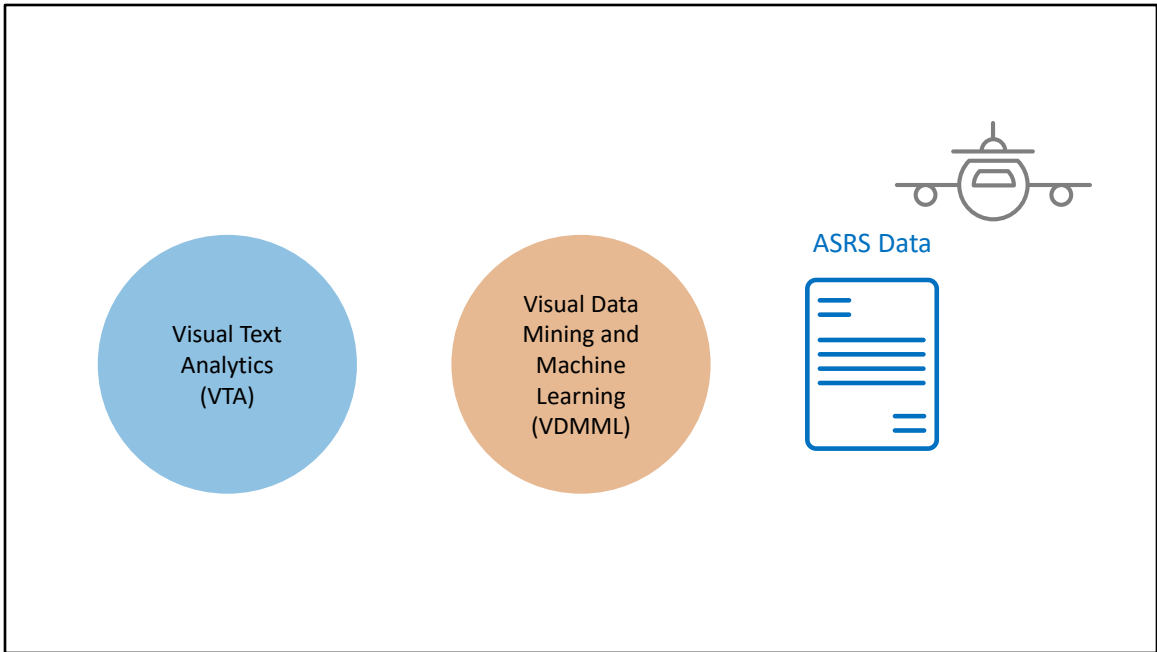
Topics (14)						
☐	Topic	Created by	Documents ↓ ?			
☐	tower, clearance, takeoff, hear, runway	System	3430			
☐	knot, gear, brake, flap, left	System	3126			
☐	flap, takeoff, knot, flight, dispatch	System	3090			
☐	approach, visual, runway, visual approach, +locate	System	3005			
☐	+foot, descend, altitude, +autopilot, cross	System	2991			
☐	engine, emergency, declare, smoke, cabin	System	2856			
☐	install, maintain, inspect, minimumequipmentlist, mechanic	System	2607			
☐	airspace, visualflightrules, classb, area, plan	System	2605			
☐	degree, head, turn, radial, degree radial	System	2515			
☐	trafficalertandcollisionavoidancesystem, resolutionadvisory, traffic, +foot, climb	System	2194			
☐	taxiway, taxiway, taxi, ramp, groundcontrol	System	2078			
☐	passenger, flightattendant, seat, captain, hi	System	2057			
☐	hold, line, short line, short, runway	System	1996			
☐	flightlevel, descend, aircraftnumber, center, +aircraft	System	1982			



ASRS Data



The ASRS safety reports generated 14 computer-derived topics. The Topics node in SAS Visual Text Analytics generates a set of numeric measurements for each document related to a set of computer derived or human generated topics. These numeric measurements can be used as inputs in a predictive model.



SAS Visual Text Analytics provides score code to generate the numeric measurements, and SAS Visual Data Mining and Machine Learning exploits these numeric measurements, using them as inputs for a predictive model that is constructed to accurately categorize the reports with respect to known safety labels.

Demo: Scoring ASRS Topics

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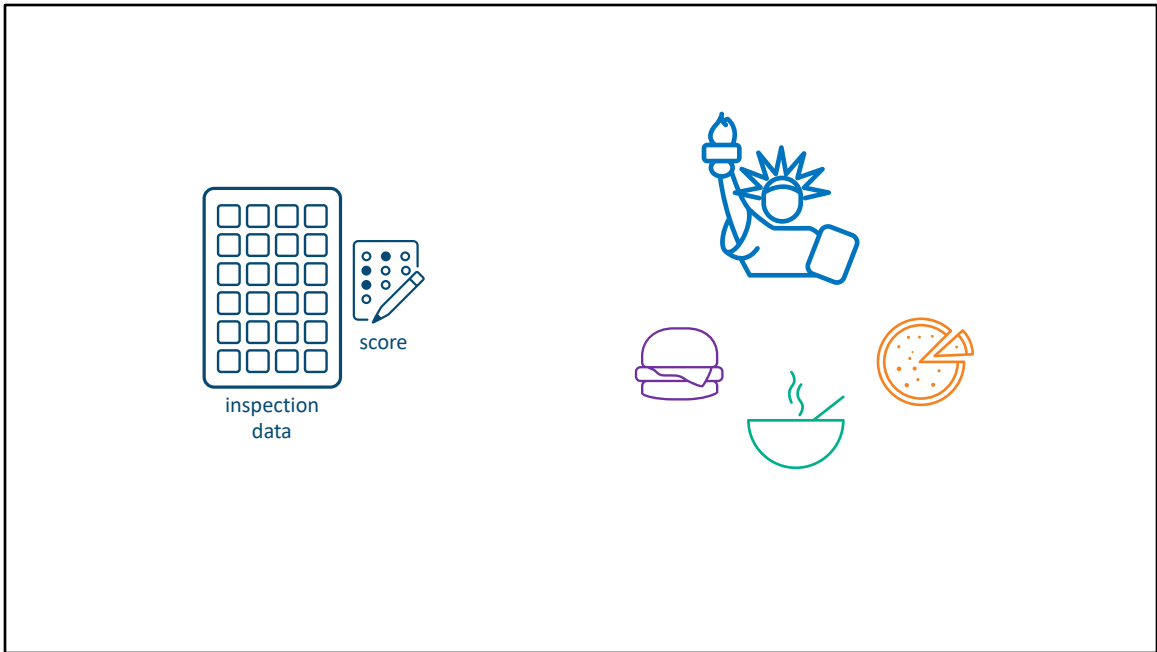
5.4 Scoring Categories



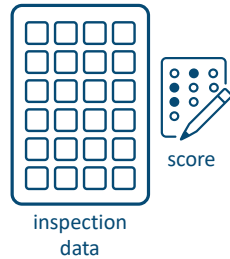
NYC Open Data Restaurant Inspections



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Now let's score a document collection with respect to categories. We'll use the New York City restaurant inspections data, which are available online as part of New York City Open Data. The data are provided by NYC Department of Health and Mental Hygiene.



- 21,544 restaurants as of January 19, 2020
- 20,530 restaurants in the analysis data set

The inspections were for 21,544 restaurants as of January 19, 2020. After cleaning up the data, only 20,530 restaurants remain in the analysis data set.

Demo: Scoring Categories, Part 1

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Demo: Scoring Categories, Part 2

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Question: Scoring Categories



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Question: Scoring Categories

Which of the following SAS Visual Analytics nodes does *not* produce score code?

- a. Concepts
- b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

Answer: Scoring Categories

Which of the following SAS Visual Analytics nodes does *not* produce score code?

- a. Concepts
- ☒ b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

b

The Text Parsing node creates the term table that is used by all subsequent nodes, but there is no score code for text parsing. The node serves as a building block for other nodes. All other nodes score documents to extract information or to classify documents with respect to categories or topics.

Lesson 6: Case Studies



6.0 Lesson Overview



Lesson Overview

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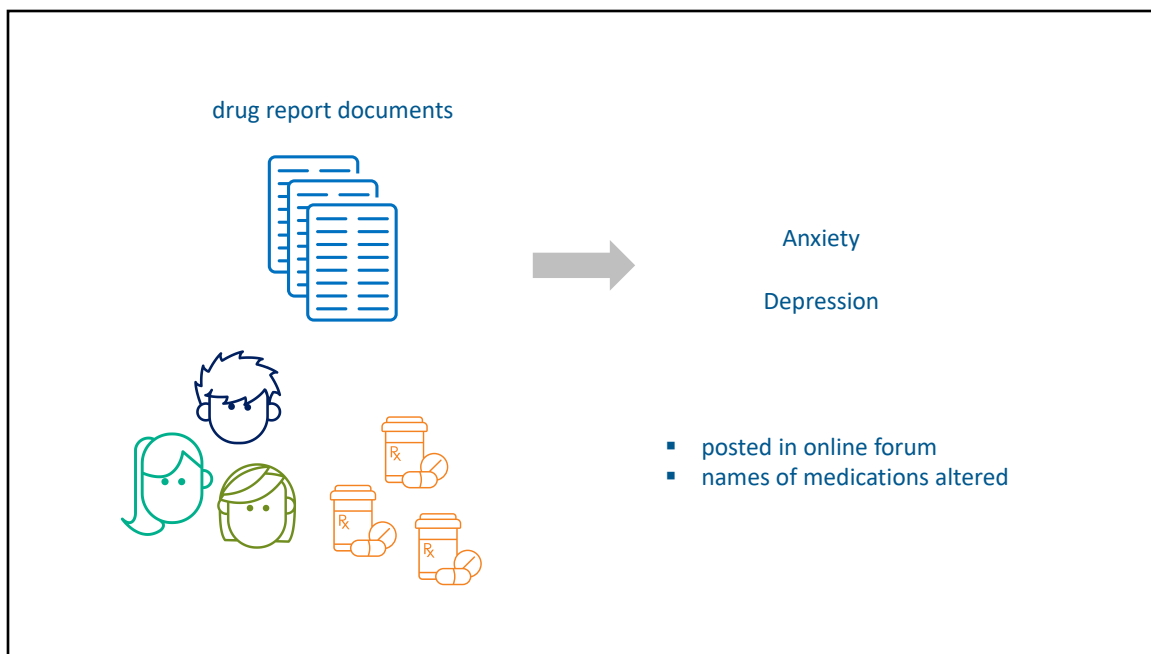
6.1 Information and Document Retrieval



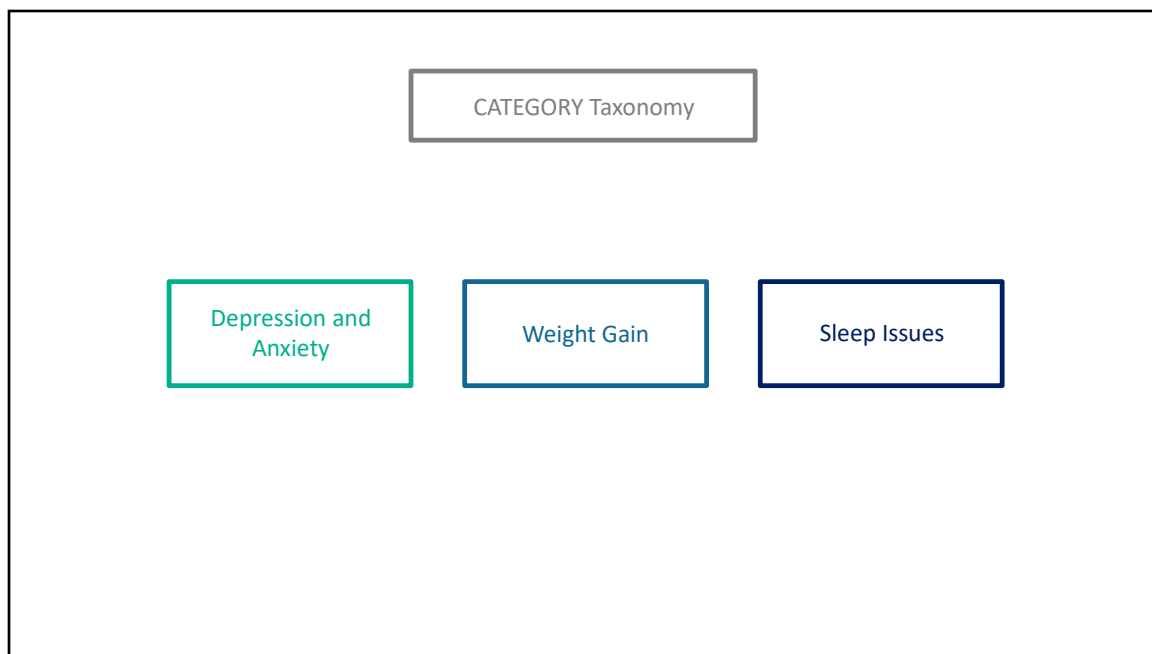
Drug Report Document Collection



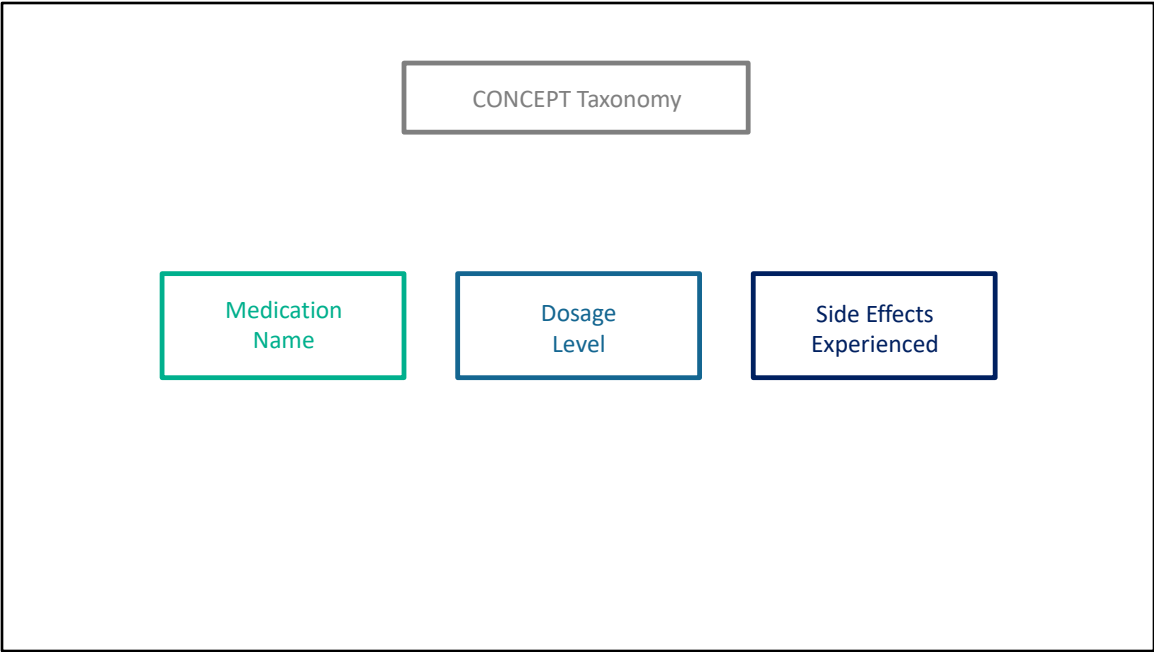
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In the first case study, we're going to retrieve information and documents about anxiety and depression from drug reports. The drug report documents contain comments from patients about prescription medications that are used for the treatment of depression. The comments were posted in an online forum, and the names of the medications that are referenced in the documents were altered. Rapid access to specific information can be enhanced through custom concepts, such as DOSAGE and PRESCRIPTION information.



Topics in the document collection include the following concerns: depression and anxiety, weight gain, and sleep issues. These will make up our Category Taxonomy.

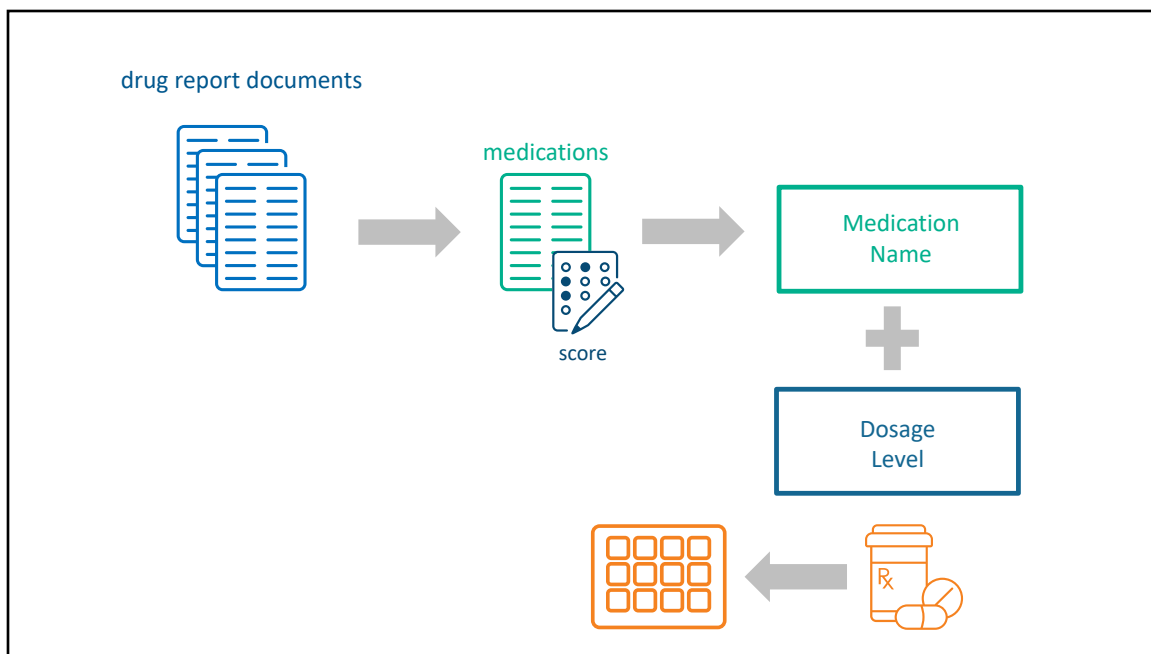


Some of the documents include the following key details: medication names, dosage levels, and side effects experienced. These will make up our Concept Taxonomy.

Concept	Examples
Medication	Abidal Escalan Cenerol
Dosage	20mg 60 MG 50 Mg
Prescription	Abidal 20mg 60 MG of Escalan 50 Mg Cenerol
SideEffects	Headache Blurred vision Nausea

Four custom concepts are defined in the Concepts node.

Medication with examples like Abidal, Escalan, Cenerol, etc. Dosage with examples like 20 mg, 60 MG, 50 Mg, etc. Prescription with examples like Abidal 20 mg, 60 MG of Escalan, 50 Mg Cenerol, etc. And SideEffects with examples like Headache, Blurred vision, Nausea, etc.



Our goal is to automatically extract information from the document collection. So for example, we'd like to obtain reports mentioning specific medications, so we have a Medication concept. When we score with respect to that concept, it's going to pull out all the medication names in the report. We will then combine that with dosage to get something called a Prescription. These concept rules will allow us to automatically parse the collection and create a table that has prescriptions in it. So we can immediately identify documents that mention a specific prescription, which would be a drug and a dosage.

Demo: Retrieving Information and Documents from Drug Reports, Part 1

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Demo: Retrieving Information and Documents from Drug Reports, Part 2

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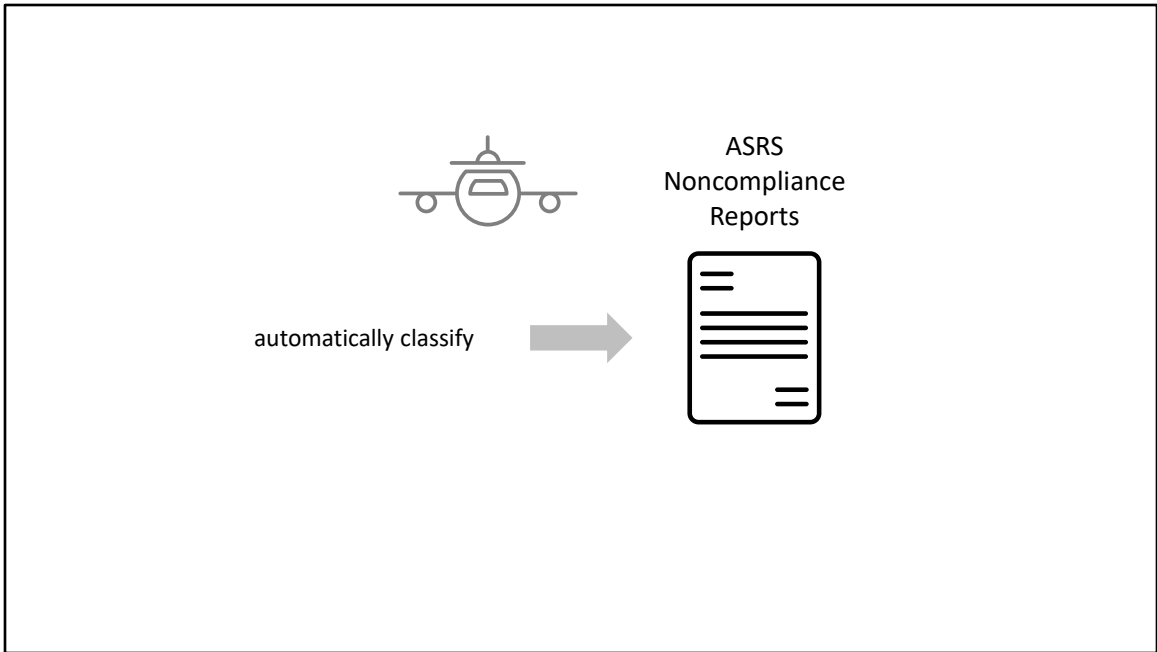
6.2 Automatic Categorization of ASRS Incident Reports



ASRS Noncompliance Reports



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We want to automatically classify ASRS procedure noncompliance reports. ASRS is the Aviation Safety Reporting System.



- fails to observe federal aviation regulation
- disregards an official notice
 - temporary runway closure
- fails to follow commands from air traffic controller
- violates temporary flight restriction
 - overflying major sporting event

The system is open to anyone who participates in aviation, including professional and private pilots, flight attendants, air traffic controllers, and airframe and powerplant mechanics. Procedure noncompliance occurs when a pilot fails to observe a federal aviation regulation, disregards an official notice such as a temporary runway closure, or fails to follow the commands given by an air traffic controller.

A common occurrence is to violate a temporary flight restriction, such as overflying a major outdoor sporting event. The ability to rapidly access specific safety information can facilitate timely action to prevent accidents.

Demo: Classifying ASRS Procedure Noncompliance Reports Automatically, Part 1

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Demo: Classifying ASRS Procedure Noncompliance Reports Automatically, Part 2

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Classifying ASRS Procedure Noncompliance Reports Automatically, Part 3

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Resources



Document Clustering



Resource

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Appendix A



Introduction

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Cheat Sheet Template: Concept Rules

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Cheat Sheet Template: Category Rules

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Appendix B



References

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